



EUROPEAN CENTRAL BANK
EUROSYSTEM

Working Paper Series

Massimo Ferrari Minesso, Daniele Siena

Private money and public debt. U.S.
Stablecoins and the global safe asset
channel

No 3174

Abstract

This paper studies the international macro-financial implications of U.S. dollar-backed payment stablecoins. These digital assets create a new *global safe asset channel* that links private money creation and global payment needs directly to U.S. public debt. By reshaping the demand for safe assets and the geography of dollar intermediation, stablecoins transform the dynamics of global financial markets, generating new trade-offs, also for the U.S.: even if they widen the dollar's global footprint and compress U.S. risk-free yields, they entail non-trivial macro-financial costs. Stablecoins dampen the domestic real effects of U.S. monetary policy and increase both U.S. and foreign exposure to cross-country shocks, making a more digital, dollar-centric reserve system less stable. These effects are limited at low adoption levels but rise non-linearly with stablecoin capitalization, reshaping the functioning of the international financial system.

Keywords: Stablecoins, Global Safe Asset, Monetary Policy, Spillovers, Financial stability

JEL Codes: G15, E42, E44, E52, F3

Non-technical summary

Recent technological innovation has made it possible for non-banks to issue new forms of private digital money known as stablecoins. These instruments are designed to maintain a stable value, typically by being backed one-to-one with safe and liquid assets such as U.S. Treasury bills. Although stablecoins originated as a tool for trading crypto-assets, their rapid expansion and the emergence of dedicated regulatory frameworks—most notably the U.S. GENIUS Act—have positioned them to move into mainstream domestic and especially cross-border payments.

The policy debate surrounding stablecoins is highly polarized. Supporters argue that they can deliver faster, cheaper and more accessible dollar liquidity across borders, potentially reducing frictions in international payments. Some policymakers emphasize the potential efficiency gains of a widely available, dollar-denominated digital instrument that combines safety and immediacy, echoing the perspective of [Waller \(2025\)](#). Others, however, underline substantial risks. A first set of concerns relates to financial stability: the absence of binding prudential requirements may amplify procyclical behavior ([Eichengreen et al., 2025](#)), increase risk-taking, and entrench global dollar dominance ([Rey, 2025](#)). A second line of critique points to technological fragilities: cyber vulnerabilities, AML/CFT, data and consumer protection and the risk of digital runs, especially during periods of market stress, as emphasized in [Cipollone \(2025\)](#).

Against this background, the transformation of stablecoins from a niche crypto tool into a globally used form of private digital money raises an important question: What happens when global demand for liquidity flows directly into U.S. public debt through privately issued digital dollars? Under the GENIUS Act, in fact, stablecoins must be fully backed by U.S. Treasury bills. As stablecoins expand, they therefore create a new channel through which global liquidity needs and global shocks influence the U.S. Treasury market. We call this mechanism the global safe asset channel.

When foreign users demand more stablecoins, issuers must buy more U.S. Treasury bills; when demand falls, they must sell them. As a result, the balance sheets of stablecoin issuers effectively connect global payment needs and portfolio decisions of households abroad with the dynamics of U.S. bond markets. This has two important implications. First, stablecoins may compress U.S. risk-free yields in the long-term, because they increase structural demand for short-term Treasuries. Second, changes in global demand

for liquidity now translate immediately into capital flows in and out of U.S. safe assets, that may amplify macroeconomic volatility. And most importantly, these flows become stronger, not weaker, as stablecoin adoption rises.

The paper studies these mechanisms in a multi-country DSGE model in which households use both money and stablecoins, issued only by the U.S., to meet liquidity needs. Stablecoins have a dual nature of both a settlement token, i.e. they are used for payments, and of a trading token, i.e. dollar-denominated store of value asset. They are however imperfect substitutes for money because households value them less when confidence falls—capturing their vulnerability to runs or technological disruptions—while issuers fully back each coin with short-term U.S. Treasuries. This setup allows us to trace how monetary policy transmission changes once stablecoins become a significant part of global liquidity creation. The model embeds stablecoins reflecting their regulatory design under the GENIUS Act: full backing, zero remuneration, and direct issuance to households. Importantly, stablecoins differ from central bank digital currencies (CBDCs) because the latter are design to avoid impacts on the demand for safe assets and therefore on the volatility or level of interest rates. In other words, CBDC are designed to be neutral from a monetary policy prospective.

Three main findings emerge.

First, stablecoins increase the pass-through of U.S. monetary policy to Treasury yields, because changes in interest rates induce stablecoin issuers to adjust their bond holdings more aggressively. At the same time the real effects of monetary policy become weaker, as capital flows associated with stablecoins dampen exchange-rate movements. The monetary authority would therefore face a sharper policy trade-off: stronger reactions to inflation may be required to achieve the same degree of macroeconomic stabilization.

Second, international spillovers strengthen in both directions. U.S. shocks transmit more forcefully abroad because stablecoin redemption or issuance triggers capital flows in and out of U.S. assets. Conversely, foreign shocks—such as shifts in payment needs or risk sentiment—transmit back into U.S. yields more directly than in the pre-stablecoin world. This two-way amplification arises because stablecoins link global liquidity demand to the U.S. Treasury market in real time.

Third, these effects are non-linear. At low levels of adoption, stablecoins behave mostly like payment tools. But as their market size grows, they increasingly act as global

financial assets, used for portfolio diversification. This greatly magnifies the portfolio adjustments required after shocks, and thus the size of flows into and out of U.S. safe assets.

Overall, the paper shows that the expansion of stablecoins can reshape global financial integration, monetary policy transmission, and the stability of the U.S. fiscal-monetary nexus. These findings are directly relevant for policymakers evaluating the international consequences of private digital money, and the design of regulatory frameworks aimed at balancing innovation with macro-financial stability.

1 Introduction

Recent technological innovation has allowed the private sector to issue a new digital form of money, so-called stablecoins.¹ For the moment, they mostly facilitate transactions between volatile crypto assets, avoiding the need for repeated conversion into FIAT money (Mizrach, 2025).² However, stablecoins are now gaining ground in mainstream finance and global payments, mainly triggered by recent regulatory decisions. While the U.S. GENIUS Act (i.e. Guiding and Establishing National Innovation for US Stablecoins), approved in July 2025, provided the necessary framework for the diffusion of U.S. dollar stablecoins, the need for a fast, low-cost, stable and around-the-clock cross-border medium of exchange is doing the rest.³ By imposing remuneration rules (no remuneration), backing requirements (U.S. safe and liquid assets), and a clear legal framework for issuance and supervision, the aim of the U.S. administration was to support dollar-stablecoin adoption worldwide, strengthening the U.S. dollar's dominant role as the world's currency. However, this innovation has the potential to radically change the dynamics of global financial markets by triggering new trade-offs. The aim of this paper is to study whether and how the growing relevance of stablecoins matters for the transmission of U.S. monetary policy, for the propagation of shocks across countries and for macroeconomic stability.

Our central insight is that U.S. dollar-backed payment stablecoins create a new *global safe asset channel* that links private money creation and global payment needs directly to U.S. public debt. This generates a new form of convenience yield due to demand for liquidity services on U.S. debt, which is held indirectly by holders of stablecoins. Using an open-economy multi-country model in which U.S. stablecoins are fully backed by U.S. Treasuries and used both as settlement tokens (payment tools) and as trading tokens (store of value), we show that their widespread adoption would reshape the dynamics of global financial markets. From a geoeconomic perspective, the global safe asset channel we

¹Throughout the paper, “stablecoins” refers to USD-pegged payment stablecoins under the U.S. GENIUS Act: unremunerated and fully backed by U.S. safe assets. More generally, stablecoins are crypto-assets pegged to a reference value and usually backed by short-term safe assets (Scotti, 2025), though some use commodities, other crypto-assets, or algorithms. Under the EU MiCAR framework, reserves may mix bonds and bank deposits.

²Only a fraction, about USD 10 billion, of the market capitalization of stablecoins is currently estimated to be used in payments outside the crypto-world.

³The average cost of retail transactions ranges between 2 and 2.6% with 24% of corridors with costs exceeding 3% with peaks of 4% in South America or Sub-Saharan Africa. 30% of retail payments also take more than one day to execute, with peaks of 100% in some regions of Africa. Sending remittances is even more expensive (up to 12% of the transacted amount), slow and more difficult to access (see Financial Stability Board (June 2025)).

identify ties the functioning of the world's payment system even more tightly to U.S. fiscal capacity and regulatory choices, with important consequences for the durability of dollar hegemony, the scope for financial sanctions, and the incentives to develop alternative payment and reserve systems (Farhi and Maggiori, 2018; Clayton et al., 2024).

The intuition of the main mechanism highlighted in this paper is straightforward. Being backed by U.S. safe assets, swings in demand for stablecoins directly affect the demand for U.S. short-term Treasuries and dollars. In steady state, this benefits the U.S. by compressing risk-free yields, more the larger is their market capitalization. But the diffusion of stablecoins also has costs. Any shock that appreciates (depreciates) the dollar creates positive (negative) valuation effects for foreign holders of dollar-backed stablecoins. This incentivises foreign holders to redeem stablecoins for two reasons: (i) to realise valuation gains on their positions (in their role as trading tokens), and (ii) because, after an appreciation, they need fewer stablecoin units to settle a given amount of domestic-currency payments (in their role as settlement tokens). As the stablecoin issuer needs to sell U.S. bonds to meet redemptions and pay back foreign holders –and, symmetrically, to buy bonds when foreign demand for stablecoins rise– changes in stablecoin demand lead to capital outflows (inflows) from (into) the U.S.. These movements become stronger the more stablecoins are held in equilibrium, as they end up serving relatively more as international financial assets than as payment instruments. As a result, in the U.S., as equilibrium holdings of stablecoins grow, the pass-through of monetary policy to yields increases –amplified by stablecoin-driven Treasury demand movements– but its effects on real variables fall, due to dampened exchange rate responses. Therefore, the U.S. monetary policy becomes less effective in steering aggregate demand, by up to 30% less, and a stronger reaction to inflation is needed to preserve macroeconomic stability. Internationally, stablecoins strengthen the cross-country transmission of shocks both from the U.S. to the rest of the world and from foreign economies to the U.S. These effects are small at low adoption levels, but increase non-linearly with stablecoin adoption.

At present, stablecoins are still far from a large adoption scenario, but their global footprint has expanded significantly. U.S. stablecoin capitalization has risen rapidly surpassing \$260 billion in November 2025 (Figure 1a). Because they are typically backed by U.S. dollar denominated Treasuries, stablecoins issuers are becoming comparable for size to sovereign investors (Ahmed and Aldasoro, 2025), surpassing countries like Norway

or Brazil (Figure 1b). If stablecoins were to deliver on their promises, their market is expected to grow up to USD 4 trillion by 2028.⁴ In the minds of U.S. regulators, the GENIUS Act would allow stablecoins to move from a fringe corner of the global economy (crypto-asset trading) to mainstream finance, becoming a vehicle for domestic and, even more, international transactions and unlocking *additional* demand for dollars.⁵ As mentioned, a crucial element of the GENIUS Act is the requirement to fully back each coin with liquid U.S.-dollar-denominated assets, ensuring that dollar assets and liabilities match. This implies that stablecoin issuers could soon become major marginal buyers of U.S. government bonds. Figure 2 shows that while Money Market Funds still hold larger amounts of U.S. debt, major stablecoin issuers are beginning to emerge among significant holders. Under the GENIUS Act, issuers are not required to be registered banks, which opens the door for large technology companies to issue stablecoins. At the same time, since stablecoins are unremunerated, they are unlikely to be viewed as perfect substitutes for bank deposits as a store of value.⁶ It is important to note that we intentionally abstract from the effects on banks (Liang et al., 2024), disintermediation and legal arbitrage that can arise from multi-issuer stablecoins (Portes, 2025); our results show that the global safe asset channel reshapes bond-market transmission and global financial linkages even under these conservative assumptions.⁷

The diffusion of stablecoins crucially hinges on their promise to improve access to dollar liquidity worldwide⁸ and to serve additional demand for dollars, much like the eurodollar market was developed for in the 1960s.⁹ Similarly to the debate that emerged

⁴J.P. Morgan early projections implied that the stablecoin market could reach \$500 billion by 2028, Coinbase estimated the market to rise to 1.2 trillion by 2028, Citigroup to 1.6 to 3.7 trillion by 2030. Similarly, Standard Chartered estimated the market to be \$2 trillion by 2028 and Bernstein \$4 trillion by 2035 (Singh, 2025).

⁵Early use of stablecoins outside crypto-asset trading for international trade transactions is limited to cases of sanction evasion, [Kremlin-backed crypto coin moves \\$6bn despite US sanctions](#), *Financial Times*, 6 October 2025.

⁶Although the GENIUS Act prohibits the payment of interest on stablecoins, several commentators have noted that issuers or affiliated platforms may attempt to offer indirect forms of remuneration—such as rewards, fee rebates, or yield products linked to the stablecoin’s reserve portfolio—thereby circumventing the spirit of the regulation.

⁷Multi-issuer stablecoins arise when Electronic Money Tokens can be issued in multiple jurisdictions by different legal entities but are de facto identical. Disintermediation risk and multi-issuer arrangements are extensively studied in the ESRB report (European Systemic Risk Board, 2025).

⁸The U.S. dollar is central to the global monetary system, comprising 50% of international loans, over half of foreign reserves, 90% of FX transactions, and 40% of trade invoicing (European Central Bank, June 2025). This dominance fuels global demand for dollar liquidity to settle liabilities and hold assets.

⁹The 1963 Interest Equalization Tax drove international borrowers to less regulated markets, making London a hub for eurodollars—dollar-denominated deposits outside the U.S. These offered tax benefits,

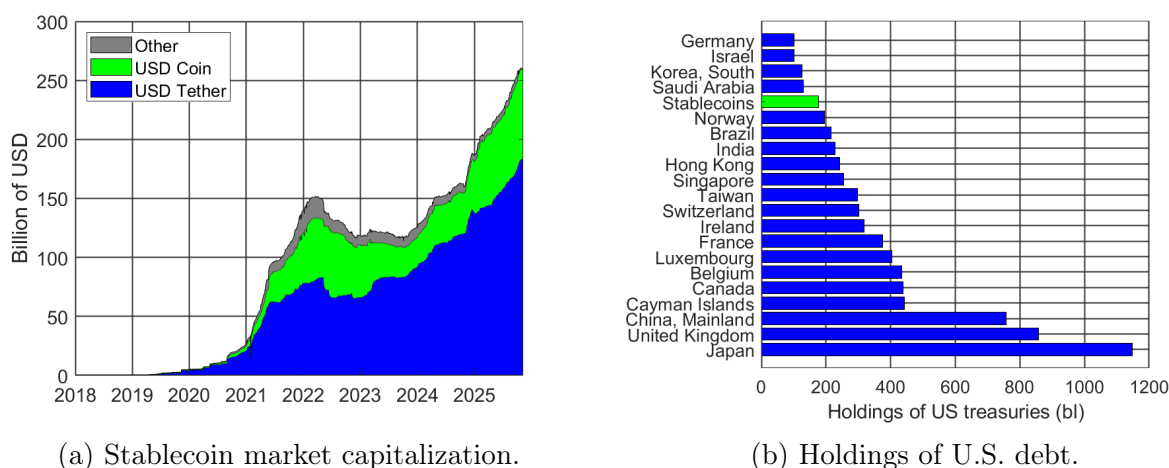


Figure 1: Stablecoin market capitalization and holdings of U.S. debt by countries and stablecoin issuers.

Notes: Panel a, billion U.S. dollars. Sources: Chainalysis, CryptoCompare, CoinGecko, IntoTheBlock. Last observation 11 November 2025. Panel b: holdings as of June 2025 in billion U.S. dollars. Sources: U.S. Treasury (TIC); “Stablecoins” include reported holdings of Tether (USDT) and USD Coin (USDC).

at that time, stablecoins now need to be studied to assess their potentially destabilizing role for the functioning of the U.S. money market. In the 1960s, some argued that eurodollar liquidity was doomed to drain funds from other segments of U.S. markets, as eurodollar banks’ deposits needed backing (Frydl, 1979). Others, like Friedman (1971), were less concerned, viewing eurodollar banks simply as another type of fractional-reserve bank.¹⁰

Similarly, the emergence of stablecoins has sparked an intense debate among economists and policymakers about their potential risks and benefits. Some warn that the absence of binding capital and liquidity requirements may generate cycles of excessive risk-taking and strengthen the dollar’s dominance (Rey, 2025; Eichengreen, 2025; Eichengreen et al., 2025), while others emphasize the possibility of cyber runs linked to technological vulnerabilities (Rogoff et al., 2025; Cipollone, 2025) or the inability to maintain the peg during stress episodes (Kosse et al., 2023). At the same time, stablecoins may offer efficiency gains by providing a dollar-denominated digital asset that combines safety and

anonymity, and freedom from controls, boosting global finance and cementing London as a financial center. During the Cold War, Soviet and Eastern Bloc countries sought dollar reserves beyond U.S. reach, while European and Asian banks met rising demand for dollar financing, fueling offshore markets and complicating global financial oversight.

¹⁰Over time, both visions proved to be partially right. On one hand, the eurodollar market became an institutionalized central mechanism for global dollar liquidity, enabling firms, governments, and financial institutions worldwide to access U.S. dollars without operating directly within the U.S. banking system. However, on the other, it proved to be an important channel for the transmission of both U.S. monetary policy and U.S. shocks across the globe (e.g. during the 2008 global financial crisis, see McCauley (2024)).

immediacy ([Waller, 2025](#); [Ahmed et al., 2025](#)) and could push down U.S. yields in the long run ([Azzimonti and Quadrini, 2025](#)). In this paper, we focus on whether and how the growing relevance of stablecoins matters for monetary policy transmission, and what implications it has for macroeconomic stability and for the cross-country propagation of shocks. These questions echo those raised in the past regarding eurodollar banks, but stablecoins differ in three significant ways.

First, while eurodollar banks eventually operated with fractional reserves in dollar-denominated assets ([Friedman, 1971](#)), stablecoins are expected to be fully backed on a one-to-one basis by assets, mostly U.S. short-term bonds. This requirement would amplify fluctuations in the demand for U.S. dollar assets in response to changes in international stablecoin demand. As a result, unlike eurodollar banking, an increase in the international role of the dollar due to stablecoins would necessarily imply a larger U.S. current account deficit. Second, stablecoins are unremunerated (both under the GENIUS Act and the EU's regulation of stablecoins, MiCAR). This means that issuers pocket the full returns from reserve assets. Consequently, the direct benefits (or costs) of shifts in U.S. financial conditions are absorbed by stablecoin issuers, and not by end-users.¹¹ However, as mentioned, non-U.S. stablecoin holders would also be affected through exchange rate movements. Third, and last, while eurodollars were primarily accessible to banks, stablecoins are designed to be issued directly to retail users and households. This could structurally increase the demand for dollars and create a direct link, via stablecoin issuers' balance sheets, between the economic cycles of foreign countries and demand for U.S. bonds. Historically, this connection was only indirect, mediated by imperfect cross-border bond markets and banks, which however only held fractional reserves of bonds.

It is important to notice that stablecoins are also different from Central Bank Digital Currencies (CBDC). The issuance of a CBDC, in fact, does not generate additional demand for safe assets. This is because agents directly swap liabilities with the central bank when acquiring CBDCs, leaving asset prices unchanged, see [Adalid et al. \(2022\)](#).¹²

¹¹This is meant to prevent them from competing directly with dollar deposits in their role as a store of value. As a result, the risks of stablecoins disintermediating banks remain limited (since deposits are remunerated), and they are more likely to substitute cash-like instruments.

¹²CBDCs can be issued through several channels: (i) in exchange for banknotes, (ii) in exchange for bank reserves, (iii) against additional borrowing by banks from the central bank, or (iv) in exchange for assets sold by banks to the central bank. The first two cases represent a substitution among existing central bank liabilities. Under the third, the central bank effectively re-injects into the banking system

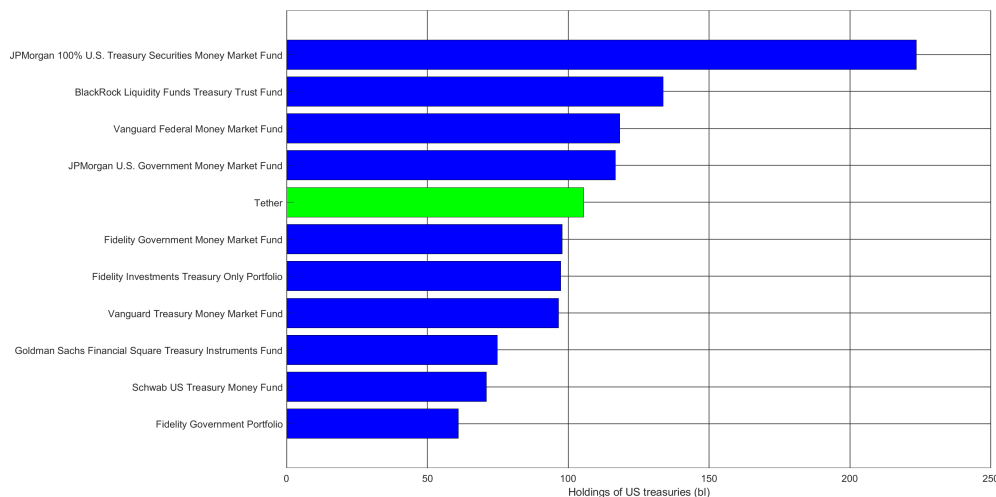


Figure 2: Stablecoin holdings of U.S. Treasury bills compared to major hedge fund holdings, 2024.

Sources: Bloomberg.

Stablecoins, on the contrary, directly impact asset prices as each new coin minted (or destroyed) requires a proportional purchasing (or selling) of backing assets to the market.

These structural differences between eurodollars, CBDC and stablecoins have profound implications, not only for the foreign transmission of U.S. shocks, but also for how business-cycle fluctuations in third countries affect the U.S.. In particular, in this paper, we aim to address four precise questions: (i) How does the presence of stablecoins change the transmission of US monetary policy, domestically and internationally? (ii) How should the Federal Reserve adapt its monetary policy strategy in a world where stablecoins play a growing role in liquidity creation and fiscal financing? (iii) Does the dependence of U.S. government debt markets on stablecoin demand introduce new volatility channels for the dollar and for Treasury yields? (iv) What happens if real Treasury bill rates turn negative, reducing issuers' profits and leading households to demand a risk premium, possibly triggering a run?

We address these questions using a multi-country DSGE model in which stablecoins enter the cash-in-advance constraint alongside money, acting as an imperfect substitute for cash as liquidity instrument. We take the perspective of a successful diffusion of stablecoins to mainstream markets, with households using them in transactions and an

the liquidity withdrawn through the conversion of deposits into CBDC. In the fourth, the transfer of assets from banks to the central bank merely alters ownership without exerting meaningful effects on asset prices.

adoption in line with current and projected numbers (Birch, 2025). In the model, households derive utility from holding stablecoins through the liquidity services they provide (i.e. stablecoins are a settlement token) but value them less than money when confidence declines, making them imperfect substitutes in the liquidity aggregator. A confidence parameter, $\mu_{sc,t}$, captures this perception of safety and reliability: when the risk of valuation losses or redemptions increases, the confidence falls, reducing the effective liquidity yield of stablecoins. At the same time, stablecoins can also be used by households to carry wealth, in dollars, from one period to another, i.e. as a trading token. On the supply side, each coin is fully backed by short-term U.S. Treasuries, linking private liquidity creation directly to the fiscal position of the U.S. government. The remainder of the model builds on Ferrari Minesso et al. (2022), Agur et al. (2022), and Niepelt (2024). This framework allows us to study how monetary policy transmission changes when the demand for Treasuries depends on stablecoin growth, and to identify conditions under which the system is prone to instability. Our contribution is to show that stablecoins amplify the fiscal-monetary nexus, generating new trade-offs for the Federal Reserve and new risks for global financial stability.

We find that stablecoins create a global safe asset channel that links private money creation directly to U.S. public debt and reshapes international spillovers. On the U.S. side, the presence of stablecoins increases the pass-through of domestic monetary policy to short-term yields but dampens its real effects. This is because foreign investors, in response to a U.S. tightening (easing), redeem (buy) stablecoins, prompting issuers to sell (buy) U.S. bonds and convert more (less) dollars into foreign currencies. This raises (lowers) U.S. yields but limits dollar appreciation (depreciation), cushioning the output decline (increase) by about 30%. On the spillover side, when a shock is generated in the foreign economy, stablecoins increase the pass-through to the U.S. economy: when payment needs abroad fall (rise), foreign agents sell (buy) stablecoins and, one-for-one, issuers sell (buy) Treasuries, transmitting shocks directly to U.S. yields. This mechanism increases the sensitivity of U.S. financial conditions to global shocks and tightens the two-way link between the U.S. and the rest of the world.

The global safe asset channel arises from this cross-country arbitrage condition introduced by stablecoin adoption. We show analytically that these effects are driven by the concavity of preferences over liquid assets: the larger the steady-state holdings of

stablecoins, the larger the portfolio adjustments required to re-equilibrate after shocks. As agents hold larger quantities of stablecoins, their use for payments declines and their role for portfolio diversification increases, making stablecoin holdings more responsive to changes in monetary conditions. Since stablecoins are fully backed by U.S. assets, these adjustments translate directly into fluctuations in Treasury demand and yields. This feature distinguishes stablecoins from CBDCs, where issuance is sterilized by the central bank and does not require one-for-one purchases of U.S. debt (Ferrari Minesso et al., 2022).¹³ By contrast, stablecoin issuers buy and sell U.S. Treasuries for each new coin minted, leading to direct changes in Treasury demand after shocks.

Related literature. This paper is linked to the growing literature on the macro-financial role of stablecoins. One strand of literature has considered the characteristics, dynamics and implications of stablecoins balance sheets. Oefele et al. (2024) and Mizrach (2025) document that typically stablecoins hold more cash than most government funds and experience flight-to-quality dynamics during episodes of stress in crypto markets, Anadu et al. (2024). Additionally, Liang et al. (2024) show that stablecoins reduce banks' deposit market power, thereby intensifying the transmission of monetary policy by making banks respond more strongly to policy rate changes, leading to higher deposit rates and greater loan contractions.

The literature has highlighted potential risks for stability and for the peg (Eichengreen et al., 2025; Kosse et al., 2023; Arner et al., 2020; d'Avernas et al., 2022; Hoang and Baur, 2024; Lyons and Viswanath-Natraj, 2023); adoption (Bertsch, 2023), and runs (Gorton et al., 2022). Lessons for today's stablecoins have also been drawn from history, for example from the free banking era in the U.S. (Luck, 2025) or even further back in time to the collapse of the Bank of Amsterdam, Bolt et al. (2024).¹⁴ Altavilla et al. (2025) shows empirically the spillovers from shocks to stablecoin demand to the real economy, in particular thorough their impact on the financial system.¹⁵

More recently stablecoins have been studied in relation to the demand for safe as-

¹³Concrete CBDC projects, like the digital euro, explicitly target neutrality with the objective of keeping CBDCs only as a payment instrument, European Central Bank (2025).

¹⁴There is also a growing literature on the economics of crypto-assets: Karau (2023) looks at the reaction of the Bitcoin price to monetary policy while other contributions (Liu and Tsyvinski, 2021; Corbet et al., 2020) consider the implications of crypto-assets for traditional asset prices.

¹⁵Specifically, in this empirical setting, stablecoin growth diverts deposits from banks, reshapes their funding and lending capacity, and ultimately weakens deposit-based while strengthening lending-based monetary policy transmission—especially if foreign-currency stablecoins dominate.

sets. There is evidence that demand for dollar liquidity suppresses Treasury yields ([Krishnamurthy and Vissing-Jorgensen, 2012](#)), including from outside the U.S. ([Ahmed and Rebucci, 2024](#)), with potential implications for financial stability, [Greenwood et al. \(2015\)](#). [Eren et al. \(2025\)](#), importantly, estimate the elasticity of U.S. yields to demand from different investor types. Demand from money market funds, in particular, is shown to influence the price of other liquid assets ([Doerr et al., 2023](#)), a mechanism that could apply to stablecoins if their adoption were to extend outside crypto markets. Recently, empirical evidence suggests that stablecoins already influence US safe asset markets. [Ahmed and Aldasoro \(2025\)](#) and [Kim \(2025\)](#) show that stablecoin inflows reduce three-month Treasury yields, while outflows have asymmetric and stronger effects. [Aldasoro et al. \(2025\)](#) document that stablecoins and money market funds respond differently to US monetary policy shocks: while prime-MMF assets tend to rise after a monetary tightening, stablecoin capitalization declines. Relatedly, [Azzimonti and Quadrini \(2025\)](#) investigate from a theoretical perspective how stablecoins could affect the centrality of the US debt in global financial markets in the long-run. By developing a multi-country model, they find that the financial demand (reserve banking) will dominate over the real demand (service substitution), structurally increasing global demand for US dollar safe assets. This dominance would lead to a decline in long-run US interest rates, greater US foreign borrowing, and ultimately reinforce the US dollar's "exorbitant privilege" in the global financial system. From a different perspective, but also focusing on the supply of safe assets from the U.S., [Barthélemy et al. \(2025\)](#) build on the international monetary system framework of [Farhi and Maggiori \(2018\)](#) and endogenize the hegemon's debt issuance in the presence of fully backed stablecoins. They show that while stablecoins are neutral for total debt issuance unless they attract new entrants or the hegemon values stablecoin profits, once these conditions are met they introduce a non-trivial trade-off: the issuer is tempted to over-issue debt to capture stablecoin rents but is also pushed-back by the need to avoid devaluation and preserve the safe-asset status of stablecoins. This has ambiguous implications for Triffin Dilemma (i.e. the conflict between domestic policy goals and the need to supply safe reserve assets to the rest of the world) and the stability of the international monetary system. These papers connect the discussion on stablecoins with the broader literature that studies the role of dominant currencies and dollar-denominated safe assets in shaping global portfolios and capital allocation ([Gabaix](#)

and Maggiori, 2015; Maggiori, 2017; Maggiori et al., 2020).

Our paper is also related to the growing literature on the macroeconomic implications of digital assets. Brunnermeier and Niepelt (2019) derive equivalence conditions under which a particular type of digital asset, a central bank digital currency (CBDC), can be economically neutral. In our case one of their conditions does not hold, namely the neutrality of the digital asset demand for equilibrium allocations. Benigno et al. (2022) show that a remunerated global digital asset would equalize asset returns across countries leading to a loss of monetary autonomy. Ikeda (2022) shows that these effects can be limited by the role of government in collecting taxes and financing public goods. Ferrari Minesso et al. (2022) find that a global CBDC would also create a new cross-country asset holding conditions that constrains policy autonomy abroad. Kumhof et al. (2023) find similar results in a different setting and experimenting with policy rules. Stablecoins are however different from both cases: they are unremunerated, fully backed by assets and not issued by a public institution which aims at “market neutrality”. Our results point therefore to slightly different macroeconomic implications of stablecoins that derive from them connecting consumption in foreign countries with the U.S. treasury market. This paper also draws on the literature on CBDC for the modelling of liquidity demand and the characterization of digital assets, like Niepelt (2024) and Burlon et al. (2024).¹⁶ Possibly more related to our work is Cova et al. (2022). This early contribution, however, studies CBDCs against a global crypto-asset which differs significantly from the current configuration of stablecoins: backing is not perfect, the issuers optimizes its holding of different assets (money, domestic and foreign bonds) to maximise profits, and more importantly the exchange rate between the crypto-asset and the domestic currency is set optimally by the issuer, i.e. it is set to maximise profits of the issuer and can therefore absorb part of the shocks. Under these characteristics, the issuer does not back fully minted coins with bonds and when the bond price soars, optimally shifts to other source of liquidity (cash) which are price insensitive. In this way, the link between foreign demand and U.S. bond markets is broken, while we think it is one of the key characteristics of the GENIUS Act. The connection of balance sheets movements with exchange rates and capital flows borrows a lot from Maggiori (2022).

¹⁶The literature on CBDC is indeed much larger and ranges from transition dynamics, Assenmacher et al. (2024), to implications for monetary operations, Assenmacher et al. (2023) Abad et al. (2024) to competition and financial stability, Andolfatto (2021) and Schilling et al. (2024).

The remainder of the paper is structured as follows. [Section 2](#) discusses the main macro-financial implications of stablecoins, emphasizing those analyzed in this study. [Section 3](#) introduces the model, describing its key features and the equilibrium dynamics associated with the role of stablecoins. [Section 4](#) presents the simulation results. It first examines the effects of U.S. monetary policy and TFP shocks within and outside the United States, and then analyzes how stablecoins transmit foreign shocks back to U.S. financial markets. [Section 5](#) provides empirical evidence on the impact of stablecoins on the transmission of U.S. and third country monetary policy shocks to U.S. Treasury yields. Finally, [Section 6](#) concludes by summarizing the macro-financial implications of this new global safe asset channel.

2 Macroeconomic Implications of stablecoins

The GENIUS Act establishes a formal regulatory foundation for U.S. dollar-backed stablecoins, enabling their transition from niche crypto-market settlement tools to broadly used digital payment instruments. In fact, the new framework was explicitly constructed to allow stablecoins to evolve into a mainstream means of U.S. digital dollar payments for trade, remittances, and cross-border transactions. The implicit goal was also to reinforce the global role of the U.S. dollar and increase demand for U.S. government debt. Importantly, issuance is now open to non-bank entities, which may accelerate its diffusion, but also raises risks, especially in economies where cross-border payments remain slow and costly. These developments carry significant macroeconomic implications, for U.S. monetary and financial conditions and for international spillovers. The discussion that follows outlines the principal channels through which expanding stablecoin use may influence the U.S. economy, global dollar dynamics, and international financial linkages. We do so by consciously being silent on all financial risks arising from international regulatory arbitrage or multi-issuer risks ([European Systemic Risk Board, 2025](#)).

We start by highlighting the **differences with other forms of dollar assets**. Stablecoins importantly differ from (i) international dollar deposits, (ii) money market fund (MMF) shares, (iii) U.S. bond holdings, and (iv) hypothetical dollar CBDC. (i) Their global, app-based accessibility and 24/7 transferability enable faster and lower-cost cross-border payments than international dollar bank deposits. Unlike bank deposits, more-

over, stablecoin units are fully backed by dollar assets. This makes stablecoin issuers potentially larger players in the Treasury market than Eurodollar banks. (ii) Because stablecoins promise on-par convertibility –backed one-for-one by dollar assets– and are not officially remunerated, their function is closer to digital money than to investment products. In fact, While MMF assets increase after a monetary policy tightening (as deposit rates lag policy rates), stablecoin market capitalization significantly declines. Additionally, unlike MMFs, whose investors bear portfolio losses, stablecoin issuers must absorb asset-side shortfalls to maintain parity, creating run-risk dynamics specific to privately issued par-money. (iii) U.S. stablecoins can be used for transactions, connecting pure U.S. debt holdings to liquidity services. (iv) Finally, stablecoins are different from CBDC. In fact, CBDCs constitute a direct central-bank liability and need not be backed by Treasuries on a one-to-one basis (Adalid et al., 2022); as a result, a CBDC would operate under a fundamentally different balance-sheet regime and would not generate the same incremental demand for U.S. government securities as stablecoins do.

Expanded global use of stablecoins would **increase demand for Treasuries but also the risk of making it more volatile**. In fact, as for other safe dollar assets, an increase in demand would compress yields, while fluctuations in stablecoin balances –especially from foreign holders– could heighten Treasury market volatility. In particular, stronger structural demand at the front end would compress short-term yields and, via expectations and portfolio-balance channels, may flatten the long end of the curve, influencing monetary transmission to long-term borrowing costs. This could generate an important direct channel of transmission from foreign demand to the U.S. bond market, amplifying two-way spillovers between the United States and the global economy. This effect amplifies non-linearly as stablecoins constitute a larger share of household portfolios.

Backing requirements favouring liquid short-term U.S. debt may intensify demand for bills, incentivize the sovereign to issue more short-term debt. This could **increase system-wide exposure to rollover and duration risks**; notably, Tether, which is issued outside the GENIUS Act, already allocates almost 80% of its portfolio to short-dated instruments. Additionally, stablecoin-driven demand for Treasury bills would **exacerbate the global scarcity of high-quality collateral**, placing further downward pressure on U.S. bond yields.

While these were implications affecting mostly the U.S. economy, consequences could also be felt in the rest of the world. In fact, in high-inflation economies, the frictionless access to dollar-backed stablecoins via mobile wallets could **accelerate dollarization**, eroding non-U.S. monetary autonomy and amplifying the international reach of U.S. monetary policy. In jurisdictions without access to Federal Reserve swap lines, this greater dependence on dollar liquidity may translate into heightened financial fragility during global stress episodes. Stablecoins may also sustain the dollar's primacy in global payments and reserves. By offering a safe, convenient digital dollar, they counter the advance of foreign digital currencies and extend U.S. monetary influence abroad. Assuming momentarily that stablecoins would increase the net world demand for U.S. assets ([Azzimonti and Quadrini, 2025](#)), this would **preserve the dollar's "exorbitant privilege"** and the associated convenience yield on U.S. liabilities.

Additionally, the widespread foreign adoption of dollar-backed stablecoins would introduce an additional transmission channel of U.S. monetary policy to foreign households, beyond traditional trade and financial linkages. An example can summarize the intuition. Higher U.S. interest rates appreciate the dollar. This generates capital gains for foreign stablecoin holders, partly offsetting contractionary financial spillovers. Symmetrically, U.S. easing imposes wealth losses that dampen the expansionary spillovers. Large-scale use of stablecoins internationally would also generate new capital flows to and from the U.S. when coins are purchased or redeemed, impacting the dollar exchange rate. This would imply also that stablecoin demand responds non-linearly to short-term interest rates: higher rates raise the opportunity cost of holding stablecoins, as they are not remunerated, yet simultaneously bolster confidence by improving issuer carry and easing redemption pressures. For foreign users, rate hikes also generate capital gains via dollar appreciation, adding a wealth channel to demand. This asymmetry can induce liquidity imbalances, with elevated demand and issuance in tightening cycles and depressed demand in easing phases. As mentioned, however, all these channels would be absent if international investors simply changed the composition of their international portfolios, substituting stablecoins with already-held dollar assets.

Other potential challenges are related to the technology used. By enabling instantaneous global portfolio rebalancing, stablecoins may **increase the speed and amplitude of capital flows**, potentially magnifying price swings in U.S. bond and foreign-exchange

markets and increasing the likelihood of short-lived instability episodes that are difficult to address with conventional lender-of-last-resort tools. Related to this, with stablecoins also heavily involved in crypto-asset trading, widespread adoption could allow **disturbances in crypto markets to transmit more directly to the real economy**, raising the likelihood of systemic spillovers. Additionally, widespread adoption of stablecoins for payments and savings can **disintermediate banks and legacy payment networks**, eroding deposit bases and incumbent revenues [Liang et al. \(2024\)](#). Competitive pressures may push traditional intermediaries to integrate crypto infrastructure, yet diminished intermediation weakens banks' capacity to absorb shocks. This shift also complicates the Federal Reserve's lender-of-last-resort function, as extending liquidity support to stablecoin users is substantially less direct. Competition among stablecoin issuers may gradually degrade collateral quality, especially when low interest rates or return pressures encourage shifts from short-term government securities toward riskier assets. Such deterioration reintroduces MMF-style run risk ([Bolt et al., 2024](#)), with confidence shocks prompting fire sales and broader market spillovers. This dynamic echoes historical episodes—such as the collapse of the Bank of Amsterdam—where weakening asset quality ultimately eroded monetary trust.

Finally, stablecoins carry significant **geopolitical implications**, potentially undermining U.S.-centric financial sanctions and global leverage by allowing settlement to migrate outside traditional CHIPS- and SWIFT-based channels.¹⁷ Maintaining a strong domestic presence is therefore critical, while widespread use of foreign tokens exposes users to extraterritorial control and data risks. Competition over standards and security in digital payments is emerging as a new dimension of global monetary geopolitics. Taken together, the expansion of dollar-backed stablecoins can strengthen the global role of the U.S. currency and reduce transaction costs, but it also introduces new sources of financial instability, fiscal vulnerability, and geopolitical tension for the international monetary system.

¹⁷Countries under sanctions are already experimenting with stablecoin-based settlement to bypass restrictions, see [European Central Bank \(June 2025\)](#).

3 Model

We construct a three-country DSGE model, based on [Eichenbaum et al. \(2021\)](#) and [Ferrari Minesso et al. \(2022\)](#), incorporating two key features: the introduction of a U.S. stablecoin used for payments across countries, and a convenience yield on U.S. bonds capturing their special role as the world's dominant safe and liquid asset. These elements jointly allow us to study how the rise of stablecoins alters the international transmission of shocks through changes in global demand for U.S. debt. Stablecoins are introduced following the framework shaped by the GENIUS Act: they are unremunerated instruments for transactions issued by a global stablecoin issuers located in the U.S. that backs each coin minted with a safe U.S. bond. Stablecoins serve households both as settlement (i.e. payment) and trading (i.e. store of value) tokens.

[Figure 3](#) visualizes the overall structure of the model from a U.S. perspective. The full framework is, however, a three-country model, where countries are indexed by $c = \{US, 1, 2\}$ and have size n_c . In the baseline calibration, the countries are symmetric and of the same size. A three-country structure is needed to capture how shocks originating in a non-U.S. economy propagate both to the stablecoin-issuing country (the U.S.) and to another non-issuing country. This feature is important for our question because it allows us to study how stablecoin adoption not only changes how foreign shocks feed back to the issuer, but also how those shocks are redistributed across non-issuing economies. It is therefore necessary to accurately capture the complexity and two-way amplification of international financial spillovers introduced by the global safe asset channel.

In each country there are households, firms, a government and a central bank; additionally, in the U.S. there is a sector of stablecoin issuers. Households derive utility from consumption, leisure, and holdings of U.S. bonds. The assumption that U.S. bonds enter households' utility functions captures the special role of U.S. dollar bonds—that they are the most liquid and safe financial asset in the world (see e.g. [Jiang et al., 2021](#)). This creates a convenience yield from holding U.S. dollar bonds, as in e.g. [Valchev \(2020\)](#) and [Bianchi and Sosa-Padilla \(2023\)](#). For this reason the U.S. pays lower interest rates on its debt in equilibrium, calibrated following [Eren et al. \(2025\)](#). Households also require liquid instruments to perform transactions, which can be either domestic money or an internationally traded stablecoin. As will be explained later, these two instruments are not perfect substitutes. Firms are standard: they are owned by households, produce in-

intermediate goods, that are sold domestically and internationally, using labor and capital. Prices have quadratic adjustment costs à-la Rotemberg.

Stablecoins are perfectly liquid and can be used by households worldwide to settle transactions, acting as an imperfect substitute for money in the liquidity aggregator. The issuer's balance sheet is simple: assets consist of U.S. Treasuries, and liabilities correspond to outstanding stablecoins. Issuers earn a spread equal to the return on Treasuries, reflecting seigniorage-like profits from private issuance of dollar-backed liquidity. Since both assets and liabilities are denominated in U.S. dollars, the issuer faces no exchange rate risk, while foreign holders of stablecoins remain exposed to FX fluctuations. Finally, because stablecoins increase aggregate demand for Treasuries, we account for the fact that this pushes down the cost of financing for the Federal Government, in line with the evidence from [Eren et al. \(2025\)](#) on the impact of foreign demand on U.S. yields.

The rest of the model follows [Eichenbaum et al. \(2021\)](#): international markets are incomplete and uncovered interest parity does not hold. Monetary policy is set in each country independently by a central bank reacting to inflation (π_c) and output (Y_c). Government expenditures (consumption of final goods and debt repayment) are financed through taxes and issuance of new debt. Taxes are set to reach a target level of deficit. Implicitly the government balance sheet determines the amount of bonds in circulation. For our purposes, this supply of U.S. debt is taken as exogenous and policy-invariant at business-cycle frequency: for a given level of government expenditure and taxation, stablecoin issuers compete with foreign households for the same quantity of U.S. bonds; this implies that prices clear the bond market, so any increase or decrease in demand by stablecoin issuers is mirrored in U.S. yield movements.¹⁸

The full model is explained in greater detail in Appendix [A](#). In the next subsections we focus on the most distinctive and novel features.

3.1 Stablecoins

As mentioned, we model stablecoins following the spirit of the GENIUS Act. They are assumed to have evolved beyond crypto markets and to be used by households as a payment instrument. We consider alternative steady-states with different degrees of

¹⁸A complementary line of work endogenizes the issuance decision of a monetary hegemon in the presence of stablecoins and studies the changes in the international monetary system ([Azzimonti and Quadrini, 2025](#); [Barthélemy et al., 2025](#))

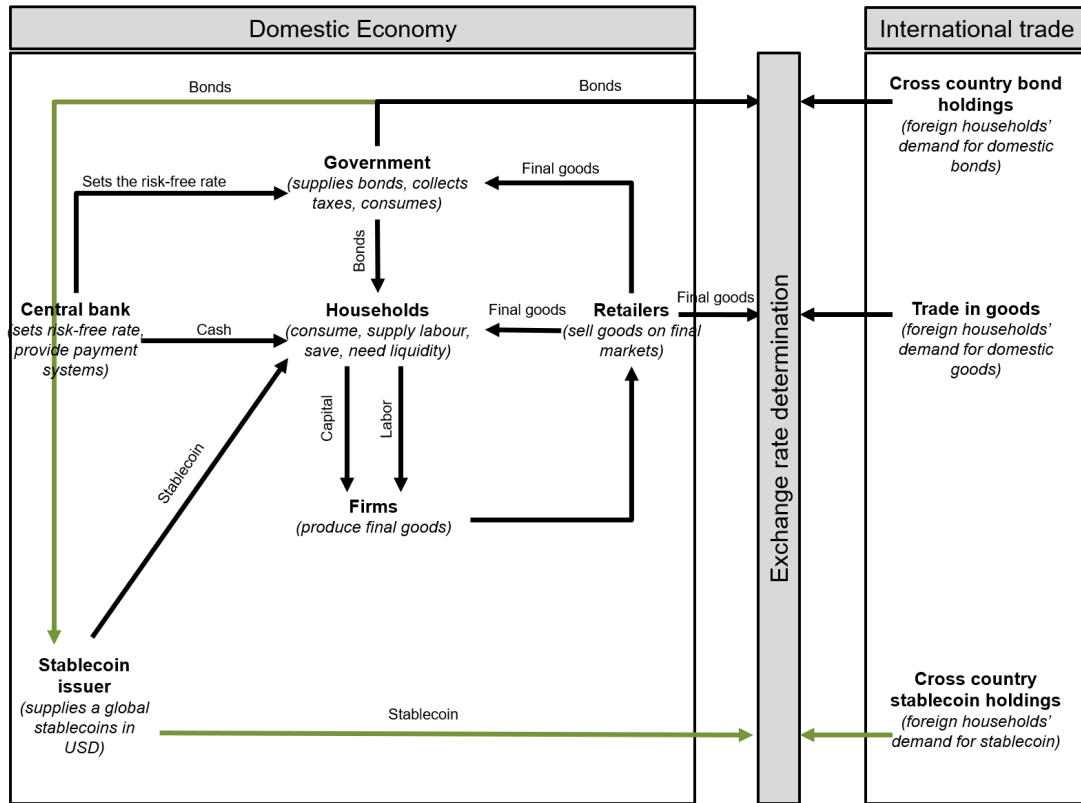


Figure 3: Description of the model.

Notes: The chart shows a description of the model from the U.S. perspective. The two foreign economies are symmetric. Green lines show the additional channels deriving from stablecoin adoption.

global adoption, from limited use (USD 500 billion) to widespread circulation with a market capitalization of around USD 2 trillion. A global issuer located in the United States backs each coin one-to-one with U.S. Treasury securities, earning the return on these assets as profit. The issuer's balance sheet satisfies:

$$B_{sc,t} = \sum_{c=\{US,1,2\}} n_c SC_{c,t}, \quad (1)$$

where $B_{sc,t}$ denotes the issuer's holdings of U.S. bonds and $SC_{c,t}$ the outstanding stablecoins held by households in country c . Each coin is fully backed by a U.S. bond, implying that any change in global stablecoin demand requires the issuer to buy or sell Treasuries, directly altering demand for the global safe asset. The issuer earns a spread equal to the return differential between U.S. bonds and the stablecoin rate, $(R_t - R_{sc,t})$, which is positive under the GENIUS Act as stablecoins are unremunerated ($R_{sc,t} = 1$). Profits are given by:

$$PSC_t = \frac{B_{sc,t-1}(R_{t-1} - R_{sc,t-1})}{\pi_t}, \quad (2)$$

and allow the issuer to accumulate a seigniorage-like income. Since both assets and liabilities are denominated in U.S. dollars, the issuer is insulated from exchange rate risk. In steady-state, the stablecoin rate is normalized to one and the issuer's profits are constant.

On the demand side, households can use stablecoins to transfer wealth, in U.S. dollars, across time (trading tokens) and as a payment instrument alongside domestic money (settlement tokens). Transactions are subject to a cash-in-advance constraint with a CES liquidity aggregator,

$$P_{c,t}C_{c,t} = \mathcal{L}_t\left(\frac{SC_{c,t}}{NER_{US,c,t}}, M_{c,t}, \mu_{sc,c,t}\right) \\ = \left[\mu_{sc,c,t} \left(\frac{SC_{c,t}}{NER_{US,c,t}}\right)^{\eta_{c,m}} + (1 - \mu_{sc,c,t}) (M_{c,t})^{\eta_{c,m}}\right]^{\frac{1}{\eta_{c,m}}}. \quad (3)$$

where $M_{c,t}$ is *fiat* money, $\eta_{c,m} \in (0, 1]$ governs the substitutability between the two instruments (notice that for $\eta_{c,m} \rightarrow 1$ Equation (3) becomes linear) and $\mu_{sc,c,t} \in [0, 1]$ measures households' confidence in using stablecoins as a payment instrument. It summarizes perceptions of safety, liquidity, and operational reliability, and hence determines the relative weight of stablecoins in the liquidity aggregator. Confidence is decreasing in three dimensions of perceived fragility: (i) valuation risk of the reserve portfolio, (ii) negative-carry risk due to interest-rate movements, and (iii) run risk associated with the payment infrastructure. Formally, we specify:

$$\mu_{sc,c,t} = \mu_{sc,c}^* - \left(\alpha_\mu \sigma_{q,t}^2 + \beta_\mu \exp\{1 - R_{US,t}\} + \gamma_\mu \rho_t\right), \quad (4)$$

where $\mu_{sc,c}^*$ is the steady-state level of confidence, $\sigma_{q,t}^2 \equiv \text{Var}\left(\frac{\Delta q_t}{q_t}\right)$ measures valuation volatility of the reserve portfolio, $R_{US,t}$ is the gross return on U.S. bonds, and ρ_t denotes the intensity of potential redemption pressures (e.g., cyber or governance shocks). The parameters $\alpha_\mu, \beta_\mu, \gamma_\mu > 0$ capture the sensitivity of confidence to each component. When reserves are stable, returns are positive, and redemption risk is low, $\mu_{sc,c,t} \simeq \mu_{sc,c}^*$ and stablecoins are close substitutes for money. Conversely, a decline in confidence (e.g., due to rising valuation uncertainty or run expectations) reduces $\mu_{sc,c,t}$, lowering the effective liquidity services provided by stablecoins and shifting demand back toward domestic money. Notice that Equation (4) endogenously dampens demand for stablecoins as they

grow: the more stablecoins are minted, the more U.S. debt the issuer needs to purchase, compressing its yields. Lower remuneration decreases the issuer's intermediation margin, making it more likely to default. Appendix B.1 shows that the main results are unchanged assuming a constant $\mu_{sc,c}$. This implies that Equation (4) is not the driver of results, but acts as a realistic amplification channel.

The first-order condition for stablecoin holdings equates the marginal liquidity value of stablecoins to their opportunity cost. Intuitively, households hold stablecoins up to the point where the liquidity services they provide equal the cost of keeping resources in an unremunerated digital dollar instrument.¹⁹ Notice that although stablecoins are not remunerated, foreigners can earn capital gain (losses) from an appreciation (depreciation) of the U.S. dollar against their domestic currency.

Finally, we deliberately abstract from the financial stability implications of stablecoins and the possibility of multi-issuer stablecoins (European Systemic Risk Board, 2025). Throughout the paper, we adopt the perspective of a regulatory framework in which stablecoins are successfully established as primarily as a means of payment, with limited implications for the banking sector. Additionally, we assume that stablecoins are issued only by U.S. entities. While these considerations are important, we leave them for future research. In the baseline model stablecoins do not charge redemption fees, which are negligible in real-world stablecoins,²⁰ but this assumption is relaxed in the Appendix.

3.2 The household problem

Consider the allocation problem of a representative household (HH) in economy c . Period utility is:

$$U_{c,t} = e_{c,t} \ln (C_{c,t} - h_c C_{c,t-1}) - \frac{\chi_{c,l}}{1 + \phi_{c,l}} L_{c,t}^{1+\phi_{c,l}} + \mathcal{F}_b \left(\frac{B_{US,c,t}}{NER_{US,c,t} P_{c,t}} \right) \quad (5)$$

where C denotes consumption, L hours worked, M domestic nominal money balances, P_c the domestic price index, $B_{US,c}$ holdings of internationally-traded U.S. bonds, $NER_{US,c}$ the nominal exchange rate (defined as dollars per unit of currency c) and e

¹⁹Our main results are unchanged if liquidity demand is modeled through money-in-utility preferences, which similarly capture micro-founded frictions behind the use of payment instruments; see Feenstra (1986).

²⁰Tether charges 0.1% redemption fee while USDC charges no fee to non institutional accounts.

is a preference shock.²¹ As in Valchev (2020) and Jiang et al. (2024), $\mathcal{F}_b(\bullet)$ defines the utility of liquidity services arising from holding international reserve assets, with $\mathcal{F}_b(0) = 0$, $\mathcal{F}'_b(\bullet) > 0$, $\mathcal{F}''_b(\bullet) < 0$. $\mathcal{F}_b(\bullet)$ generates an international premium on U.S. bonds. Households optimize utility subject to an intertemporal budget constraint and a liquidity cash-in-advance constraint, that captures demand for liquid assets (money and stablecoins) to conduct transactions as in Equation (3). We explicitly distinguish the non-pecuniary benefits of U.S. bonds, which do not serve as a medium of exchange, from cash and stablecoins, whose transactional role is captured through a cash-in-advance constraint. This distinction highlights how stablecoins' introduction makes part of the U.S. bond supply synthetically more liquid, namely the portion held as reserve by the issuer, strengthening the global demand for U.S. assets. The budget constraint is:

$$\begin{aligned}
& P_{c,t}C_{c,t} + B_{c,c,t} + \sum_{l \neq c} NER_{l,c,t}B_{l,c,t} + P_{c,t}I_{c,t} + M_{c,t} + \frac{SC_{c,t}}{NER_{US,c,t}} \\
& \leq w_{c,t}L_{c,t} + R_{c,t-1}B_{c,c,t-1} + \sum_{l \neq c} NER_{l,c,t}R_{l,t-1}B_{l,c,t-1} - \sum_{l \neq c} \frac{\phi_c^B}{2} \left(\frac{B_{l,c,t}NER_{l,c,t}}{P_{c,t}} \right)^2 P_{c,t} \\
& \quad + P_{c,t}R_{c,t}^k K_{c,t-1} + M_{c,t-1} + \frac{R_{sc,t-1}SC_{c,t-1}}{NER_{US,c,t}} + \Pi_{c,t}
\end{aligned} \tag{6}$$

where SC are stablecoin holdings, R , R_{sc} , R^k are respectively nominal returns on bonds, on stablecoins and on capital holdings and K are capital investments by households (who fully own firms). We assume international bond markets are incomplete because of transactions costs proportional to ϕ^B . As mentioned, $NER_{l,c}$ is the nominal exchange rate, defined as units of currency l for currency c : an increase in $NER_{l,c}$ is an appreciation of currency l against currency c . Stablecoins are assumed to bring no remuneration, so $R_{sc,t} = 1 \forall t$. $\mu_{sc,t}$ is endogenous and varies according to the confidence that HHs attach to stablecoins. Equation (3) is characterised also by decreasing returns to scale as in standard macro-theory. In other words, households' need for payment instruments decreases in the total amount of payment assets they hold. Hence any additional purchase of payment assets has less value in terms of the payment it brings to households. Therefore $\mathcal{L}(0) = 0$, $\mathcal{L}'(\bullet) > 0$ and $\mathcal{L}''(\bullet) < 0$. First order conditions for U.S.dollar denominated

²¹In other words, for country c , a fall in $NER_{US,c,t}$ is a depreciation of the their currency. For the U.S., $NER_{US,US} = 1$.

internationally traded assets, bonds and stablecoins, are:

$$-\frac{\lambda_{c,t}}{NER_{US,c,t}} + \beta_c E_t \left(\frac{\lambda_{c,t+1}}{NER_{US,t+1}} \frac{R_{US,t}}{\pi_{c,t+1}} \right) - \frac{\phi^B \lambda_{c,t}}{NER_{US,t}} \left(\frac{B_{US,c,t}}{P_{c,t} NER_{US,c,t}} \right) + \frac{\mathcal{F}'_b(\bullet)}{NER_{US,c,t}} = 0. \quad (7)$$

where $\{\lambda_{c,t}\}_{t=0}^{\infty}$ is the sequence of Lagrangian multipliers associated to the budget constraint of the household. We can rewrite Equation (7) as:

$$-\frac{\lambda_{c,t}}{NER_{US,c,t}} + \beta_c E_t \left(\frac{\lambda_{c,t+1}}{NER_{US,t+1}} \frac{R_{US,t}}{\pi_{c,t+1}} \right) + \mathcal{P}_{US,t} = 0.$$

to highlight the role of the premium on U.S.bonds. $\mathcal{P}_{US,t} = \frac{\mathcal{F}'_b(\bullet)}{NER_{US,c,t}} - \frac{\phi^B \lambda_{c,t}}{NER_{US,t}} \left(\frac{B_{US,c,t}}{P_{c,t} NER_{US,c,t}} \right)$ denotes the premium given by the liquidity minus transaction costs. The first order condition for stablecoins is:

$$-\frac{\lambda_{c,t}}{NER_{US,c,t}} + \beta_c E_t \left(\frac{\lambda_{c,t+1}}{NER_{US,c,t+1}} \frac{R_{sc,t}}{\pi_{c,t+1}} \right) + \gamma_{c,t} \frac{\mathcal{L}'(\bullet)}{NER_{US,c,t}} = 0 \quad (8)$$

where $\{\gamma_{c,t}\}_{t=0}^{\infty}$ is the sequence of Lagrangian multipliers associated to the cash-in-advance constraint.²²

Combining Equation (7) and Equation (8), conditional on the issuance constraint of the stablecoin holder of Equation (1), leads to the key equation of the model:

$$\beta_c E_t \left[\frac{\lambda_{c,t+1}}{NER_{US,c,t+1} \pi_{c,t+1}} \right] [R_{US,t} - R_{sc,t}] = \gamma_{c,t} \frac{\mathcal{L}'(\bullet)}{NER_{US,c,t}} - \mathcal{P}_{US,t} \quad (9)$$

It can be shown analytically, see Appendix A.7, that the elasticity of stablecoins holdings to changes in the interest rates depends on steady-state holdings. Log-linearizing Equation (9) leads to the following elasticity of the demand for stablecoins to the interest rate differential:

$$(\eta_{c,m} - 1) \frac{\mu_{sc,c,ss} \gamma_{c,ss} \left(\frac{SC_{c,ss}}{C_{c,ss}} \right)^{\eta_{c,m}-1}}{\gamma_{c,ss} \mu_{sc,c,ss} \left(\frac{SC_{c,ss}}{C_{c,ss}} \right)^{\eta_{c,m}-1} - \chi_{c,b} B_{US,c,ss}^{-\sigma_{c,b}} \Gamma_{ss}} \frac{1}{\Gamma_{ss}} \quad (10)$$

ss indicates steady-state holdings, Γ is a positive scalar that depends on $\mu_{sc,c}$ and on the

²²Notice that for a U.S. household first order conditions would be identical, with the only difference that $NER_{US,US,t} = 1 \forall t$

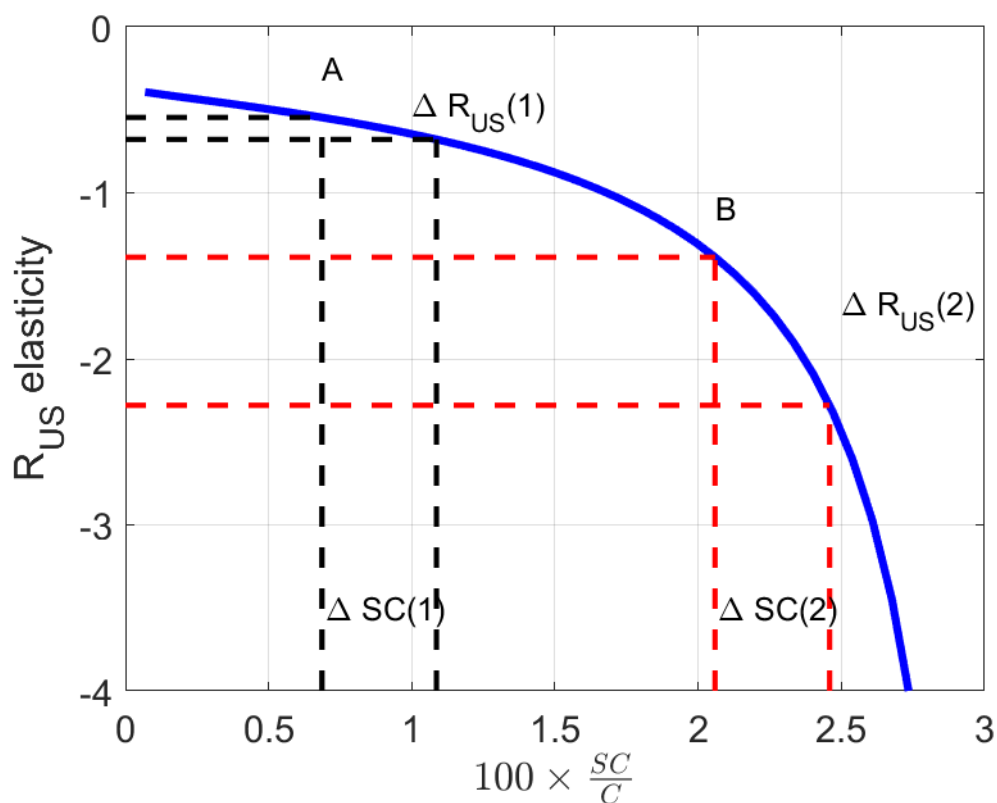


Figure 4: Demand for stablecoins against returns differential.

Notes: the chart shows the elasticity of U.S. yields (vertical axis) against stablecoin demand (horizontal axis). The elasticity is computed as in Equation (10) by keeping all other variables constant and changing values of stablecoins. $\Delta SC(1)$ and $SC(2)$ show the change in U.S. yields for the same change in stablecoin holdings.

steady-state U.S. risk-free rate. It is strictly positive under a wide range of calibrations.²³ Under the baseline calibration, the sign of Equation (10) is negative because $\eta_{c,m} - 1 < 0$ for $\eta_{c,m} \in (0, 1]$. For low stablecoin preferences, $\mu_{sc,c} \rightarrow 0$, the elasticity tends to zero, meaning that low stablecoin demand is almost unaffected by U.S. yields and vice-versa. In other words, unsurprisingly, if preferences for stablecoins are low, they have no aggregate effects. Similarly, if stablecoins and cash are linear substitute, that is $\eta_{c,m} \rightarrow 1$, the elasticity is also zero. This is intuitive as households would hold only one payment instrument cash: there would be a corner solution. In practice, it is unlikely that fiat money would lose completely its convenience relative to other forms of money, if nothing because of the role of government taxes and its role in aggregate demand, see Ikeda (2022). Data from existing surveys also convey this intuition, see Li (2023).

Dynamics might change if stablecoins are high in demand. While the numerator is

²³See Figure A.2 and Appendix A.7 for the full derivation.

larger than zero by construction, $\mu_{c,sc}\gamma_{c,ss}\left(\frac{SC_{c,ss}}{C_{c,ss}}\right)^{\eta_{c,m}-1} > 0$, the denominator might change size. Equation (10) can turn negative if $\gamma_{c,ss}\mu_{sc,c,ss}\left(\frac{SC_{c,ss}}{C_{c,ss}}\right)^{\eta_{c,m}-1} > \chi_{c,b}B_{US,c,ss}^{-\sigma_{c,b}}$. That can happen if stablecoins holdings are particularly high, as $\left(\frac{SC_{c,ss}}{C_{c,ss}}\right)^{\eta_{c,m}-1}$ (i.e. the marginal value of stablecoins for payments) decreases the more stablecoins are held by households. In other words as $\left(\frac{SC_{c,ss}}{C_{c,ss}}\right)^{\eta_{c,m}-1}$ declines households place limited value on additional stablecoins as payment instruments. In that extreme case, the relationship would be opposite: an increase in the rate differential would generate more demand for stablecoins, leading to macro instability. This happens because stablecoins would be held more and more as a financial asset, therefore households would react more aggressively to changes in financial conditions, see Figure A.2 in the Appendix.

Indeed Equation (9) is similar to the CBDC holding condition derived in Ferrari Minesso et al. (2022) but with one important difference. CBDC in that model are issued elastically by the central bank, which in the background sterilizes the effects of CBDCs on financial markets. Stablecoin issuers, on the contrary, must follow the budget constraint in Equation (1). This constraint implies that any fluctuation in stablecoins issuance is balanced by a proportional change in holdings of U.S. safe assets. These shifts are not sterilized by monetary authorities and lead to real effects from stablecoin demand. That creates the new channel of transmission for shocks, as stablecoin demand amplifies the impact on U.S. financial markets from changes in returns on U.S. assets or households' consumption and intertemporal preferences in foreign countries (i.e. λ_c).

Importantly, the financial market impact of stablecoin adoption is amplified by the amount of stablecoins minted. The larger are stablecoin holdings in country c , the lower is $\mathcal{L}'$ and therefore the more stablecoins households in country c will need to liquidate (purchase) to balance the two sides of Equation (9). Notably, a similar relation to Equation (9) can be derived for domestic bonds and money in country c , amplifying the global spillovers of domestic shocks in c through U.S. bond markets, see Section A for full derivation.

3.3 Calibration

The standard parameters of the model are calibrated as Eichenbaum et al. (2021) and Ferrari Minesso et al. (2024), with a sensitivity to inflation in the Taylor rule set to 2;

the reaction to output to 0.6 and the persistence of monetary policy to 0.75. The home bias in trade is set to 0.75. All parameters are reported in [Table A.1](#) in the Appendix. In the baseline configuration, all countries are symmetric and have the same size (1/3). The three shocks (monetary policy, TFP and stablecoin preference) follow an AR(1) process, with autoregressive parameter set to 0.2 for the monetary policy shock process and to 0.5 for the others. The standard deviation of shocks is calibrated to 0.01. We calibrate α_μ , $\sigma_{q,t}^2$, β_μ and γ_μ to 0.1, 0.05, 0.01 and 1 respectively. We set the volatility of the stablecoin shock (ρ_t) to 0.01 and the persistence of its process to 0.5. The steady-state value of $\mu_{sc,c}$ is calibrated to reach a market capitalization of USD 500 billion (0.05) and USD 2 trillion (0.1). In the absence of stablecoins, the premium on U.S. yields is calibrated to 1% annualized; we follow the quantitative results of [Eren et al. \(2025\)](#) increasing it by 30 bps when stablecoins market capitalization reaches 2 tr.

4 Results

In this section, we assess how the introduction of stablecoins affects both the issuing and foreign economies. We first examine implications for monetary stability and business-cycle dynamics in the issuing country; we then analyze how stablecoins modify the transmission of shocks originating abroad; finally, we study how shocks to stablecoin demand propagate across countries. The analysis considers three calibrations: (i) no stablecoins; (ii) moderate adoption with a steady-state market capitalization of USD 500 billion (about USD 200 billion above the level at the time of writing, $\approx 1.3\%$ of total U.S. debt); and (iii) large-scale adoption of USD 2 trillion ($\approx 5.3\%$ of U.S. debt), consistent with the upper range of investment-bank projections.

At the core of the results is a global safe asset channel. When the dollar appreciates (depreciates) there are valuation effects for foreign holders of dollar-backed stablecoins. Foreign holders are incentivized to redeem (purchase) stablecoins due to the two specific features of stablecoins: to i) realise valuation gains (as trading tokens) and ii) because they need fewer units to settle transactions (as settlement tokens). But because payment stablecoins are fully backed by short-term U.S. Treasuries, portfolio flows into and out of stablecoins map one-for-one into changes in Treasury demand, as the issuers sell (buy) bonds to meet redemptions. Redeemed dollars are then converted into foreign currencies

and transferred abroad, leading to a dollar depreciation (the opposite in case of flows into stablecoins). This ties private money creation directly to public debt and makes U.S. yields react not only to domestic policy but also to global payment needs and portfolio rebalancing.

Overall, stablecoins modify the transmission of U.S. monetary policy through three bond-market mechanisms. First (level effect, steady-state): by adding structural demand for T-bills, stablecoins lower equilibrium risk-free yields but also require central banks to be more aggressive in responding to inflation because digital capital flows limit the ability of central banks to steer aggregate demand. Second (domestic pass-through): as a growing share of Treasuries is held indirectly via foreign stablecoin demand, the pass-through of U.S. policy shocks to short maturities is amplified, as issuers sell Treasuries when foreign households realise capital gains and need fewer stablecoins to settle transactions if the dollar appreciates. Third (foreign sensitivity): by strengthening the link between foreign payment needs and U.S. bond markets, external shocks exert a larger influence on U.S. financial conditions, amplifying two-way international spillovers. These effects are small when adoption is modest relative to the size of the U.S. bond market, but scale non-linearly with market size, creating sharper trade-offs for monetary and financial stability as stablecoin capitalization rises.

4.1 Macro-Financial Stability: The Risks of the Global Safe Asset Channel

We begin by examining the implications of stablecoin introduction for macro-financial stability. Figure 5 presents a stability map in which we vary the response coefficient to inflation in the U.S. central bank's Taylor rule, θ_π in Equation (11), (vertical axis) and the steady-state demand for stablecoins (horizontal axis), expressed as a share of total U.S. debt.

$$\ln R_{c,t} = (1 - \varrho) \ln R_{c,t-1} + \varrho \left[\ln R_{c,ss} + \theta_\pi \ln \pi_{c,t} + \theta_y (\ln Y_{c,t} - \ln Y_{c,ss}) \right] + \varepsilon_{c,t}. \quad (11)$$

The blue area denotes parameter combinations yielding a stable equilibrium while the red area indicates instability.²⁴ The vertical black lines mark levels of stablecoin holdings

²⁴The model is linearized to first order. Stability is assessed using the Blanchard–Kahn conditions.

corresponding to total market capitalizations of USD 500 billion and USD 2 trillion, respectively.

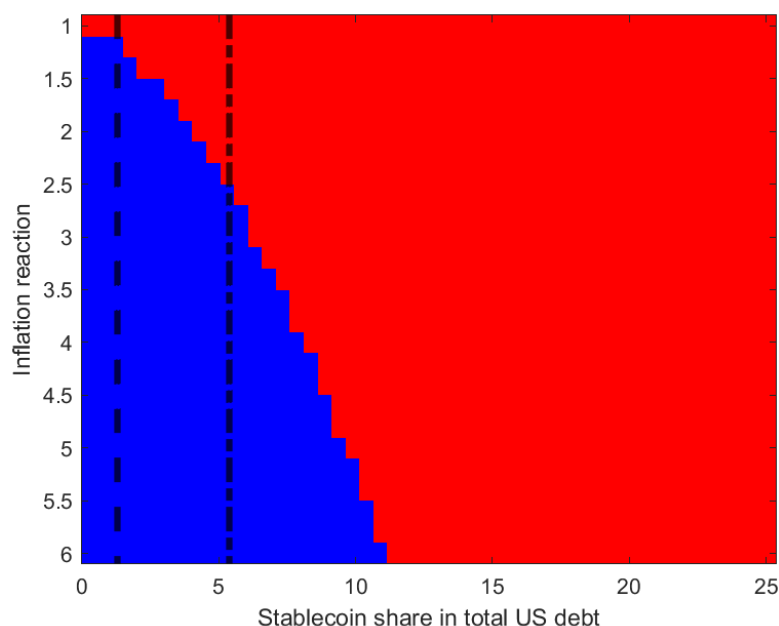


Figure 5: Stability map.

Notes: the chart shows combinations of values for the U.S. Taylor rule inflation reaction parameter (θ_π) and the stablecoin preference parameter (μ_{sc}). Blue areas indicate that the model is stable and red areas that it is unstable. Vertical lines indicate a stablecoin total market cap of USD 500 bln and 2 Tr.

For low levels of stablecoin adoption (below USD 500 billion) the conventional Taylor principle holds: monetary stability requires the policy rate to respond to inflation with a coefficient greater than one. As stablecoin adoption increases, this relationship changes. Higher stablecoin capitalization necessitates a stronger policy response to inflation for the model to remain stable. This result stems from the new linkage stablecoins create between U.S. bond markets, the dollar exchange rate and foreign holdings. With widespread stablecoin use across countries, U.S. yields and the dollar exchange rate become increasingly influenced by foreign demand for stablecoins. As a result, aggregate demand reacts less to domestic monetary policy, requiring a stronger reaction to inflation to stabilize prices and the business cycle.

These effects are non-linear, as illustrated in Figure 5. Even under substantial, though plausible, stablecoin adoption in line with market expectations, the model remains stable for standard parametrizations of the inflation response, typically between 1 and 2.5. The situation changes once stablecoin capitalization exceeds USD 2 trillion, as monetary policy

The model is solved using Dynare, [Adjemian et al. \(2024\)](#).

must react disproportionately strongly to inflation to preserve stability. For instance, if stablecoins account for 10% of the U.S. debt market (twice the upper bound of current projections) the inflation coefficient must rise to around 6. This finding underscores the potential for stablecoins to reshape the fundamental drivers of the U.S. bond market and, consequently, the behavior of U.S. yields.

This outcome follows directly from Equation (9). When the steady-state demand for stablecoins is high, their marginal value as a means of payment declines. Consequently, following a shock, households rapidly change their stablecoin holdings, strengthening the reaction of yields and exchange rates.

Figure 6 replicates the exercise while varying the elasticity of substitution between different types of liquid assets ($\eta_{c,m}$) and the extent to which stablecoins lower U.S. yields in the steady-state.²⁵ The results are broadly consistent with our previous results, but it is interesting to see two regularities across calibrations: (i) the destabilising force of stablecoins increases as their impact on the U.S. steady-state yield increases. Larger reductions in yields is amplifying the global safe asset channel of stablecoins. This means that larger swings in stablecoins' demand across the globe imply a necessarily stronger monetary policy reaction to stabilize the economy. (ii) As stablecoins become closer substitutes for money in providing liquidity services ($\eta_m \rightarrow 1$), the economy becomes more stable for a given monetary policy response. This occurs because stablecoin demand becomes less volatile, as part of the adjustment in payment needs can be absorbed by money. In turn, this mitigates the strength of the global safe asset channel. In the extreme case of $\eta_m = 1$, as shown in Equation (9), the impact would be zero.

4.2 Impulse-responses

Turning to the impulse responses, we analyse three scenarios. The first considers two shocks originating in the U.S.: a one-standard-deviation contractionary U.S. monetary policy tightening and a one-standard deviation expansionary TFP shock. The second simulates the same shocks in a non-issuer stablecoin country. The third introduces a negative preference shock to stablecoins, designed to mimic a sudden loss of confidence and a sharp drop in demand for stablecoins across both domestic and foreign markets.

²⁵We do this by varying the risk-premium parameter controlling the steady-state reduction in U.S. yields imputed by the demand for stablecoins. Additionally, notice that as $\eta_{c,m} \rightarrow 1$, preferences become linear.

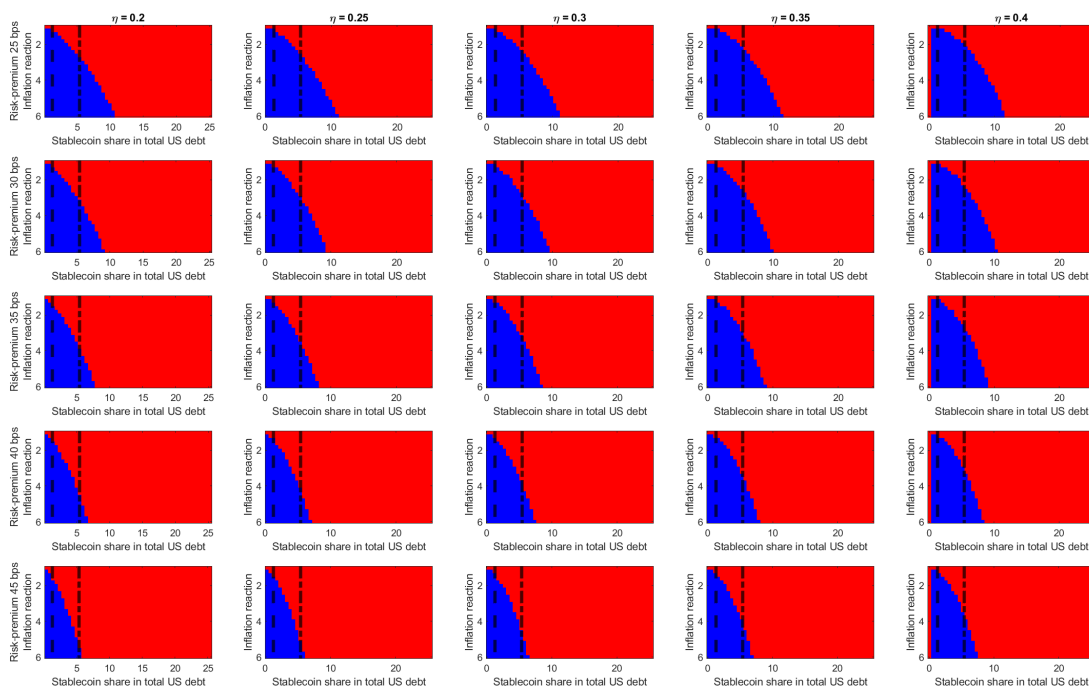


Figure 6: Stability map.

Notes: the chart shows combinations of values for the U.S. Taylor rule inflation reaction parameter (θ_π) and the elasticity of substitution between different types of liquid assets ($\eta_{c,m}$). Blue areas indicate that the model is stable and red areas that it is unstable. Vertical lines indicate a stablecoin total market cap of USD 500 bl and 2 Tr.. Different calibrations for the U.S. convenience yield and the CES liquidity aggregator parameter are considered. Vertical lines indicate a stablecoin total market cap of USD 500 bl and 2 Tr.

4.2.1 U.S. shocks: Monetary Policy and TFP

Figure 7 presents the effects of a U.S. monetary policy shock under the three adoption assumptions: (i) no stablecoins (black line), (ii) a steady-state stablecoin market capitalization of USD 500 billion (green line), and (iii) USD 2 trillion in steady-state stablecoin capitalization (red line). Consistent with the stability results, the low-adoption scenario has little impact on real outcomes relative to the baseline without stablecoins. The effects on domestic output and the interest rate are similar, reflecting the limited role of stablecoins in U.S. debt market dynamics at this scale.

However, macroeconomic consequences become important as stablecoins' capitalization increases (compare the red dots to the green line in Figure 7). Interestingly, output losses decrease when the economy is characterized by higher steady-state holdings of stablecoins. This occurs because when the dollar appreciates it creates positive valuation effects for foreign holders of dollar-backed stablecoins. This incentivises foreign holders to redeem stablecoins to realise valuation gains and because they need fewer units to settle

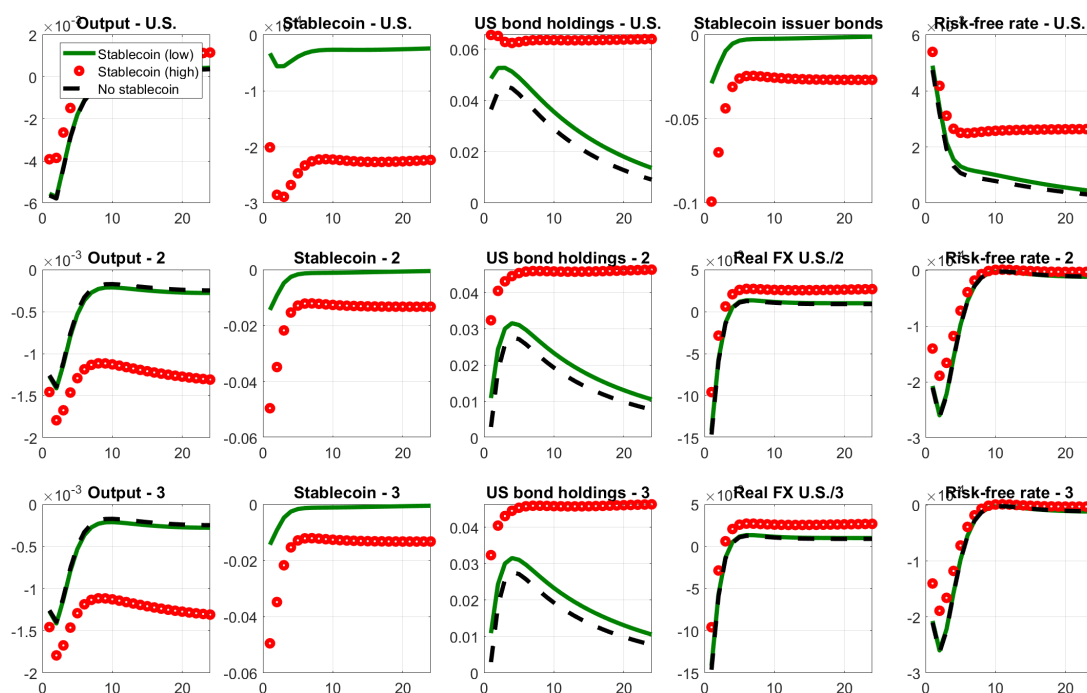


Figure 7: Impulse responses to a U.S. monetary policy shock.

Notes: the chart shows the impulse response to a 1 standard deviation U.S. monetary policy shock. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market).

transactions. As a consequence, the stablecoin issuer needs to sell U.S. dollar bonds and to convert proceeds from the sales into foreign currency to pay for those redemptions. The bond price falls, amplifying the contractionary effects of the monetary policy shock, consistently with Equation (9). Although higher U.S. rates also make Treasuries more attractive for foreign households, this additional direct demand is more than offset by the bond sales of the issuer, so that the net effect of the shock is a stronger rise in U.S. yields when stablecoins are present. Therefore, despite higher yields, redemption flows from the U.S. to the foreign country (and corresponding exchanges of dollars into foreign currency) dampen the appreciation of the dollar. Consequently, U.S. yields rise more in response to a shock of a given size (top-right panel of Figure 7), as the liquidation of bonds by stablecoin issuers depresses bond prices and raises returns. At the same time, the U.S. dollar appreciates less against its trading partners (fourth column of Figure 7), supporting trade. This effect is economically significant: U.S. output declines by roughly 30% less. This result explains why central banks need to react more strongly to inflation as stablecoins become widely adopted: with large holdings of stablecoins monetary policy

is less effective in steering aggregate demand because of (digital) capital flows.

Higher stablecoin holdings also amplify volatility in international demand for U.S. bonds, as foreign investors arbitrage between bonds and stablecoins. This effect strengthens with greater stablecoin adoption. Turning to foreign countries, the contractionary effects of U.S. tightening shocks are amplified the higher is stablecoin adoption because of exchange rate movements: the higher is the steady-state demand for stablecoins, the less the dollar appreciates, the more foreign countries lose competitiveness against the U.S.

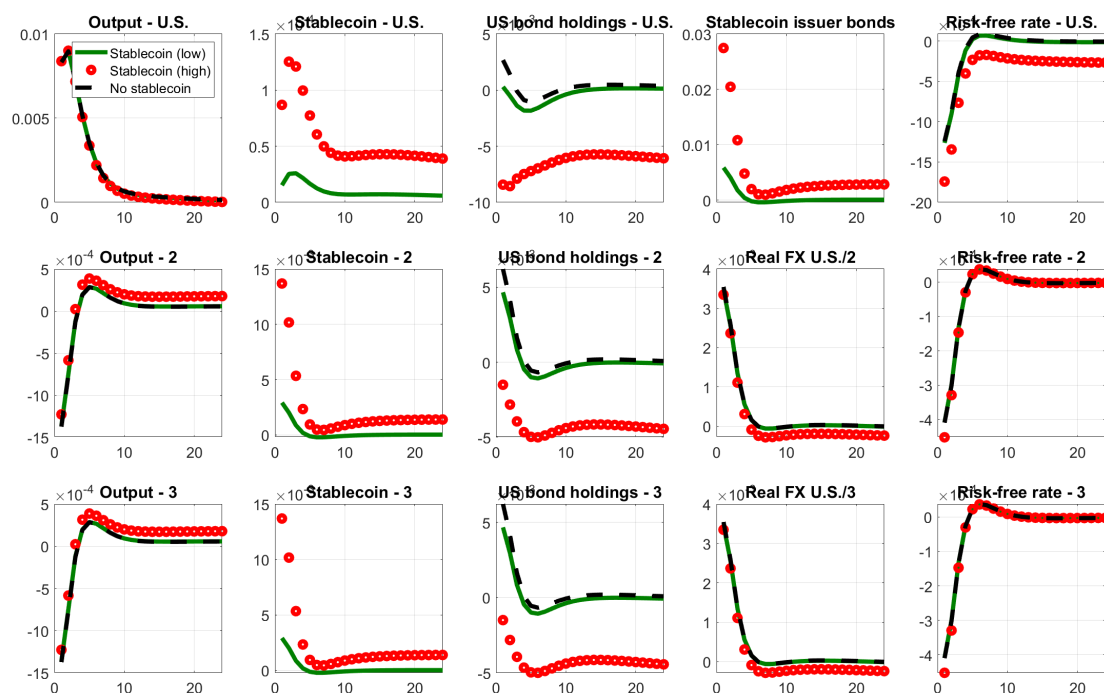


Figure 8: Impulse responses to a U.S. TFP shock.

Notes: the chart shows the impulse response to a 1 standard deviation U.S. TFP shock. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market).

The dynamics are similar when considering a U.S. TFP shock, Figure 8. In this case, however, aggregate effects are smaller, as the shock is real rather than financial. Following a positive TFP shock, U.S. yields decline more sharply when stablecoins are present. This occurs because foreign households purchase stablecoins, which are appreciating assets. Moreover, while U.S. bonds experience a decline in returns, stablecoin remuneration remains constant. This increases the incentive for households to hold stablecoins rather than U.S. bonds, even if this increase is muted by the decrease in confidence coming from

an endogenous fall in $\mu_{c,t}$, due to the higher issuer risk. In the absence of stablecoins, foreign households would have rebalanced into domestic bonds or cash. These flows are partially offset by purchases of bonds by the stablecoin issuer. As demand for stablecoins rises, the issuer must buy additional bonds, leading to a decline in U.S. yields, about 50% larger than in the absence of stablecoins. This fall in yields helps offset the appreciation pressures resulting from inflows into dollar-denominated stablecoins. Foreign spillovers also decrease, although the difference between low and high stablecoin adoption is smaller than for a monetary policy shock. Exchange rate movements are amplified as well, with the dollar depreciating more in response to the shock, consistent with standard theory, and the magnitude of this effect rises with the level of stablecoin adoption. Figure A.3 and Figure A.4 in the Appendix show impulse responses for a broader set of variables. Notably, the inclusion of stablecoins generates more fluctuations in the convenience yield on U.S. bonds, as now bonds compete with stablecoins in delivering non-pecuniary benefits to households. Unsurprisingly, money demand moves in the opposite direction to stablecoin demand in all countries as the two are (imperfect) substitutes as liquid assets.

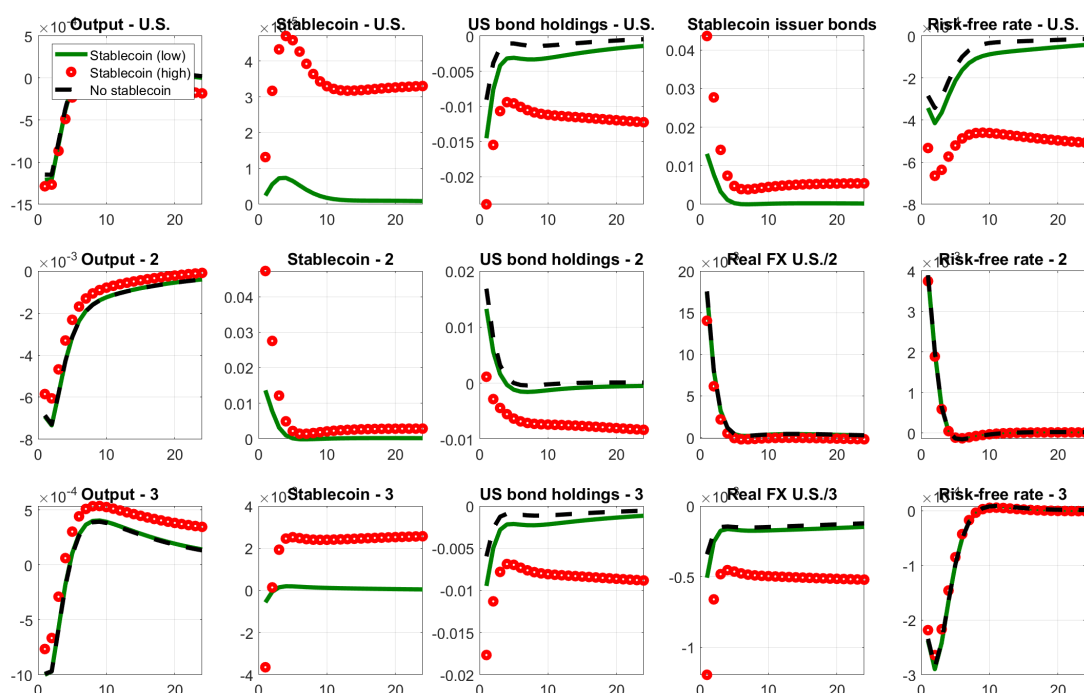


Figure 9: Impulse responses to a monetary policy shock from country 2.

Notes: the chart shows the impulse response to a 1 standard deviation monetary policy shock from a non-issuer country (country 2). The black line shows the responses of the model without stablecoins, the green line shows the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market).

4.2.2 Non-issuer country shock: Monetary Policy and TFP

Figure 9 examines spillovers to the U.S. from a monetary policy tightening in a non-issuing country, Country 2. Domestic variables in this country respond similarly with or without stablecoins; the main differences appear in the demand for foreign assets, particularly stablecoins and U.S. bonds: stablecoins demand increases while direct U.S. bond holdings decreases.

Spillovers to the U.S., however, are amplified. As U.S. yields fall, households in both the U.S. and Country 2 rebalance their portfolios toward stablecoins, generating additional demand for U.S. bonds through the stablecoin issuer. This demand further reduces U.S. yields compared to a scenario without stablecoins. At the same time, increased demand for dollars in Country 2 limits the depreciation of the U.S. dollar against that country's currency. Similarly, when agents in Country 3 liquidate stablecoins, the dollar appreciates against their currency. Taken together, these forces alter the spillovers of foreign shocks to both the U.S. and third countries. In the U.S., foreign shocks become more pronounced, with output contracting by roughly 10% more, as the dollar remains stronger due to inflows into stablecoins from both Country 2 and domestic agents. In Country 3, however, stablecoin flows have a stabilizing effect: the stronger dollar dampens the decline in trade, and output contracts by approximately 15% less when stablecoins are widely used. Figure A.5 shows all the IRFs.

Country 2 TFP shocks have broadly similar effects on macroeconomic variables across different stablecoin configurations. The main difference is that higher stablecoin market shares generate somewhat larger fluctuations in the demand for U.S. bonds and therefore in the U.S. convenience yield, see Figure A.6.

4.2.3 Negative shock to stablecoin demand

Finally, we simulate a negative preference shock to stablecoins across all countries, that is, a decline in $\mu_{sc,c,t}$. Figure 10 presents the results. This shock affects households in both the United States and the other two countries. Following the shock, stablecoin holders sell their positions. Sales by foreign investors lead to a depreciation of the U.S. dollar, which in turn makes holding dollar-denominated stablecoins even less attractive due to capital losses. The dollar depreciation benefits the U.S. by improving export competitiveness. At the same time, the stablecoin issuer sells U.S. bonds held as reserves, putting upward

pressure on U.S. interest rates.

Overall, the net effect on U.S. output is positive: the expansionary impact of a weaker dollar outweighs the contractionary effect of higher rates. However, higher interest rates reduce domestic consumption and aggregate demand, even if external demand supports total output (see Figure A.7).

Economic effects in foreign economies are negative. The dollar depreciation reduces the competitiveness of foreign exports, leading to a loss of trade shares and a decline in aggregate output. In addition, as households liquidate their stablecoin holdings, part of the proceeds are redirected toward consumption, increasing domestic demand and inflation. The resulting rise in prices prompts a monetary policy tightening. Higher foreign interest rates further depress output and contribute to an additional appreciation of foreign currencies relative to the U.S. dollar.

Also in this case, stronger steady-state stablecoin holdings magnify the spillover from stablecoins shocks.

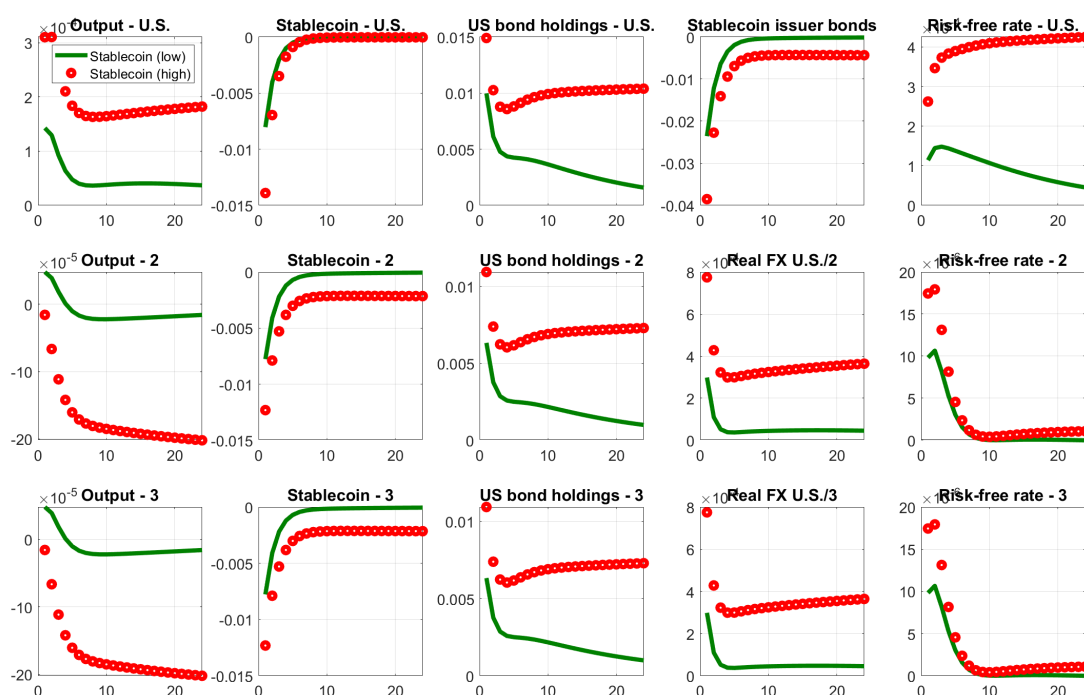


Figure 10: Impulse responses to a stablecoin preference shock.

Notes: the chart shows the impulse response to a 1 standard deviation shock to stablecoin preferences ($\mu_{sc,c}$). The green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market).

4.3 Welfare Implications

To assess the welfare impact of stablecoin adoption for the U.S. and the other countries, we compare the model with and without stablecoins. [Figure A.8](#) in the Appendix reports the resulting welfare changes as a function of the assumed effect of stablecoin demand on equilibrium U.S. yields. The model is solved at second order with pruning and welfare is defined recursively: $\mathcal{W}_{c,t} = U_{c,t} + \beta_c E_t \mathcal{W}_{c,t+1}$, with U_c denoting the period utility function. When stablecoin demand does not reduce the U.S. risk-free rate ($\Delta R_{US,ss} = 0$), welfare effects are small: at low adoption the U.S. is slightly negatively affected while foreign countries gain slightly, and at high adoption all three countries experience modest welfare gains, stronger for non U.S. economies. As the assumed impact on equilibrium rates increases, welfare becomes more skewed towards the U.S.: once $\Delta R_{US,ss} \geq 0.2$, U.S. welfare rises monotonically with the size of the yield compression and with the level of adoption, whereas both foreign countries start to lose, with losses growing in both dimensions. For example, when the steady-state U.S. yield falls by 1 percentage point and stablecoin adoption is high, U.S. welfare gains are around 1.5%, while each foreign country loses close to 2%.

This pattern reflects the trade-off generated by the global safe asset channel. The U.S. benefits from cheaper public financing and a lower average cost of capital, while the attenuation of domestic monetary transmission documented above has only a second-order effect on its welfare. Foreign countries, instead, ultimately bear the cost of lower safe yields and stronger exposure to U.S.-centered financial conditions. Overall, it is necessary to assume a sufficiently large impact of stablecoin demand on the equilibrium risk-free rate to generate sizable welfare gains for the U.S. and corresponding losses for the rest of the world.

4.4 Extensions and Robustness

We consider several extensions of the model to assess the robustness of our results.

First, we contrast two benchmark cases in which stablecoins are issued exclusively in the United States or exclusively abroad. The corresponding results are reported in [Section B.3](#). These simulations highlight the central mechanism: the global safe-asset channel amplifies the propagation of shocks in the presence of stablecoins. When issuance

is restricted to the domestic economy, the model's dynamics and stability properties are equivalent to those without stablecoins. In contrast, when stablecoins are held internationally, their adoption alters the transmission of shocks through shifts in demand for dollar-denominated bonds and thereby constrains the autonomy of domestic monetary policy.

Second, we remove the cash-in-advance constraint and instead adopt a money-in-the-utility specification. As shown by [Feenstra \(1986\)](#), this approach allows for a broader set of assumptions regarding the demand for liquid assets. Our results remain robust under this alternative formulation, indicating that the intuition derived from [Equation \(9\)](#) is general and does not depend on specific functional forms. The corresponding results are reported in [Appendix B.2](#).

Third, we relax the symmetry assumptions in the calibrated model and employ estimated parameters and weights to conduct simulations. Specifically, we use parameters from the estimated three-country model (U.S., euro area and rest of the world) of [Ferrari Minesso and Pagliari \(2023\)](#). The main intuition again proves robust, as shown in [Appendix B.4](#).

Fourth, we include redemption fees on stablecoin holdings. While redemption fees are very small (0.1% for Tether) or completely absent (USDC) for existing stablecoins, the GENIUS act does allow issuers to impose them. [Section B.5](#) simulates the model under a quite large, 2%, redemption fee. While the presence of a fee reduces redemptions, thus dampening the impact of stablecoins on yields and the exchange rate, the key results of the model are preserved.

5 Empirical Validation

Stablecoins are a relatively recent phenomenon in monetary history, so the limited availability of long time series poses a challenge for empirically validating the model. Nevertheless, financial market data provide a useful alternative for testing the predictions of our theoretical framework.

In this section, we examine how U.S. yields respond to monetary policy shocks in the U.S. and the euro area, depending on the share of stablecoins' market capitalization. We implement a local projection framework and estimate:

$$y_{t+k} = \alpha_k + \beta_k^1 MPS_t + \beta_k^2 MPS_t SC_{t-1} + \beta_k^3 SC_{t-1} + \sum_{l=1}^L \gamma' X_{t-l} + T_k + \alpha_{year,k} + \varepsilon_{t+k} \quad (12)$$

where MPS is a measure of monetary policy shock, SC is the log of the market capitalization of dollar stablecoins in the total U.S. bond market, proxied by the capitalization of USDT and USDC which constitute almost the entire supply of dollar stablecoins until now. X_{t-l} includes lagged variables to control for global risk factors and macro developments: the dependent variable, SC , the log of the USD Nominal Effective Exchange Rate and the S&P500. The lag-length is $L = 4$. Year dummies ($\alpha_{year,k}$) and a time trend (T_k) are also included. We use weekly data from 2018 to September 2025.

As dependent variable we focus on the U.S. 3-month yield –as stablecoin issuers are constrained to invest in short-term debt– and use high-frequency monetary policy shocks for both the U.S. and the euro area, identified on the same maturity. The coefficients of interest are β^1 and β^2 : according to theoretical results, a stronger footprint of stablecoins should amplify the reaction of U.S. yields to U.S. monetary policy shocks and increase also the impact of foreign shocks.

Figure 11 reports the estimated coefficients (β^1 in the upper row and β^2 in the lower row). Consistent with the evidence in Ahmed and Aldasoro (2025) and Kim (2025), stablecoin demand appears to affect U.S. yields. Higher stablecoin capitalization amplifies the impact of U.S. monetary policy shocks on yields, in line with the predictions of the theoretical model. The evidence is less clear for spillovers of euro area shocks to the U.S. Although the sign of the coefficient aligns with the theoretical prediction—higher stablecoin capitalization amplifies spillovers to the U.S.—the estimates are volatile and not statistically significant. This likely reflects the still limited adoption of U.S. stablecoins in the euro area.

Figure 12 reports the response of the entire yield curve estimated using Equation (12). We focus on three horizons: on impact ($t + 0$), after two month ($t + 8$), and after one quarter ($t + 15$). The results broadly align with prior intuition: stablecoins increase the pass-through of monetary policy shocks to the U.S. yield curve as a whole. The effect is initially concentrated at shorter maturities, consistent with the composition of assets held by stablecoin issuers. Over time, the shock gradually propagates to longer maturities of

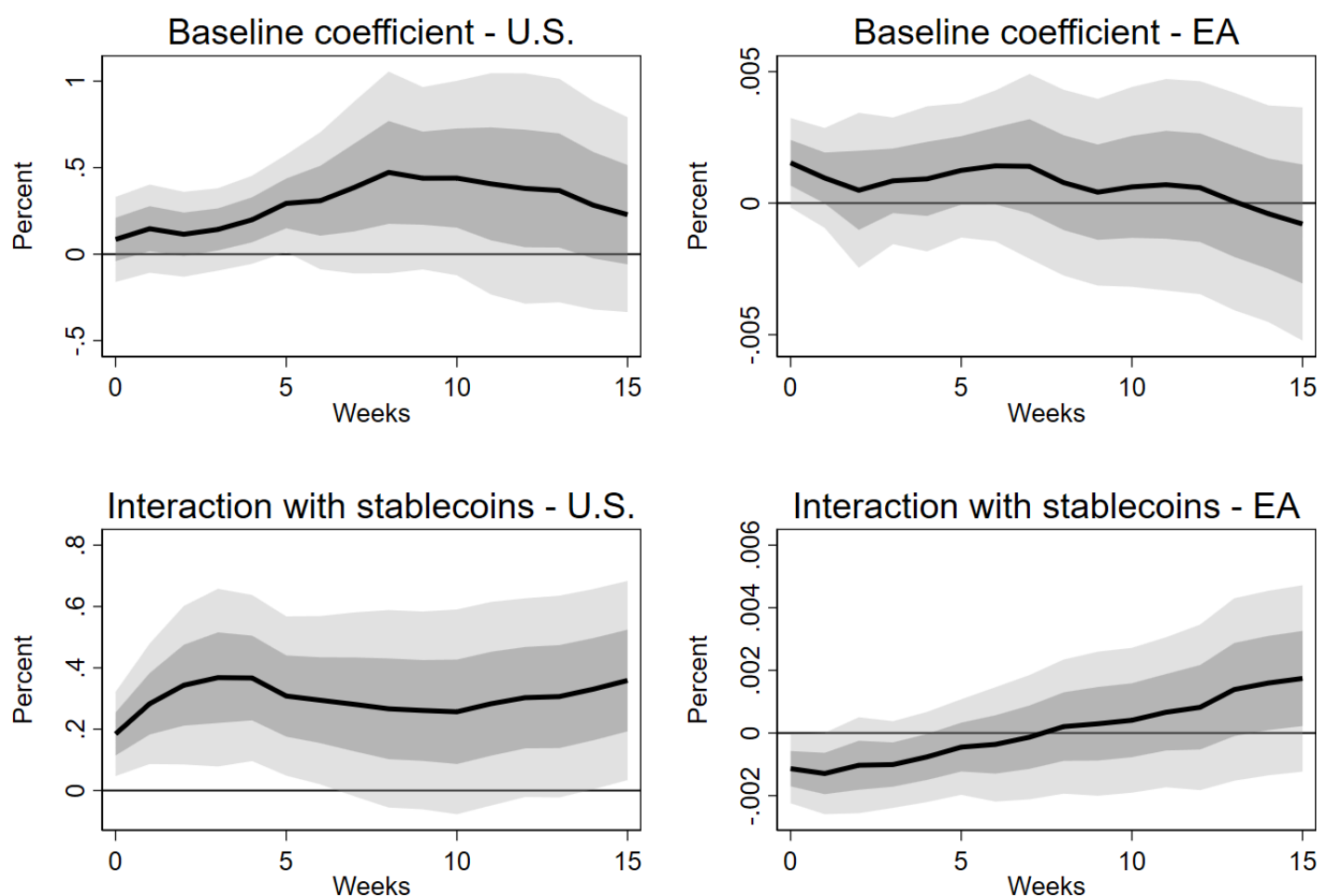


Figure 11: Response of U.S. yields to U.S. and euro area monetary policy shocks.

Notes: the chart shows the impulse response of U.S. 3-month yields to a 25 basis points U.S. (left column) and EA monetary policy shock. The upper panel shows the baseline coefficient and the lower panel the interaction with the stablecoin share in U.S. debt, β^1 and β^2 respectively in Equation (12). Local projection control for 4 lags of the dependent variable, the (log) U.S. dollar nominal effective exchange rate, the (log) U.S. stock price, the log of stablecoins market capitalization (proxied by USDT and USDC market capitalization), a time trend and year dummies.

U.S. government debt.

Finally, Figure A.9 in the Appendix reports the response of the stablecoin market capitalization to U.S. monetary policy shocks. In line with the model's predictions and the results by Ahmed and Aldasoro (2025), the market capitalization of stablecoins contracts after a U.S. monetary policy tightening.

6 Conclusion

This paper studies the macro-financial implications of dollar-backed stablecoins within a three-country DSGE framework. We show that the emergence of fully collateralized,

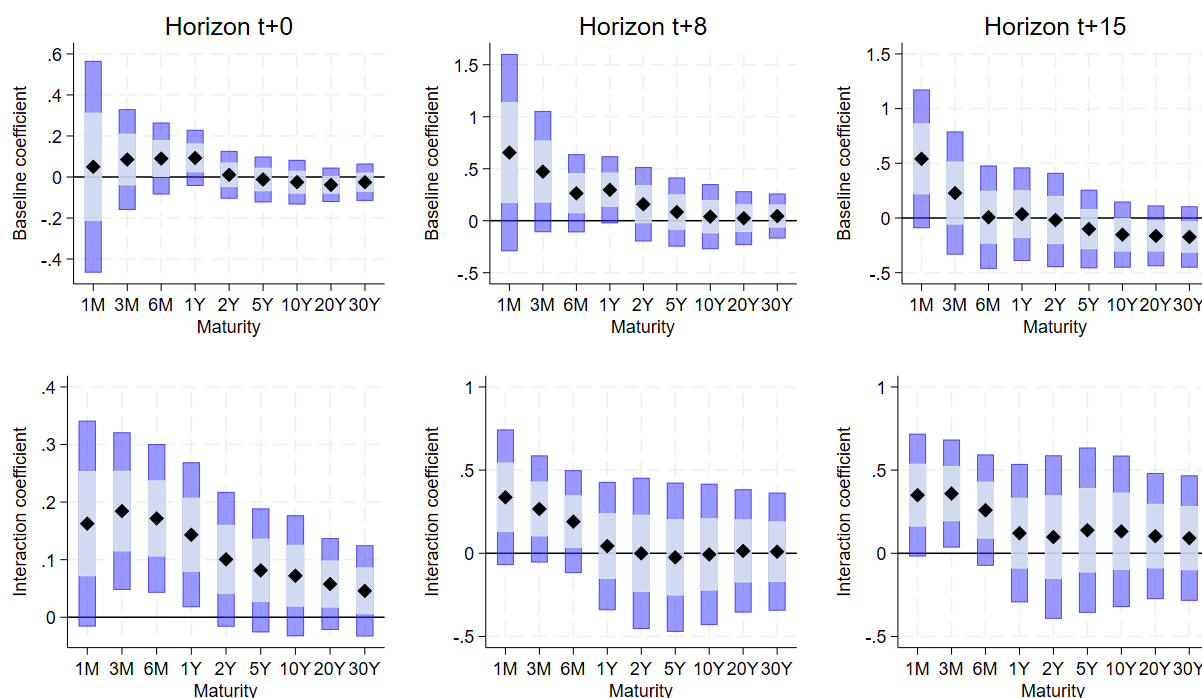


Figure 12: Response of U.S. yields at different maturities to U.S. monetary policy shocks. **Notes:** the chart shows the impulse response of the U.S. yield curve to a 25 basis points U.S. monetary policy shock. Responses are estimated with Equation (12) and the figure shows coefficients for $k = 0, 4, 8$. The upper panel shows the baseline coefficient and the lower panel the interaction with the stablecoin share in U.S. debt, β^1 and β^2 respectively in Equation (12). Local projection control for 4 lags of the dependent variable, the (log) U.S. dollar nominal effective exchange rate, the (log) U.S. stock price, the log of stablecoins market capitalization (proxied by USDT and USDC market capitalization), a time trend and year dummies.

unremunerated stablecoins, used as settlement (i.e. in payments) and trading tokens (i.e. as a saving instrument), strengthens the fiscal–monetary nexus by tying private money creation and global payment needs directly to U.S. public debt.

Through this new global safe asset channel, stablecoins lower equilibrium U.S. Treasury yields. At the same time, they also impact the transmission of U.S. monetary policy and global shocks due to the new capital flows generated by stablecoin demand. If stablecoins are largely adopted, U.S. monetary policy shocks transmit more strongly to U.S. yields but less to the real economy. Holders of stablecoins would in fact sell the digital assets after a tightening shock, forcing the issuer to liquidate bonds held as reserves. That pushes up further U.S. yields but also weakens the dollar as foreigners exchange the stablecoin for domestic currency - limiting the real effect of the policy shock. As a result, U.S. monetary policy becomes less effective in steering aggregate demand and needs to respond more aggressively to inflation to stabilize the economy. For a similar channel, stablecoin adoption increases the international spillovers of shocks both from the U.S.

to other countries and from other countries to the U.S., making the U.S. economy more exposed to the global business cycle.

We show analytically that these effects depend on the demand for stablecoins. At moderate adoption levels, their effects on stability are limited; at larger scales, however, foreign demand for stablecoins magnifies bond market volatility and weakens monetary control. This happens because as stablecoin holdings increase, they are more used as a dollar financial assets than as a pure transaction instrument. Consequently, households adjust their stablecoin positions more actively in response to changes in global financial conditions. The larger are stablecoin holdings the stronger are these effects up to the point that central banks need to react more aggressively to preserve macroeconomic stability. As stablecoins become largely adopted, in fact, monetary policy is less effective in steering aggregate demand because of (digital) capital flows. Empirically, we find that greater stablecoin capitalization dampens the response of short-term yields to policy shocks, consistently with the theoretical findings.

Overall, stablecoins extend the global reach of the dollar and the depth of U.S. debt markets, but at the cost of tighter fiscal–monetary linkages and heightened exposure to global financial cycles. In this sense, the *global safe asset channel* could reshape also the way global actors interface with the dollar system: reserve managers, sovereign wealth funds and private intermediaries face an environment in which payment technologies, safe-asset demand and policy shocks are more tightly intertwined. A natural next step is to study how this architecture coexists with attempts to diversify away from the dollar, to build alternative payment and reserve arrangements, and to use financial regulation, sanctions or swap lines as tools of economic stability. Our framework provides a tractable starting point to analyse these questions in settings with heterogeneous issuers of digital money and richer geopolitical shocks.

References

- Abad, J., Nuño, G., and Thomas, C. “CBDC and the operational framework of monetary policy”. *Banco de Espana Working Paper*, (2404), 2024.
- Adalid, R., Álvarez-Blázquez, Á., Assenmacher, K., Burlon, L., Dimou, M., López-Quiles, C., Martín Fuentes, N., Meller, B., Muñoz, M., and Radulova, P. “Central bank digital currency and bank intermediation”. Occasional Paper Series 293, European Central Bank, 2022.
- Adjemian, S., Juillard, M., Karame, F., Mutschler, W., Pfeifer, J., Ratto, M., Rion, N., and Villemot, S. “Dynare: Reference Manual, Version 6”. Dynare Working Papers 80, CEPREMAP, Feb. 2024.
- Agur, I., Ari, A., and Dell’Ariccia, G. “Designing central bank digital currencies”. *Journal of Monetary Economics*, 125:62–79, 2022.
- Ahmed, R. and Aldasoro, I. “Stablecoins and safe asset prices”. BIS Working Paper 1270, Bank for International Settlements, May 2025.
- Ahmed, R. and Rebucci, A. “Dollar reserves and us yields: Identifying the price impact of official flows”. *Journal of International Economics*, 152:103974, 2024.
- Ahmed, R., Clouse, J. A., Natalucci, F., Rebucci, A., and Sun, G. “Stablecoins: A revolutionary payment technology with financial risks”. Working Paper 34475, National Bureau of Economic Research, November 2025.
- Aldasoro, I., Cornelli, G., Ferrari Minesso, M., Gambacorta, L., and Habib, M. M. “Stablecoins, money market funds and monetary policy”. *Economics Letters*, 247:112203, 2025.
- Altavilla, C., Boucinha, M., Burlon, L., Adalid, R., Fortes, R., and Maruhn, F. “Stablecoins and monetary policy transmission”. *Mimeo*, 2025.
- Anadu, K., Azar, P., Huang, C., Cipriani, M., Eisenbach, T. M., La Spada, G., Landoni, M., Macchiavelli, M., Malfroy-Camine, A., and Wang, J. C. “Runs and flights to safety: Are stablecoins the new money market funds?”. Technical Report 1073, 2024.

- Andolfatto, D. “Assessing the impact of central bank digital currency on private banks”. *The Economic Journal*, 131(634):525–540, 2021.
- Arner, D., Auer, R., and Frost, J. “Stablecoins: Risks, potential and regulation”. *BIS Working Papers*, 2020.
- Assenmacher, K., Bitter, L., and Ristiniemi, A. “CBDC and business cycle dynamics in a New Monetarist New Keynesian model”. (2811), 2023.
- Assenmacher, K., Minesso, M. F., Mehl, A., and Pagliari, M. S. “Managing the transition to central bank digital currency”. (2907), 2024.
- Azzimonti, M. and Quadrini, V. “Digital economy, stablecoins, and the global financial system”. NBER Working Papers 34066, National Bureau of Economic Research, Inc, Jul 2025.
- Barthélemy, J., Pedrono, J., and Schmidt, J. “The international monetary system: Are stablecoins game changers?”, November 2025.
- Benigno, P., Schilling, L. M., and Uhlig, H. “Cryptocurrencies, currency competition, and the impossible trinity”. *Journal of international economics*, 136:103601, 2022.
- Bertsch, C. “Stablecoins: Adoption and fragility”. *Sveriges Riksbank Working Paper Series*, (423), 2023.
- Bianchi, J. and Sosa-Padilla, C. “International Sanctions and Dollar Dominance”. *NBER Working Papers*, (31024), 2023.
- Birch, D. “Stablecoins are a necessary, but interim, development”. *OMFIF Blog*, 2025.
- Bolt, W., Frost, J., Shin, H. S., and Wierds, P. “The bank of amsterdam and the limits of fiat money”. *Journal of Political Economy*, 132(12):3919–3941, 2024.
- Brunnermeier, M. K. and Niepelt, D. “On the equivalence of private and public money”. *Journal of Monetary Economics*, 106:27–41, 2019.
- Burlon, L., Muñoz, M. A., and Smets, F. “The optimal quantity of cbdc in a bank-based economy”. *American Economic Journal: Macroeconomics*, 16(4):172–217, 2024.

- Cipollone, P. “Preparing the future of payments and money: the role of research and innovation”. *Keynote speech at The future of payments conference*, September 2025.
- Clayton, C., dos Santos, A., Maggiori, M., and Schreger, J. “International currency competition”, Nov. 2024. SSRN working paper.
- Corbet, S., Hou, Y. G., Hu, Y., Larkin, C., and Oxley, L. “Any port in a storm: Cryptocurrency safe-havens during the covid-19 pandemic”. *Economics letters*, 194:109377, 2020.
- Cova, P., Notarpietro, A., Pagano, P., and Pisani, M. “Monetary policy in the open economy with digital currencies”. *Bank of Italy Temi di Discussione (Working Paper) No*, 1366, 2022.
- d’Avernas, A., Maurin, V., and Vandeweyer, Q. “Can stablecoins be stable?”. *University of Chicago, Becker Friedman Institute for Economics Working Paper*, (2022-131), 2022.
- Doerr, S., Eren, E., and Malamud, S. “Money market funds and the pricing of near-money assets”. *Swiss Finance Institute Research Paper*, (23-04), 2023.
- Eichenbaum, M. S., Johannsen, B. K., and Rebelo, S. T. “Monetary Policy and the Predictability of Nominal Exchange Rates”. *The Review of Economic Studies*, 88(1): 192–228, 2021.
- Eichengreen, B. “Will stablecoins preserve dollar dominance?”. *Project Syndicate*, October 2025.
- Eichengreen, B., Nguyen, M. T., and Viswanath-Natraj, G. “Stablecoin devaluation risk”. *The European Journal of Finance*, pages 1–28, 2025.
- Eren, E., Schrimpf, A., and Xia, F. D. “The demand for government debt”. *Management Science*, *forthcoming*, 2025.
- European Central Bank. “Progress on the preparation phase of a digital euro - closing progress report”. 2025.
- European Central Bank. “The international role of the euro report”. June 2025.

- European Systemic Risk Board. “Crypto-assets and financial stability risks”. Technical report, European Systemic Risk Board, October 2025. URL https://www.esrb.europa.eu/pub/pdf/reports/esrb.report202510_cryptoassets.en.pdf. ESRB Report.
- Farhi, E. and Maggiori, M. “A model of the international monetary system”. *The Quarterly Journal of Economics*, 133(1):295–355, None 2018.
- Feenstra, R. C. “Functional Equivalence Between Liquidity Costs and the Utility of Money”. *Journal of Monetary Economics*, 17(2):271–291, 1986.
- Ferrari Minesso, M. and Pagliari, M. S. “No country is an island. International cooperation and climate change”. *Journal of International Economics*, 145:103816, 2023.
- Ferrari Minesso, M., Mehl, A., and Stracca, L. “Central bank digital currency in an open economy”. *Journal of Monetary Economics*, 127:54–68, 2022.
- Ferrari Minesso, M., Krahne, T., Mehl, A., and Vansteenkiste, I. “Seizing central bank assets?”. *CEPR Discussion Papers*, (19186), 2024.
- Financial Stability Board. “Annual progress report on meeting the targets for cross-border payments: 2024 report on key performance indicators”. June 2025.
- Friedman, M. “The euro-dollar market: Some first principles”. *Federal Reserve Bank of St. Louis Review*, (July 1971), 1971.
- Frydl, E. “The debate of regulating the eurocurrency markets”. *Federal Reserve Bank of New York Quarterly Review*, 1979.
- Gabaix, X. and Maggiori, M. “International liquidity and exchange rate dynamics”. *The Quarterly Journal of Economics*, 130(3):1369–1420, None 2015.
- Gorton, G. B., Klee, E. C., Ross, C. P., Ross, S. Y., and Vardoulakis, A. P. “Leverage and stablecoin pegs”. *Journal of Financial and Quantitative Analysis*, pages 1–65, 2022.
- Greenwood, R., Landier, A., and Thesmar, D. “Vulnerable banks”. *Journal of financial economics*, 115(3):471–485, 2015.

- Hoang, L. T. and Baur, D. G. “How stable are stablecoins?”. *The European Journal of Finance*, 30(16):1984–2000, 2024.
- Ikeda, D. “Digital Money as a Medium of Exchange and Monetary Policy in Open Economy”. *IMES Discussion Paper Series*, 2022-E-10, 2022.
- Jiang, Z., Krishnamurthy, A., and Lustig, H. “Foreign Safe Asset Demand and the Dollar Exchange Rate”. *The Journal of Finance*, 76(3):1049–1089, 2021.
- Jiang, Z., Krishnamurthy, A., Lustig, H., and Sun, J. “Convenience Yields and Exchange Rate Puzzles”. *NBR Working Papers*, (32092), 2024.
- Karau, S. “Monetary policy and bitcoin”. *Journal of International Money and Finance*, 137:102880, 2023.
- Kim, S. R. “Macro-financial impact of stablecoin’s demand for treasuries”. *Available at SSRN 5259528*, 2025.
- Kosse, A., Glowka, M., Mattei, I., and Rice, T. “Will the real stablecoin please stand up?”. *BIS Papers*, 2023.
- Krishnamurthy, A. and Vissing-Jorgensen, A. “The aggregate demand for treasury debt”. *Journal of Political Economy*, 120(2):233–267, 2012.
- Kumhof, M., Pinchetti, M., Rungcharoenkitkul, P., and Sokol, A. “CBDC Policies in Open Economies”. *CEPR Discussion Papers*, (17982), 2023.
- Li, J. “Predicting the demand for central bank digital currency: A structural analysis with survey data”. *Journal of Monetary Economics*, 134:73–85, 2023.
- Liang, P., Sampaio, M., and Sarkisyan, S. “Digital payments and monetary policy transmission”. Working Paper Series 2024-14, Ohio State University, Charles A. Dice Center for Research in Financial Economics, Aug 2024.
- Liu, Y. and Tsyvinski, A. “Risks and returns of cryptocurrency”. *The Review of Financial Studies*, 34(6):2689–2727, 2021.
- Luck, S. “A historical perspective on stablecoins”. *Liberty Street Economics*, (October 1), 2025. URL <https://doi.org/10.59576/lse.20251001>. Liberty Street Economics Blog.

- Lyons, R. K. and Viswanath-Natraj, G. “What keeps stablecoins stable?”. *Journal of International Money and Finance*, 131:102777, 2023.
- Maggiore, M. “Financial intermediation, international risk sharing, and reserve currencies”. *American Economic Review*, 107(10):3038–3071, October 2017.
- Maggiore, M. “International macroeconomics with imperfect financial markets”. CEPR Discussion Papers 17197, C.E.P.R. Discussion Papers, Apr 2022.
- Maggiore, M., Neiman, B., and Schreger, J. “International currencies and capital allocation”. *Journal of Political Economy*, 128(6):2019–2066, June 2020.
- McCauley, R. N. “The offshore dollar and us policy”. *FRB Atlanta Policy Hub Paper*, (2024-02), 2024.
- Mizrach, B. “Stablecoins: Survivorship, transactions costs, and exchange microstructure”. *Journal of Alternative Investments*, 27(4):82–109, 2025.
- Niepel, D. “Money and banking with reserves and cbdc”. *The Journal of Finance*, 79(4):2505–2552, 2024.
- Oefele, N., Baur, D. G., and Smales, L. A. “Are stablecoins the money market mutual funds of the future?”. *Journal of Empirical Finance*, 79:101557, 2024.
- Portes, R. “The stablecoin loophole that could expose the eu”. *Financial Times*, July 2025. Opinion article.
- Rey, H. “Stablecoins, tokens, and global dominance”. *F&D Magazine*, September 2025.
- Rogoff, K., Johnson, S., Rey, H., and Varoufakis, Y. “Trump’s stablecoin gamble”. Project Syndicate, The Big Picture (OnPoint), August 2025.
- Schilling, L., Fernández-Villaverde, J., and Uhlig, H. “Central bank digital currency: When price and bank stability collide”. *Journal of Monetary Economics*, 145:103554, 2024.
- Scotti, C. “Stablecoins in the payments ecosystem: Reflections on responsible innovation”. In *Economics of Payments XIV Conference*, Rome, September 2025. Banca d’Italia.

- Singh, R. “J.p. morgan wary of stablecoin’s trillion-dollar growth bets, cuts them by half”. Reuters, July 2025.
- Valchev, R. “Bond Convenience Yields and Exchange Rate Dynamics”. *American Economic Journal: Macroeconomics*, 12(2):124–166, 2020.
- Waller, C. J. “Technological advancements in payments”. *Speech at the Wyoming Blockchain Symposium 2025*, August 2025.

Online appendix

A Model

A.1 Model setup

There are three economies in the model, indexed by $c \in \{US, S, F\}$, each with size n_c . Within each country, households consume, save, hold real balances, supply labor to firms and invest in firms' capital. Households also need liquidity to finance transactions, which we model as an aggregator of money and stablecoins. Firms produce undifferentiated final goods which are bundled together by retailers and sold on final markets with monopoly power. Under Rotemberg pricing, firms face a quadratic cost proportional to the parameter $\phi_{c,r}$ to update prices. We set the Rotemberg parameter to match a frequency of price adjustment of 0.6 under the equivalent Calvo pricing formalism of the problem. Goods and bonds are traded across countries and determine the exchange rate. We assume that bond markets are incomplete, hence the UIP condition does not hold across countries, and that U.S. bonds deliver a convenience yield. For this reason, in steady-state, U.S. bonds have lower returns.

Central banks set the interest rate and the government the level of public consumption. It taxes and borrows on national and international markets. We assume a convenience yield on US bonds, the value of the convenience yield is calibrated as in [Eren et al. \(2025\)](#). Finally, we assume there is a global stablecoin issuer, owned by US households, that issues a dollar-denominated stablecoin to households in all three countries. Stablecoins bear no remuneration and are fully backed by US safe assets. Intermediation profits from stablecoin issuance are rebated to US households.

A.2 Households

Households derive utility from consumption, holdings of real balances (domestic cash and US bonds) and leisure:

$$U_{c,t} = e_{c,t}^C \ln(C_{c,t} - h_c C_{c,t-1}) - \frac{\chi_{c,l}}{1 + \phi_{c,l}} L_{c,t}^{1+\phi_{c,l}} + \frac{\chi_{c,b}}{1 - \sigma_{c,b}} \left(\frac{B_{US,c,t}}{P_{c,t} NER_{US,c,t}} \right)^{1-\sigma_{c,b}} \quad (\text{A.1})$$

with c the country index. The budget constraint is:

$$\begin{aligned}
& P_{c,t}C_{c,t} + B_{c,c,t} + \sum_{l \neq c} NER_{l,c,t}B_{l,c,t} + P_{c,t}I_{c,t} + M_{c,t} + \frac{SC_{c,t}}{NER_{US,c,t}} \\
& \leq w_{c,t}L_{c,t} + R_{c,t-1}B_{c,c,t-1} + \sum_{l \neq c} NER_{l,c,t}R_{l,t-1}B_{l,c,t-1} - \sum_{l \neq c} \frac{\phi_c^B}{2} \left(\frac{B_{l,c,t}NER_{l,c,t}}{P_{c,t}} \right)^2 P_{c,t} \\
& \quad + P_{c,t}R_{c,t}^k K_{c,t-1} + M_{c,t-1} + \frac{R_{sc,t-1}SC_{c,t-1}}{NER_{US,c,t}} + \Pi_{c,t}
\end{aligned} \tag{A.2}$$

C_c is aggregate consumption, L_c aggregate labor and $e_{c,t}^C$ is a consumption preference shock. I_c , K_c , R_c^k , w_c are investments, capital, capital returns, nominal wages. P_c is an aggregate price level. $NER_{l,c,t}$ is the nominal exchange rate between country c and country l (expressed in units of currency of l per currency of c), $B_{l,c,t}^F$ are bonds issued in country l and held in country c (i.e. foreign bond holdings) and $R_{l,t}$ is the (foreign) interest rate on bonds issued in country l . M_c are money balances; $\frac{\chi_{c,b}}{1-\sigma_{c,b}} \left(\frac{B_{US,c,t}}{NER_{US,c,t}} \right)^{1-\sigma_{c,b}}$ captures the utility value of holding U.S. bonds, that reflects, for example, liquidity services. SC_c are stablecoin holdings expressed in U.S. dollars while R_{sc} is the stablecoin remuneration that we set to 1 in the baseline calibration. Finally, Π_c includes lump-sum taxes as well as profits transferred from retailers to households. The cash-in-advance constraint takes the form of:

$$C_{c,t} = \left[\mu_{sc,c,t} \left(\frac{SC_{c,t}}{NER_{US,c,t}} \right)^{\eta_m} + (1 - \mu_{sc,c,t}) (M_{c,t})^{\eta_m} \right]^{\frac{1}{\eta_m}} \tag{A.3}$$

with $\mu_{sc,c,t} = \mu_{sc,c}^* - \left(\alpha_\mu \sigma_{q,t}^2 + \beta_\mu \exp\{1 - R_{US,t}\} + \gamma_\mu \rho_t \right)$, where $\alpha_\mu \sigma_{sc}$ capture the change in stablecoin preferences due to volatility, $\beta_\mu \exp\{1 - R_{US,t}\}$ due to risks of stablecoin issuers incurring in negative profits and $\gamma_\mu \rho_t$ is a stablecoin demand shock. The law of motion of capital in each sector is:

$$K_{c,t+1} = \left\{ (1 - \delta_c)K_{c,t} + I_{c,t} \left[1 - \frac{\phi_c^K}{2} \left(\frac{I_{c,t}}{I_{c,t-1}} - 1 \right)^2 \right] \right\} \tag{A.4}$$

δ_c the depreciation rate of capital and ϕ_c^K the capital adjustment cost. Households optimize utility (Equation (A.1)) under the budget constraint (Equation (A.2)), liquidity constraint (Equation (A.3)) and the law of motion of capital (Equation (A.4)). First

order conditions are:

$$\frac{e_{c,t}^C}{C_{c,t} - hC_{c,t-1}} - E_t \frac{\beta_c e_{c,t+1}^C h_c}{C_{c,t+1} - h_c C_{c,t}} = \lambda_{c,t} \quad (\text{A.5})$$

$$\chi_c L_{c,t}^{\phi_c} = \lambda_{c,t} \frac{w_{c,t}}{P_{c,t}} \quad (\text{A.6})$$

$$\beta_c E_t \left(\frac{R_{c,t} \lambda_{c,t+1}}{\pi_{c,t+1}} \right) = \lambda_{c,t} \quad (\text{A.7})$$

$$\lambda_{c,t} \left[1 + \phi_c^B \left(\frac{RER_{l,c,t} B_{l,c,t}}{P_{l,t}} \right) \right] = \beta_c E_t \left(\frac{\lambda_{c,t+1}}{\pi_{c,t+1}} R_{l,t} \frac{NER_{l,c,t+1}}{NER_{l,c,t}} \right) \quad (\text{A.8})$$

$$\begin{aligned} Q_{c,t} \left\{ \left[1 - \frac{\phi_c^K}{2} \left(\frac{I_{c,t}}{I_{c,t-1}} - 1 \right)^2 \right] - \frac{I_{c,t}}{I_{c,t-1}} \phi_c^K \left(\frac{I_{c,t}}{I_{c,t-1}} - 1 \right) \right\} + \\ + \beta_c E_t \left[Q_{c,t+1} \phi_c^K \left(\frac{I_{c,t}}{I_{c,t-1}} - 1 \right) \left(\frac{I_{c,t+1}}{I_{c,t}} \right)^2 \right] = \lambda_{c,t} \\ \beta_c E_t [Q_{c,t+1} (1 - \delta_c) + \Lambda_{c,t+1} R_{c,t+1}^k] = Q_{c,t} \end{aligned} \quad (\text{A.9})$$

$$-\frac{\lambda_{c,t}}{NER_{US,c,t}} + \beta_c E_t \left(\frac{\lambda_{c,t+1}}{NER_{US,c,t+1}} \frac{R_{sc,t}}{\pi_{c,t+1}} \right) + \gamma_{c,t} \mu_t \left(\frac{SC_{c,t}}{NER_{US,c,t} P_t C_{c,t}} \right)^{\eta_m - 1} \frac{1}{NER_{US,c,t}} = 0 \quad (\text{A.10})$$

$$-\lambda_{c,t} + \beta_c E_t \left(\frac{\lambda_{c,t+1}}{\pi_{c,t+1}} \right) + \gamma_{c,t} \left(\frac{M_{c,t}}{C_{c,t}} \right)^{\eta_m - 1} = 0 \quad (\text{A.11})$$

Q_c is the price of capital in country c and $\{\lambda_{c,t}\}_{t=0}^{\infty}$, $\{\gamma_{c,t}\}_{t=0}^{\infty}$ the sequences of Lagrangian multipliers associated to the optimization problem. Equation (A.5) is the Euler equation for consumption, defining the marginal utility of consumption (λ_c). Equation (A.6) defines labor supply which equals the value, in terms of consumption, of the real wage. Equation (A.7) determines the demand for domestic bonds while Equation (A.8) for foreign bonds. The liquidity premium $\left(\frac{\chi_{c,b}}{\lambda_{c,t}} \left(\frac{B_{US,c,t}}{RER_{US,c,t}} \right)^{-\sigma_{c,b}} \right)$ applies only to holdings of U.S. bonds outside the U.S. Notice that bond markets are not complete (because of cross-border transactions costs for $\phi_c^B > 0$), therefore combining Equation (A.7) and Equation (A.8) does not lead to perfect UIP. Moreover, as each country has two trade partners, there is one Equation (A.8) for each of them. Finally, Equation (A.8) is the Tobin's Q equation, Equation (A.10) is the stablecoin demand equation and Equation (A.11) is the money demand equation. First order conditions are exactly symmetric

in each country excluding the U.S. for which the Euler condition on domestic bonds is:

$$\beta_c E_t \left(\frac{R_{c,t} \lambda_{c,t+1}}{\pi_{c,t+1}} \right) + \chi_{c,m} \left(\frac{B_{c,c,t}}{P_{c,t}} \right)^{-\sigma_{c,b}} = \lambda_{c,t} \quad (\text{A.12})$$

A.3 Production

In each country there is a *continuum* of perfectly competitive firms, indexed by k . Firms' production function is:

$$X_{c,t}(k) = A_{c,t} (K_{c,t}(k))^{\alpha_c} (L_{c,t}(k))^{1-\alpha_c} \quad (\text{A.13})$$

where A_c is a total factor productivity shock. Total output is the sum of output consumed domestically and exported, formally $X_t(k) = \sum_{l=US,S,RoW} X_{c,l,t}^b(k)$ with domestic demand being for $X_{c,c,t}^b(k)$. Cost minimization implies:

$$R_{c,t}^k(k) = A_{c,t} MC_{c,t}(k) \alpha_c (K_{c,t}(k))^{\alpha_c-1} (L_{c,t}(k))^{1-\alpha_c} \quad (\text{A.14a})$$

$$W_{c,t}(k) = A_{c,t} MC_{c,t}(k) (1 - \alpha_c) (K_{c,t}(k))^{\alpha_c} (L_{c,t}(k))^{-\alpha_c} \quad (\text{A.14b})$$

where MC_c is the Lagrangian multiplier associated to the optimization problem of firms and W_c real wages.

A.4 Retailers and aggregation

Retailers aggregate intermediate goods and transform them into final goods. Define $Y_{c,c}$ and $Y_{l,c}$ as domestic demand and export, to country l , of final goods. We adopt the following aggregators for domestic demand:

$$C_{c,c,t} + I_{c,c,t} + G_{c,c,t} = Y_{c,c,t} \quad (\text{A.15})$$

where $C_{c,c}$, $I_{c,c}$, $G_{c,c}$ denote final consumption, investment and government spending in country c of goods produced in country c . Exports aggregators are defined as:

$$\sum_{l \neq c} C_{l,c,t} + I_{l,c,t} + G_{l,c,t} = Y_{l,c,t} \quad (\text{A.16})$$

where $C_{l,c}$, $I_{l,c}$, $G_{l,c}$ denote final consumption, investment and government demand in country l of goods produced in country c . In other terms, these are total exports from country c to country l . $Y_{c,c}$ and $Y_{l,c}$ are produced aggregating across undifferentiated intermediate goods produced by domestic and foreign firms, respectively. The demand function for these goods are:

$$Y_{c,c,t} = \left[\int_0^1 X_{c,c,t}(k)^{\frac{\nu_c-1}{\nu_c}} dk \right]^{\frac{\nu_c}{\nu_c-1}}, \quad Y_{l,c,t} = \left[\int_0^1 X_{l,c,t}(k)^{\frac{\nu_c-1}{\nu_c}} dk \right]^{\frac{\nu_c}{\nu_c-1}} \quad (\text{A.17})$$

where ν_c is the elasticity of substitution across different goods produced by country c .

Price aggregators are:

$$P_{c,c,t} = \left[\int_0^1 P_{c,c,t}(k)^{1-\nu_c} dk \right]^{\frac{1}{1-\nu_c}}, \quad P_{l,c,t} = \left[\int_0^1 P_{l,c,t}(k)^{1-\nu_c} dk \right]^{\frac{1}{1-\nu_c}} \quad (\text{A.18})$$

with $P_{c,c,t}$ and $P_{l,c,t}$ being the prices of domestically consumed and exported goods. Demand functions for individual varieties are:

$$X_{c,c,t}(k) = \left[\frac{P_{c,c,t}(k)}{P_{c,c,t}} \right]^{-\nu_c} Y_{c,c,t}, \quad X_{l,c,t}(k) = \left[\frac{P_{l,c,t}(k)}{P_{l,c,t}} \right]^{-\nu_c} Y_{l,c,t} \quad (\text{A.19})$$

Final consumption goods are created by combining goods from each country. Aggregate consumption C_c then is:

$$C_{c,t} = \left\{ \sum_l \omega_{c,l} (C_{c,l,t})^{\rho_c} \right\}^{\frac{1}{\rho_c}} \quad (\text{A.20})$$

$C_{c,l}$ is intermediate consumption consumed in country c and produced in l , so that for example $C_{c,c}$ is domestic consumption of domestically produced goods. $\omega_{c,l} \in [0, 1]$ captures the share of goods produced in country l in total consumption, with $\omega_{c,c}$ the home bias. Similarly, aggregate government spending and investment are:

$$G_{c,t} = \left\{ \sum_l \omega_{c,l} (G_{c,l,t})^{\rho_c} \right\}^{\frac{1}{\rho_c}} \quad (\text{A.21})$$

$$I_{c,t} = \left\{ \sum_l \omega_{c,l} (I_{c,l,t})^{\rho_c} \right\}^{\frac{1}{\rho_c}} \quad (\text{A.22})$$

where $G_{c,c}$ ($I_{c,c}$) is government consumption (investment) of domestically produced

goods while $G_{c,l}$ ($I_{c,l}$) is government consumption (investment) of goods produced in country l . Cost minimization defines the demand function for consumption of domestic and imported goods:

$$\begin{aligned} C_{c,c,t} &= \left(\frac{P_{c,c,t}}{P_{c,t}} \right)^{\frac{1}{\varrho_c-1}} \omega_{c,c} C_{c,t}, & C_{c,l,t} &= \left(\frac{P_{c,l,t}}{P_{c,t}} \right)^{\frac{1}{\varrho_c-1}} \omega_{c,l} C_{c,t} \\ G_{c,c,t} &= \left(\frac{P_{c,c,t}}{P_{c,t}} \right)^{\frac{1}{\varrho_c-1}} \omega_{c,c} G_{c,t}, & G_{c,l,t} &= \left(\frac{P_{c,l,t}}{P_{c,t}} \right)^{\frac{1}{\varrho_c-1}} \omega_{c,l} G_{c,t} \\ I_{c,c,t} &= \left(\frac{P_{c,c,t}}{P_{c,t}} \right)^{\frac{1}{\varrho_c-1}} \omega_{c,c} I_{c,t}, & I_{c,l,t} &= \left(\frac{P_{c,l,t}}{P_{c,t}} \right)^{\frac{1}{\varrho_c-1}} \omega_{c,l} I_{c,t} \end{aligned} \quad (\text{A.23})$$

whereby, demand in country c for goods produced in country l depends on their price relative to the aggregate price level and by total consumption in country c . The aggregate price level is then obtained by substituting Equation (A.23) into Equation (A.20):

$$P_{c,t} = \left\{ \sum_l \omega_{c,l} (P_{c,l,t}^b)^{\frac{\rho_c}{1-\rho_c}} \right\}^{\frac{\rho_c-1}{\rho_c}} \quad (\text{A.24})$$

A.5 Monopolists

Goods are sold on the final market by monopolists who set prices with some degree of market power. We assume that there are frictions in price setting à-la Rotemberg. Formally, monopolists optimize:

$$E_t \sum_{d=0}^{\infty} (\beta_c)^d \Lambda_{c,t+d} \left\{ \frac{NER_{l,c,t} P_{l,c,t}}{P_{c,t}} X_{l,c,t}(k) - TC_{c,t} - \frac{\phi_{c,r}}{2} \left(\frac{P_{l,c,t}}{P_{l,c,t-1}} - 1 \right)^2 \right\} \quad (\text{A.25})$$

where TC are total costs and $\phi_{c,r}$ the Rotemberg parameter. $l = c$ is the case of domestic goods sold in the domestic economy and the exchange rate $NER_{c,c,t}$ is 1. First order conditions are:

$$\begin{aligned} (1 - \nu_c) + \frac{MC_{c,t} \nu_c}{P_{l,c,t} NER_{l,c,t}} - \phi_{c,r} (\pi_{l,c,t} - 1) \frac{\pi_{l,t}}{P_{l,c,t-1} NER_{l,c,t}} + \\ \beta_c E_t \left[\frac{\lambda_{t+1}}{\lambda_t} \frac{Y_{l,c,t+1}}{Y_{l,c,t}} \phi_{c,r} (\pi_{l,c,t+1} - 1) \frac{\pi_{l,c,t+1}}{P_{l,c,t} NER_{l,c,t}} \right] = 0 \end{aligned} \quad (\text{A.26})$$

Inflation rates are $\pi_{c,l,t} = \frac{P_{c,l,t}}{P_{c,l,t-1}}$ and $\pi_{c,t} = \frac{P_{c,t}}{P_{c,t-1}}$. Notice that the Rotemberg pa-

parameter $\phi_{c,r}$ can be written as a function of the probability of updating prices ξ_c as:

$$\phi_{c,r} = \frac{(\nu_c - 1)\xi_c}{(1 - \xi_c)(1 - \beta_c \xi_c)}$$

A.6 Public sector & stablecoin issuer

The central bank sets the policy rate and manages its own balance sheet. The policy rate follows a Taylor-type rule of the form:

$$\ln R_{c,t} = (1 - \varrho) \ln R_{c,t-1} + \varrho \left[\ln R_{c,ss} + \theta_\pi \ln \pi_{c,t} + \theta_y (\ln Y_{c,t} - \ln Y_{c,ss}) \right] + \varepsilon_{c,t}. \quad (\text{A.27})$$

where Y_c is total output.

The government sets public consumption exogenously and follows a tax rule aimed at stabilizing debt as share of GDP relative to the steady-state ($\frac{Debt_{c,t}}{Y_{c,t}} - \frac{Debt_{c,ss}}{Y_{c,ss}}$):

$$\ln \left(\frac{T_{c,t}}{T_{c,ss}} \right) = \varrho_T \ln \left(\frac{T_{c,t-1}}{T_{c,ss}} \right) + (1 - \varrho_T) \kappa_T \left[\frac{Debt_{c,t}}{Y_{c,t}} - \frac{Debt_{c,ss}}{Y_{c,ss}} \right] \quad (\text{A.28})$$

where $\varrho_T \in [0, 1]$ measures the persistency of taxation and κ_T the strength of the fiscal adjustment. The government budget is:

$$G_{c,t} + Debt_{c,t-1} R_{c,t} = T_{c,t} + Debt_{c,t} \quad (\text{A.29})$$

Total debt is the sum of outstanding debt:

$$Debt_{c,t} = \sum_c n_c B_{l,c,t} + B_{sc,t} \quad (\text{A.30})$$

in the case of the U.S. it also includes central bank purchases of U.S. securities as reserve assets.

The stablecoin issuer purchases government bonds using seigniorage from stablecoins minting. Stablecoins are backed one-to-one by US assets, are not remunerated and produced with negligible costs. The balance sheet of the issuer is:

$$B_{sc,t} = \sum_{c=\{US,F,S\}} n_c SC_{c,t} \quad (\text{A.31})$$

profits from arbitrage are: $\Pi_{sc,t} = B_{sc,t-1} R_{US,t}$.

All shocks follow an AR(1) process.

A.7 Log-linearization

Consider Equation (9), which under the parametrization of the previous section takes the form of:

$$\beta_c E_t \left[\frac{\lambda_{c,t+1}}{NER_{US,c,t+1} \pi_{c,t+1}} \right] (R_{US,t} - R_{sc,t}) = \gamma_{c,t} \left(\frac{SC_{c,t}}{C_{c,t}} \right)^{\eta_m - 1} \frac{\mu_{sc,c,t}}{NER_{US,c,t}} - \frac{\chi_{c,b}}{NER_{US,c,t}} B_{US,c,t}^{-\sigma_{c,b}} \quad (\text{A.32})$$

for simplicity consider negligible cross-border transaction costs, $\phi_c^B \rightarrow 0$. Multiply on both sides for the nominal exchange rate against the dollar and call $\beta_c E_t \left[\frac{NER_{US,c,t} \lambda_{c,t}}{NER_{US,c,t+1} \pi_{c,t+1}} \right] = D_{c,t}$ and $(R_{US,t} - R_{sc,t}) = \mathcal{S}_t$. Taking the natural logs the previous become:

$$\ln(D_{c,t}) + \ln(\mathcal{S}_t) = \ln \left[\gamma_{c,t} \mu_{sc,c,t} \left(\frac{SC_{c,t}}{C_{c,t}} \right)^{\eta_m - 1} - \chi_{c,b} B_{US,c,t}^{-\sigma_{c,b}} \right] \quad (\text{A.33})$$

linearize around the steady-state (denoted by ss), calling $\hat{x}$ variables in percent deviation from the steady-state:

$$\begin{aligned} \hat{D}_{c,t} + \frac{R_{US,ss}}{\mathcal{S}_t} R_{US,t} &= \frac{1}{\gamma_{c,ss} \mu_{sc,c,ss} \left(\frac{SC_{c,ss}}{C_{c,ss}} \right)^{\eta_m} - \chi_{c,b} B_{US,c,ss}^{-\sigma_{c,b}}} \left\{ \left[\gamma_{c,ss} \mu_{sc,c,ss} \left(\frac{SC_{c,ss}}{C_{c,ss}} \right)^{-\eta_m} \right] \hat{\gamma}_{c,t} + \right. \\ &(\eta_m - 1) \left[\gamma_{c,ss} \mu_{sc,c,ss} \left(\frac{SC_{c,ss}}{C_{c,ss}} \right)^{\eta_m - 1} \right] S \hat{C}_{c,t} - (\eta_m - 1) \left[\gamma_{c,ss} \mu_{sc,c,ss} \left(\frac{SC_{c,ss}}{C_{c,ss}} \right)^{\eta_m - 1} \right] \hat{C}_{c,t} + \\ &+ \left[\gamma_{c,ss} \mu_{sc,c,ss} \left(\frac{SC_{c,ss}}{C_{c,ss}} \right)^{\eta_m - 1} \right] \hat{\mu}_{sc,c,t} + \left[\gamma_{c,ss} \left(\frac{SC_{c,ss}}{C_{c,ss}} \right)^{\eta_m - 1} \beta_\mu \exp \{1 - R_{US,ss}\} R_{US,ss} \right] R_{US,t} \hat{U}_{c,t} + \\ &\left. - \left[\gamma_{c,ss} \left(\frac{SC_{c,ss}}{C_{c,ss}} \right)^{\eta_m - 1} \gamma_\mu \right] \hat{\rho}_t + \chi_{c,b} \sigma_{c,b} B_{US,c,ss}^{-\sigma_{c,b}} \hat{B}_{c,t} \right\} \quad (\text{A.34}) \end{aligned}$$

the elasticity of stablecoin holdings to U.S. yields ($R_{US,t}$) is:

$$(\eta_m - 1) \frac{\left[\gamma_{c,ss} \mu_{sc,c,ss} \left(\frac{SC_{c,ss}}{C_{c,ss}} \right)^{\eta_m - 1} \right]}{\gamma_{c,ss} \mu_{sc,c,ss} \left(\frac{SC_{c,ss}}{C_{c,ss}} \right)^{\eta_m - 1} - \chi_{c,b} B_{US,c,ss}^{-\sigma_{c,b}}} \frac{1}{\Gamma_{ss}} \quad (\text{A.35})$$

with $\Gamma_{ss} \equiv \frac{R_{US,ss}}{\mathcal{S}_t} - \frac{\left[\gamma_{c,ss} \left(\frac{SC_{c,ss}}{C_{c,ss}} \right)^{\eta_m - 1} \beta_\mu \exp \{1 - R_{US,ss}\} R_{US,ss} \right]}{\gamma_{c,ss} \mu_{sc,c,ss} \left(\frac{SC_{c,ss}}{C_{c,ss}} \right)^{\eta_m - 1} - \chi_{c,b} B_{US,c,ss}^{-\sigma_{c,b}}}$ the denominator of the previous function is always positive, while the numerator can change sign depending on the

share of stablecoins in total consumption. For large stablecoin holdings, $\left(\frac{SC_{c,ss}}{C_{c,ss}}\right)^{\eta_m-1} \rightarrow 0$, hence $\gamma_{c,ss}\mu_{sc,c,ss}\left(\frac{SC_{c,ss}}{C_{c,ss}}\right)^{\eta_m-1} - \chi_{c,b}B_{US,c,ss}^{-\sigma_{c,b}}$ becomes negative.

Table A.1: Calibration

Parameter	Description	Value	Parameter	Description	Value
h_c	Habit formation	0.65	$\gamma_{c,r}$	Interest rate smoothing	0.75
ϕ_c	Inverse of Frish elasticity of labor	1	$\theta_{c,Y}$	Interest rate sensitivity to output	0.6
β_c	Discount factor	0.9926	$\theta_{c,\pi}$	Interest rate sensitivity to inflation	2
$\sigma_{c,m}$	Elasticity of money	10.62	$\kappa_{c,T}$	Tax sensitivity to deficit	0.48
$\sigma_{c,b}$	Elasticity of bonds	10.62	$\varrho_{c,T}$	Persistency of taxation	0.5
ϕ_c^K	Investment costs	1.728	$\frac{G_{ss}}{Y_{ss}}$	steady-state gov. spending over output	0.2
$\omega_{c,c}$	Home bias	0.75	$\rho_{c,R}$	Persistency of monetary shocks	0.2
ϱ_c	Elasticity of substitution across goods	1/3	$\rho_{c,A}$	Persistency of TFP shocks	0.5
δ_c	Capital depreciation rate	0.025	$\sigma_{c,r}$	Volatility of monetary shocks	0.01
c	Prob. of price update	0.6	$\sigma_{c,A}$	Volatility of TFP shocks	0.01
ν_c	Demand elasticity	6	ϕ_c^B	Cross-country bond holding cost	0.001
α_c	Capital share in production	0.3			

A.8 Additional figures

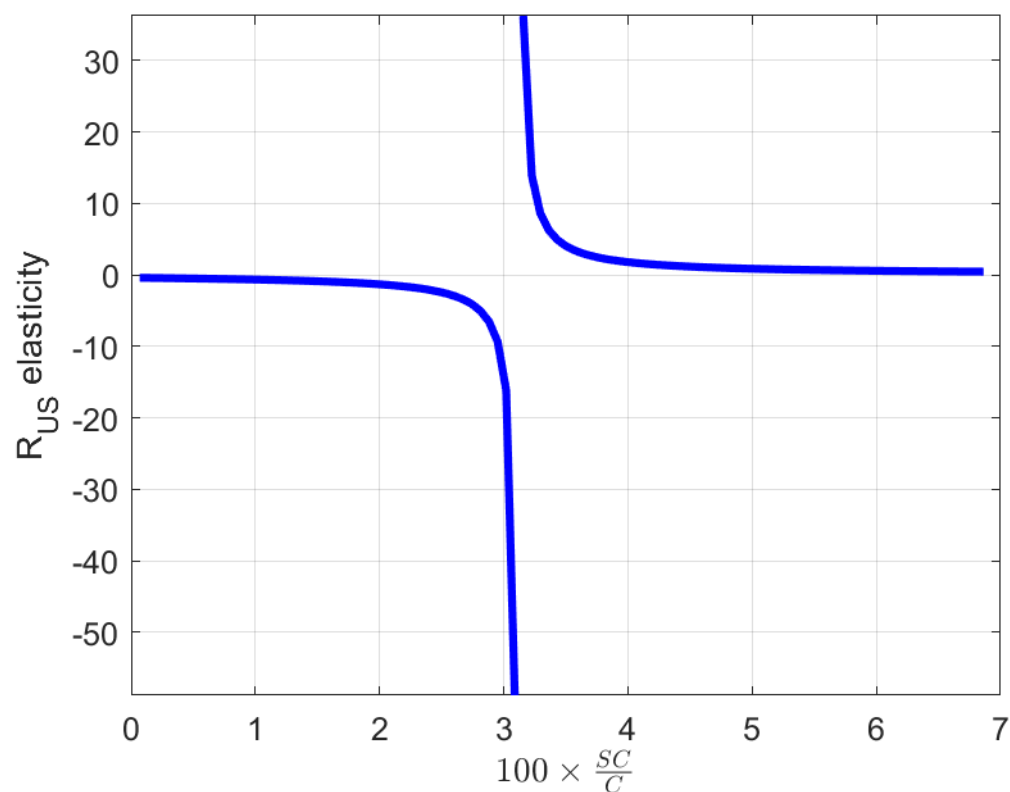


Figure A.1: Demand for stablecoins against returns differential – extreme holdings.

Notes: the chart shows the elasticity of U.S. yields (vertical axis) against stablecoin demand (horizontal axis). The elasticity is computed as in [Equation \(10\)](#) by keeping all other variables constant and changing values of stablecoins. Holdings of stablecoins are allowed to increase up until the elasticity is reversed.

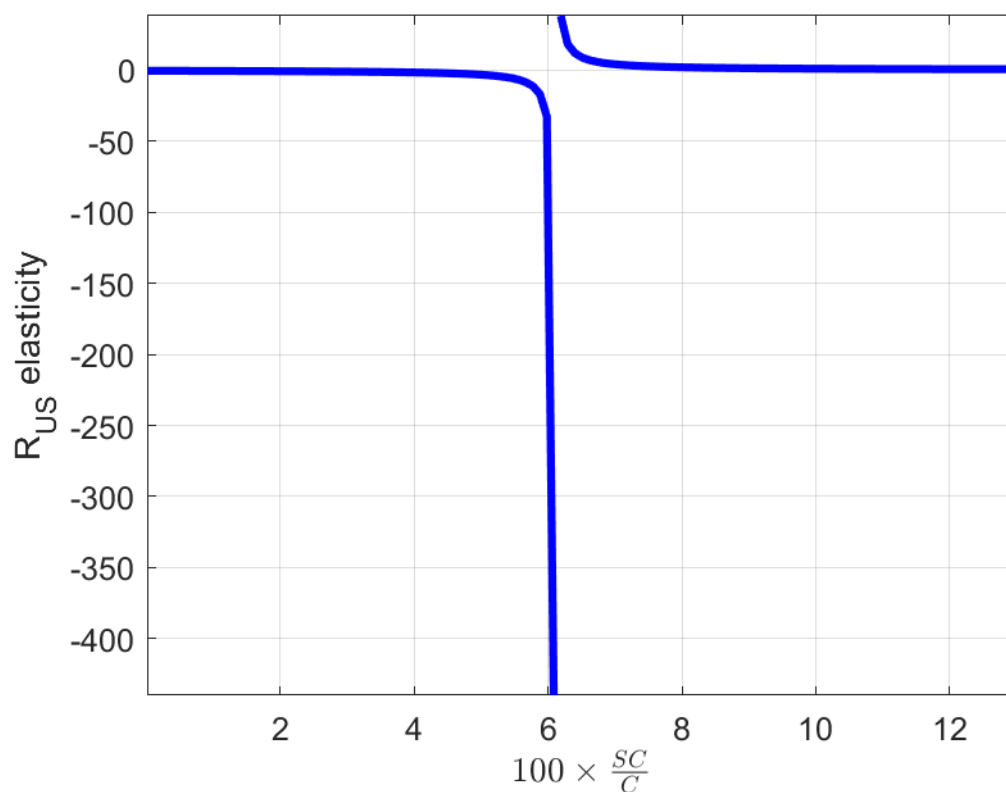


Figure A.2: Demand for stablecoins against returns differential – full solution.

Notes: the chart shows the elasticity of U.S. yields (vertical axis) against stablecoin demand (horizontal axis). The elasticity is computed as in [Equation \(10\)](#) by solving the model for different values of μ_{sc} . Holdings of stablecoins are allowed to increase up until the elasticity is reversed.

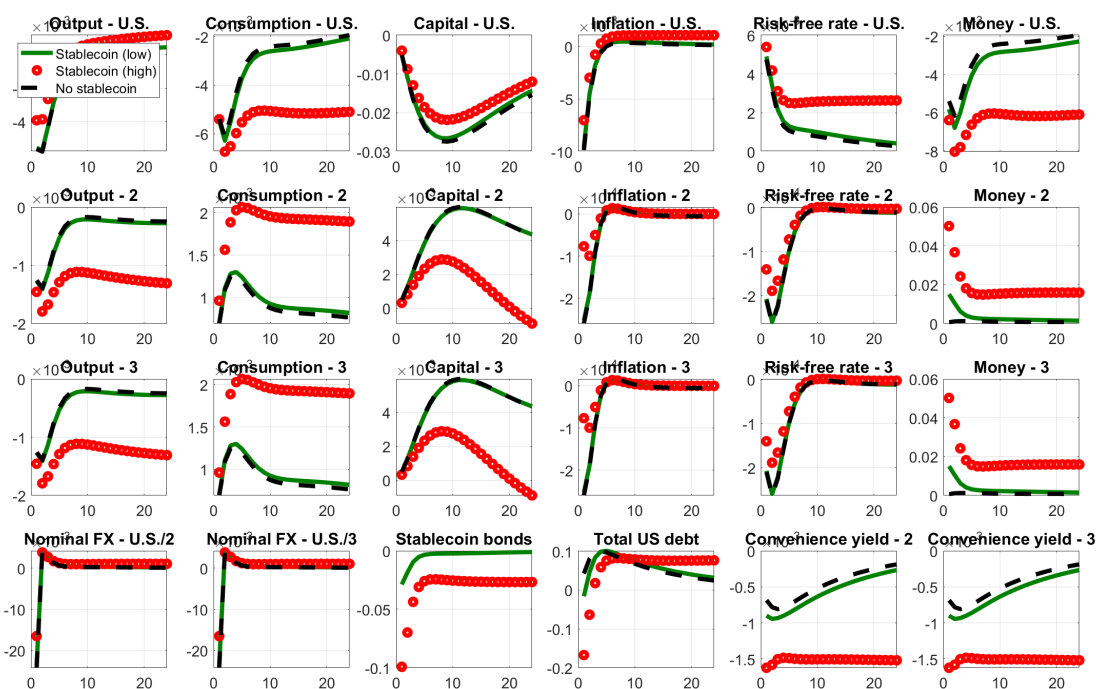


Figure A.3: Impulse responses to a U.S. monetary policy shock.

Notes: the chart shows the impulse response to a 1 standard deviation U.S. monetary policy shock. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market).

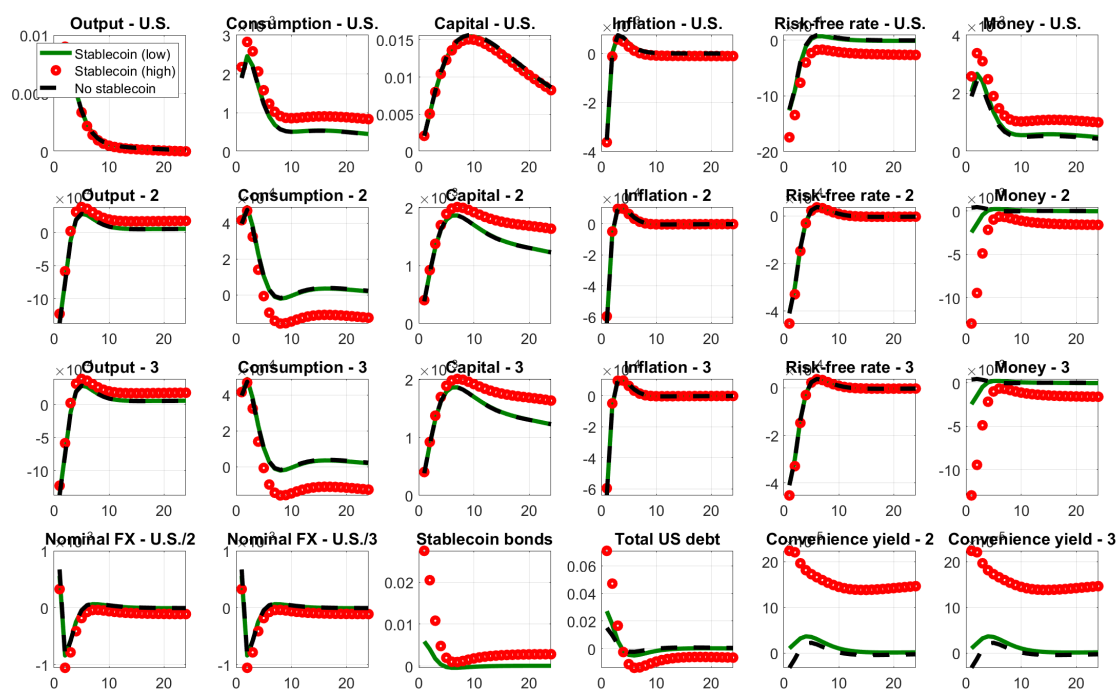


Figure A.4: Impulse responses to a U.S. TFP shock.

Notes: the chart shows the impulse response to a 1 standard deviation U.S. TFP shock. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market).

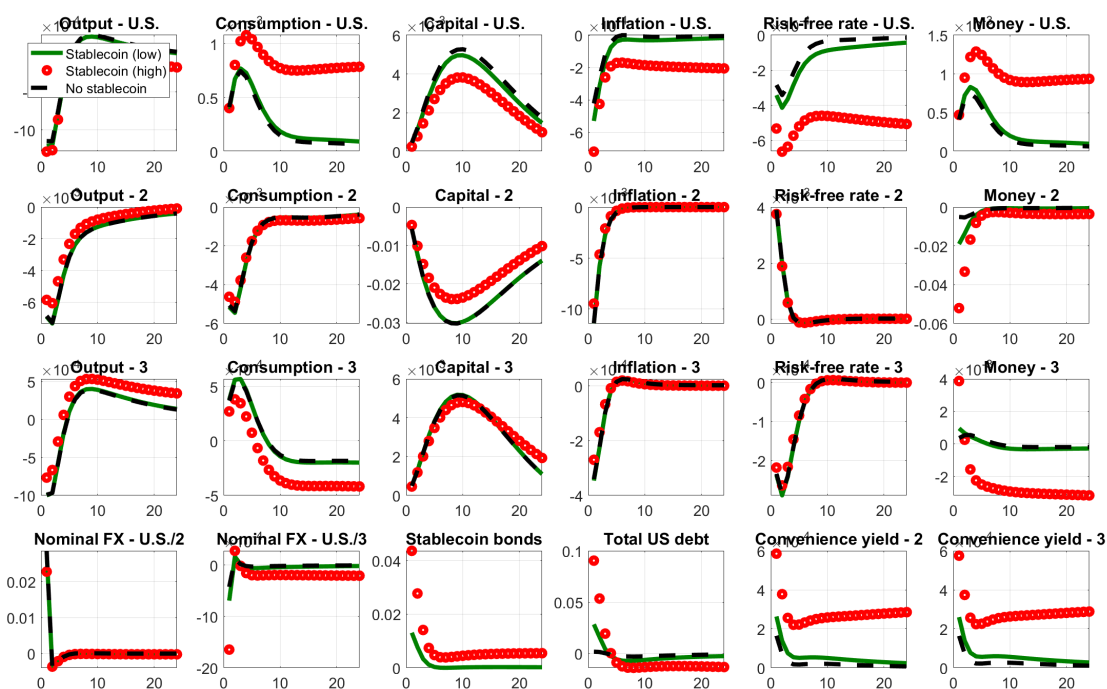


Figure A.5: Impulse responses to a monetary policy shock from country 2.

Notes: the chart shows the impulse response to a 1 standard deviation monetary policy shock from country 2. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market).

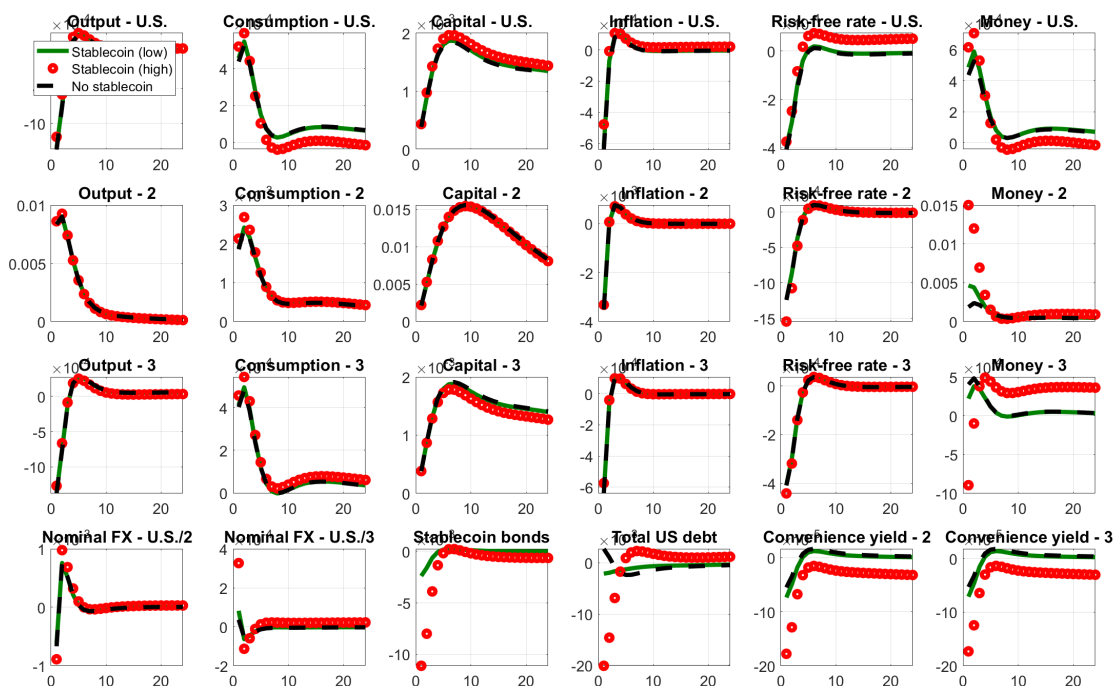


Figure A.6: Impulse responses to a TFP shock from country 2.

Notes: the chart shows the impulse response to a 1 standard deviation TFP shock from country 2. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market).

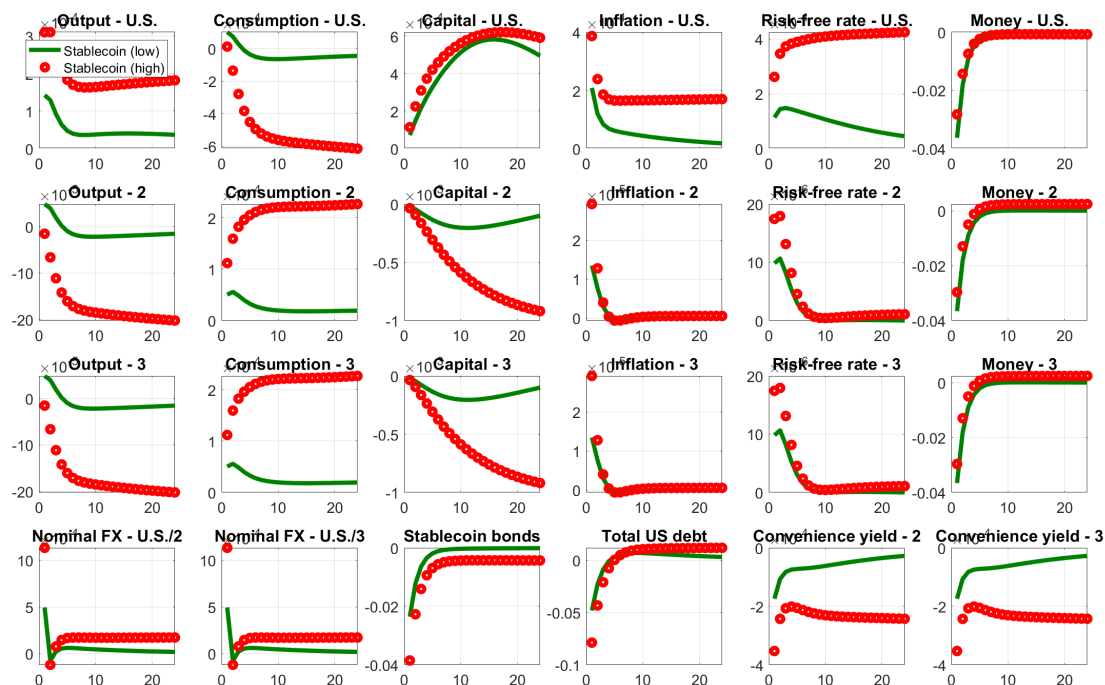


Figure A.7: Impulse responses to stablecoin preference shock.

Notes: the chart shows the impulse response to a stablecoin preference shock. The the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market).

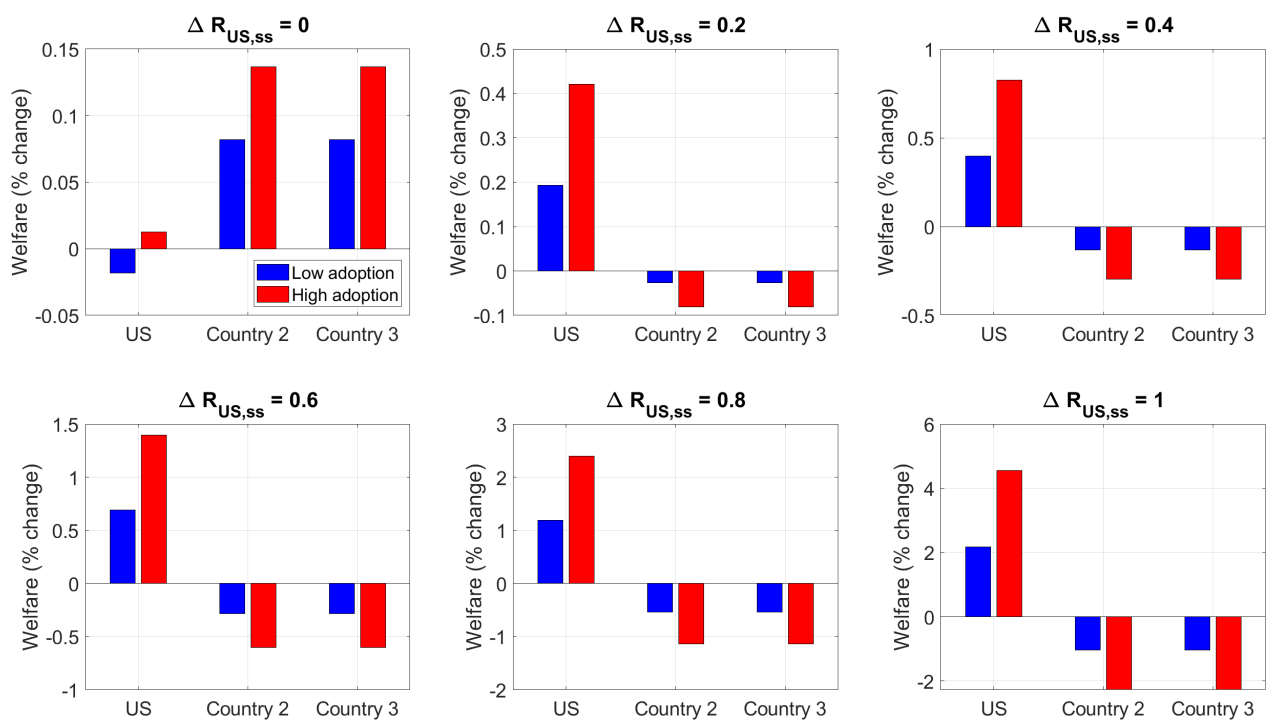


Figure A.8: Welfare changes relative to the model without stablecoins for different level of U.S. yield reduction.

Notes: the chart shows the percent change in welfare relative to the model without stablecoins for the U.S. and the other two country depending on how much stablecoin adoption reduces equilibrium U.S. yields. The model is solved at second order with pruning. Welfare is defined recursively, i.e. $\mathcal{W}_{c,t} = U_{c,t} + \beta_c E_t \mathcal{W}_{c,t+1}$, with U_c the period utility function.

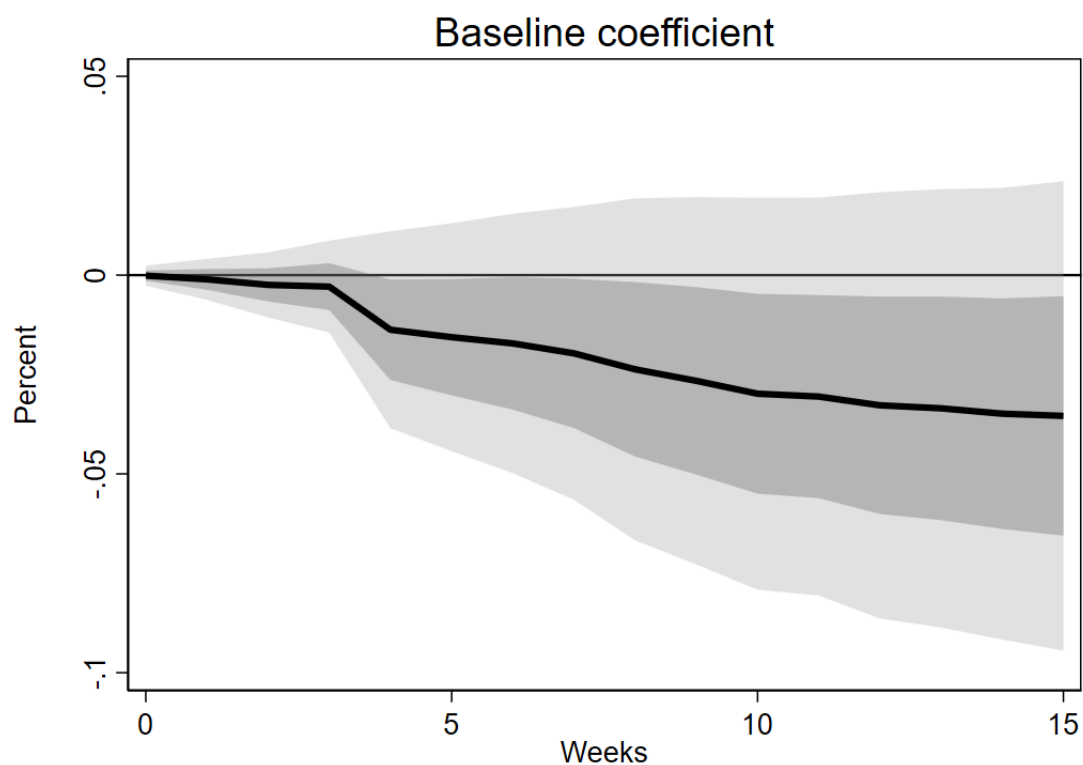


Figure A.9: Response of stablecoin market capitalization to U.S. monetary policy shocks. **Notes:** the chart shows the impulse response of the stablecoin market capitalization (proxied by the sum of USDT and USDC) to a 25 basis points U.S. Local projection control for 4 lags of the dependent variable, the (log) U.S. dollar nominal effective exchange rate, the (log) U.S. stock price, the log of stablecoins market capitalization (proxied by USDT and USDC market capitalization), a time trend and year dummies. Equation (12) is estimated without interaction terms.

B Alternative specifications

B.1 Constant preferences for stablecoins

This section simulates the model under constant preferences for stablecoins $\mu_{sc,c,t} = \mu_{sc,ss}$. In this case, stablecoin preferences do not change with higher or lower interest rates, implicitly assuming that households do not consider the possibility of a default of the stablecoin issuer. Under this specification results are qualitatively the same, only slightly smaller in magnitude, showing the amplification effect of the riskiness channel.

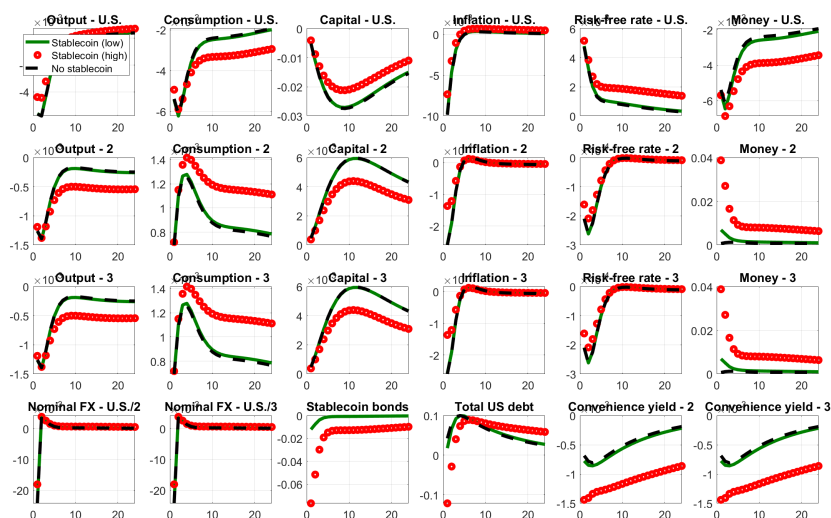


Figure B.1: Impulse responses to a U.S. monetary policy shock – constant $\mu_{sc,c}$.

Notes: the chart shows the impulse response to a 1 standard deviation U.S. monetary policy shock. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). $\mu_{sc,c}$ is constant.

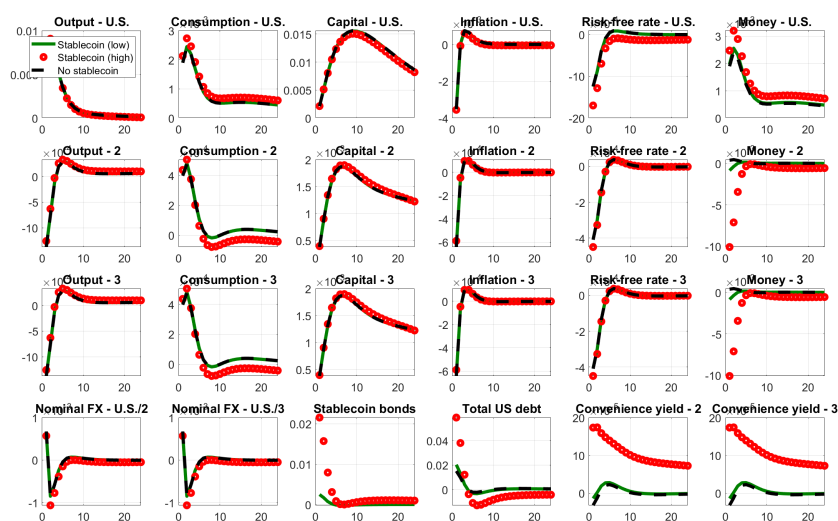


Figure B.2: Impulse responses to a U.S. TFP shock – constant $\mu_{sc,c}$.

Notes: the chart shows the impulse response to a 1 standard deviation U.S. monetary policy shock. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). $\mu_{sc,c}$ is constant.

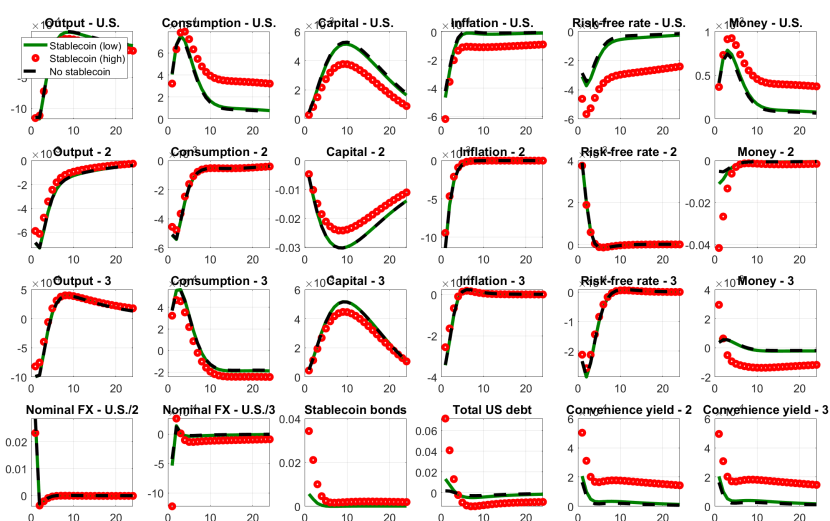


Figure B.3: Impulse responses to a monetary policy shock from country 2 – constant $\mu_{sc,c}$.

Notes: the chart shows the impulse response to a 1 standard deviation monetary policy shock from country 2. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). $\mu_{sc,c}$ is constant.

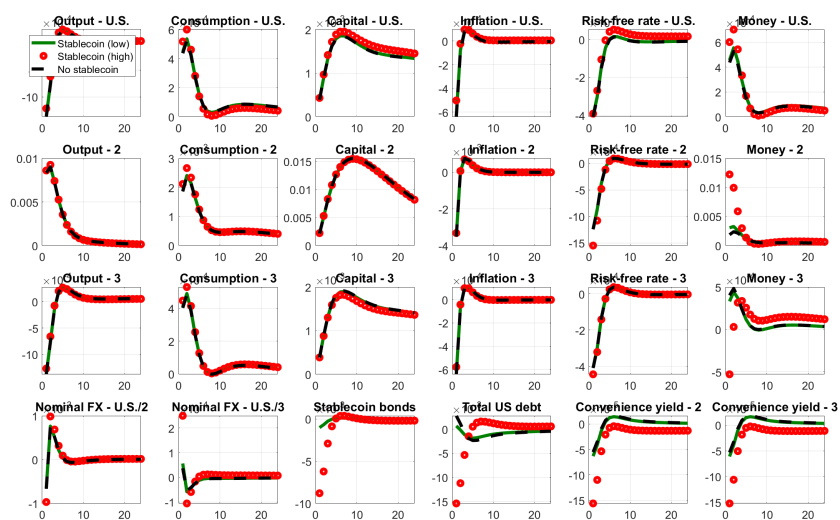


Figure B.4: Impulse responses to a TFP shock from country 2 – constant $\mu_{sc,c}$.

Notes: the chart shows the impulse response to a 1 standard deviation TFP shock from country 2. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). $\mu_{sc,c}$ is constant.

B.2 Money in the utility

We consider an alternative specification to generate demand for liquid instrument in the model, the money-in-utility framework. As shown by [Feenstra \(1986\)](#), money in utility captures a wide range of micro-founded frictions that generate demand for payment instruments, such as cash-in-advance or shopping-time constraints. Specifically, we assume that households directly derive utility from holding liquid instruments:

$$U_{c,t} = e_{c,t}^C \ln(C_{c,t} - h_c C_{c,t-1}) - \frac{\chi_{c,l}}{1 + \phi_{c,l}} L_{c,t}^{1+\phi_{c,l}} + \frac{\chi_{c,l}}{1 - \sigma_{c,l}} (\mathcal{L}_{c,t})^{1-\sigma_{c,l}} + \frac{\chi_{c,b}}{1 - \sigma_{c,b}} \left(\frac{B_{US,c,t}}{P_{c,t} NER_{US,c,t}} \right)^{1-\sigma_{c,b}} \quad (\text{B.1})$$

where total liquidity $\mathcal{L}$ is: $\left[\mu_{sc,c,t} \left(\frac{SC_{c,t}}{NER_{US,c,t}} \right)^{\eta_m} + (1 - \mu_{sc,c,t}) (M_{c,t})^{\eta_m} \right]^{\frac{1}{\eta_m}}$.

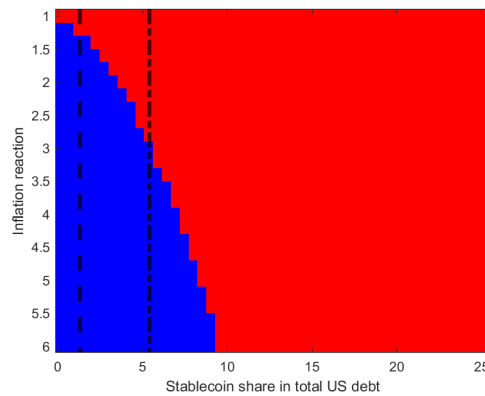


Figure B.5: Stability map.

Notes: the chart shows combinations of values for the US Taylor rule inflation reaction parameter (θ_π) and the stablecoin preference parameter (μ_{sc}). Blue areas indicate that the model is stable, red areas that it is unstable, yellow areas that the model is stable, but wealth effects dominate leading to expansionary monetary policy tightening. Liquidity demand is introduced as money-in-utility.

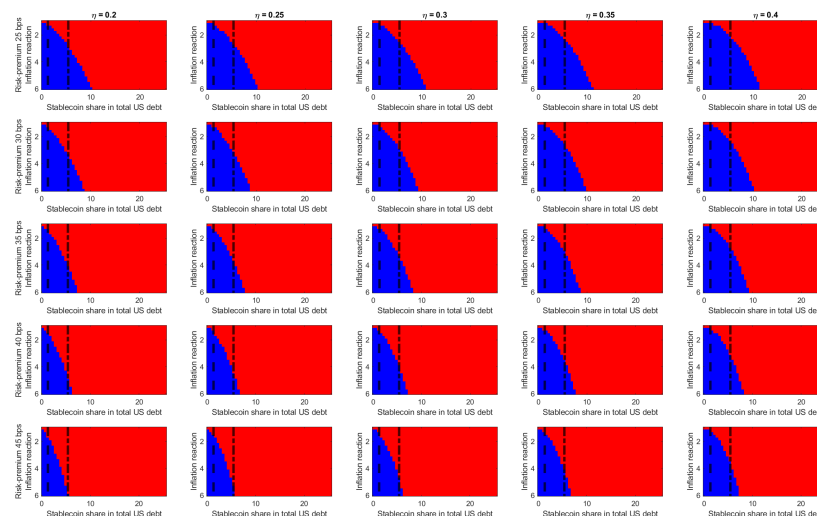


Figure B.6: Stability map.

Notes: the chart shows combinations of values for the US Taylor rule inflation reaction parameter (θ_π) and the stablecoin preference parameter (μ_{sc}). Blue areas indicate that the model is stable, red areas that it is unstable, yellow areas that the model is stable, but wealth effects dominate leading to expansionary monetary policy tightening. Different calibrations for the US convenience yield and the CES liquidity aggregator parameter are considered. Liquidity demand is introduced as money-in-utility.

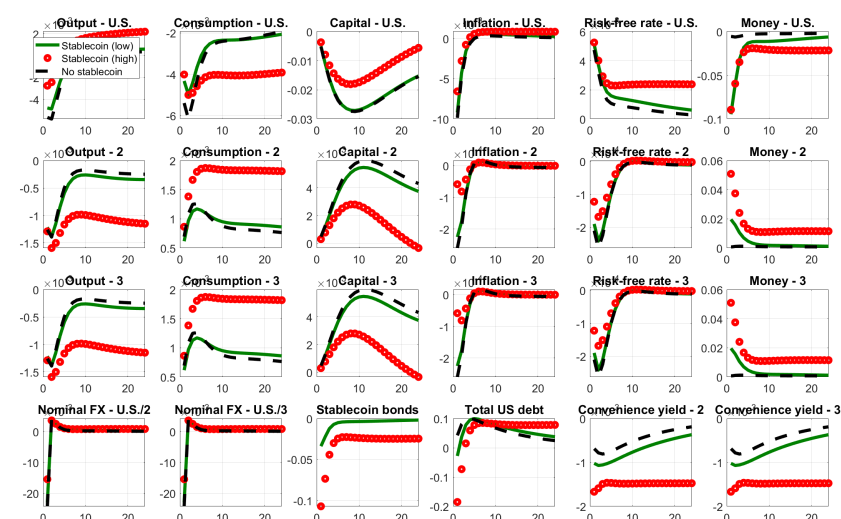


Figure B.7: Impulse responses to a U.S. monetary policy shock – money-in-utility.

Notes: the chart shows the impulse response to a 1 standard deviation U.S. monetary policy shock. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). Demand for payment instruments is generated through money-in-utility framework.

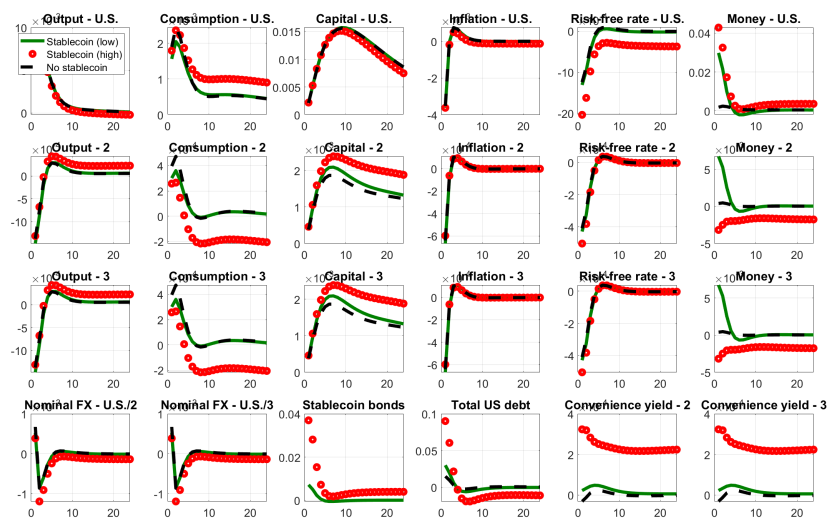


Figure B.8: Impulse responses to a U.S. TFP shock – money-in-utility.

Notes: the chart shows the impulse response to a 1 standard deviation U.S. monetary policy shock. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). Demand for payment instruments is generated through money-in-utility framework.

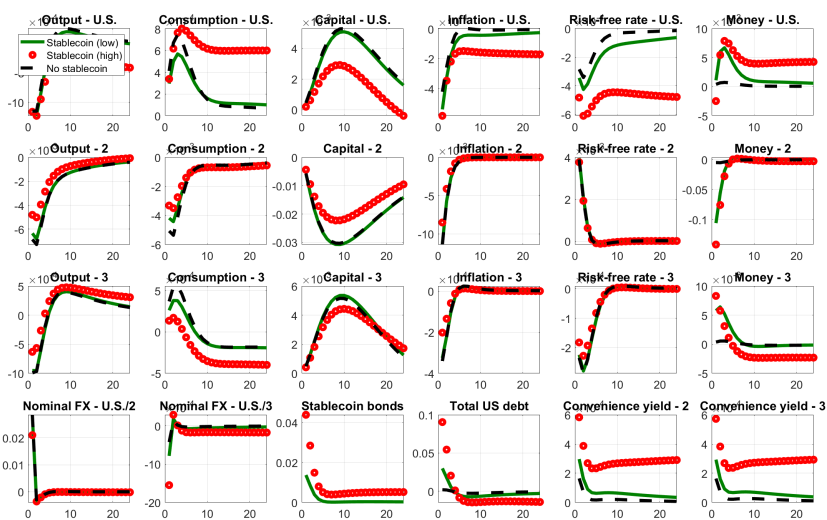


Figure B.9: Impulse responses to a monetary policy shock from country 2 – money-in-utility.

Notes: the chart shows the impulse response to a 1 standard deviation monetary policy shock from country 2. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). Demand for payment instruments is generated through money-in-utility framework.

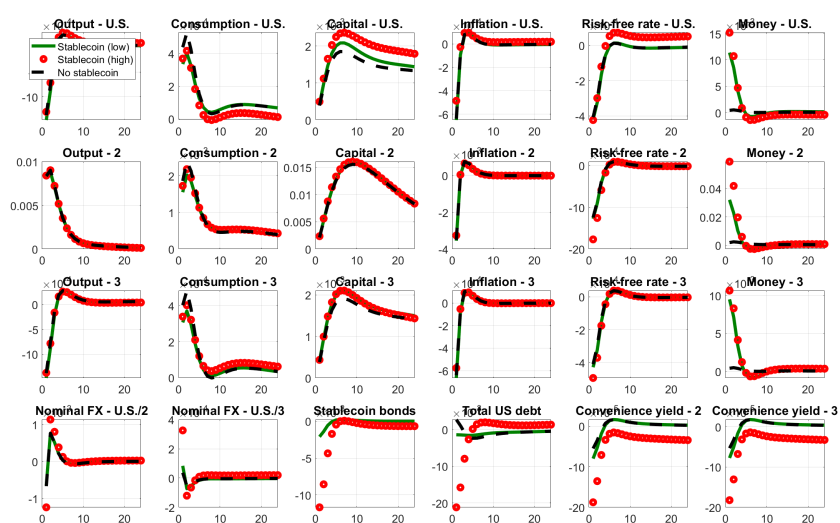


Figure B.10: Impulse responses to a TFP shock from country 2 – money-in-utility.

Notes: the chart shows the impulse response to a 1 standard deviation TFP shock from country 2. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). Demand for payment instruments is generated through money-in-utility framework.

B.3 Stablecoins only in the domestic or foreign economy

In this section we compare two calibrations of the model one in which stablecoins are issued only domestically and one in which they are available only to foreign households. Figure B.11 shows the stability plot for both cases. As the figure shows, macroeconomic instability arises only from foreign demand for stablecoins. Impulse responses confirm this intuition: when stablecoins are held only in the U.S. macroeconomic spillovers are limited. On the contrary, most of the dynamics are determined by international demand for stablecoins. This highlights the role of the global safe asset channel, which is reinforced through international demand for stablecoins, as a multiplier of shocks.

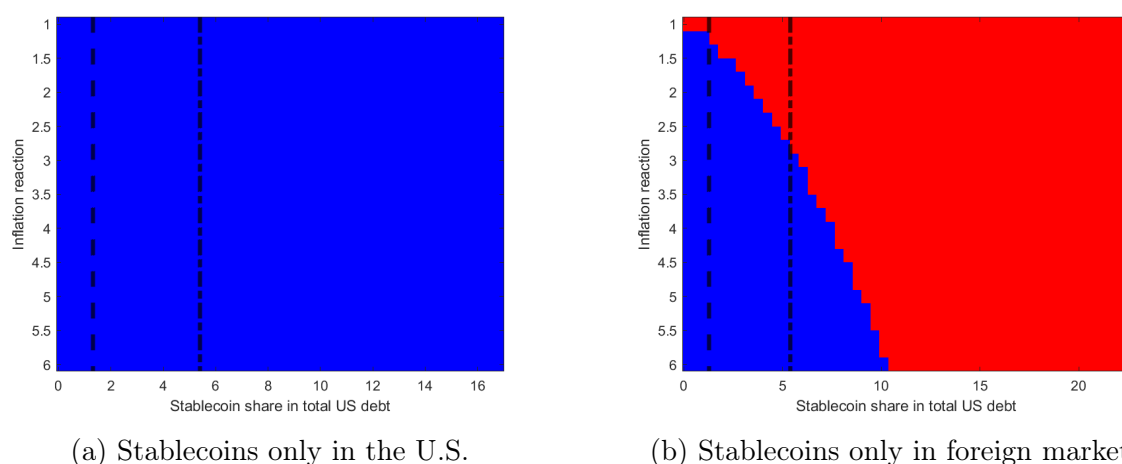


Figure B.11: Stability map when stablecoins are issued only domestically or only internationally.

Notes: Panel a, stablecoins are issued only in the domestic economy (U.S.). Panel b, stablecoins are issued only in the foreign economies. The chart shows combinations of values for the US Taylor rule inflation reaction parameter (θ_π) and the stablecoin preference parameter (μ_{sc}). Blue areas indicate that the model is stable and red areas that it is unstable.

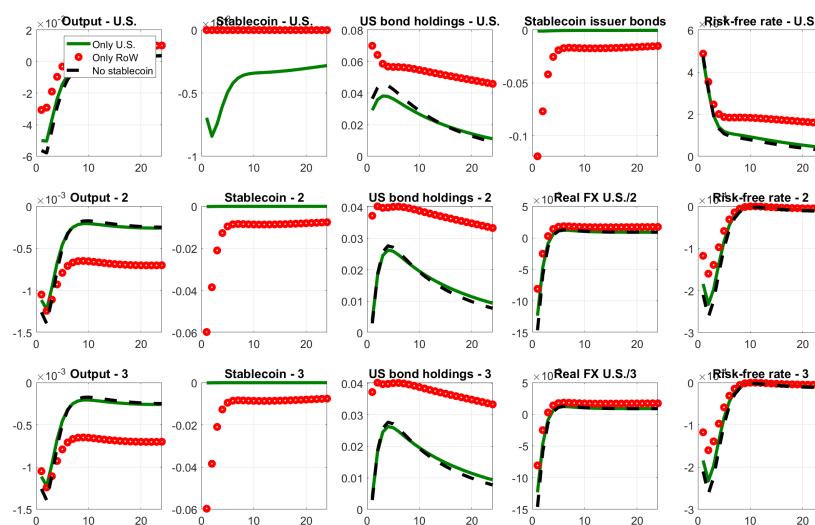


Figure B.12: Impulse responses to a U.S. monetary policy shock – U.S. vs foreign issuance.
Notes: the chart shows the impulse response to a 1 standard deviation U.S. monetary policy shock. The black line shows the responses of the model without stablecoins, the green line the responses of the model with stablecoins issued only in the U.S. while the red dots show the response of the model with stablecoins issued only in foreign countries. Both models are simulated for high steady-state holdings of stablecoins.

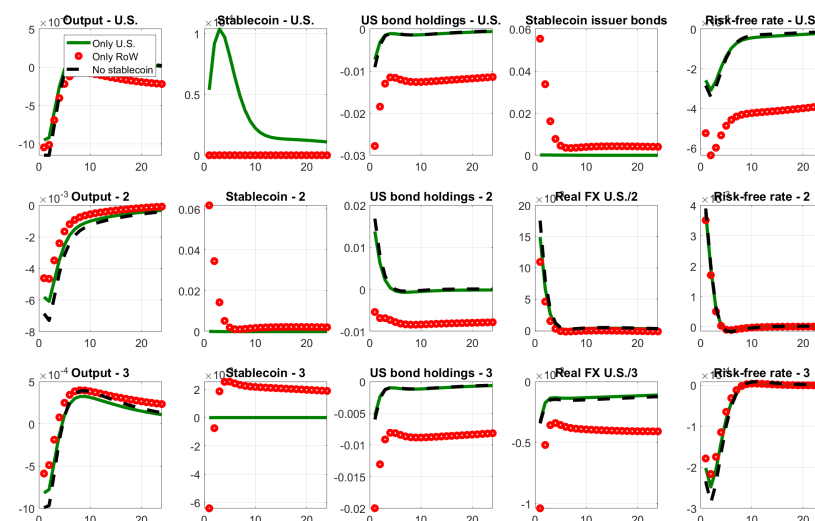


Figure B.13: Impulse responses to a monetary policy shock from country 2 – U.S. vs foreign issuance.

Notes: the chart shows the impulse response to a 1 standard deviation monetary policy shock from country 2. The black line shows the responses of the model without stablecoins, the green line the responses of the model with stablecoins issued only in the U.S. while the red dots show the response of the model with stablecoins issued only in foreign countries. Both models are simulated for high steady-state holdings of stablecoins.

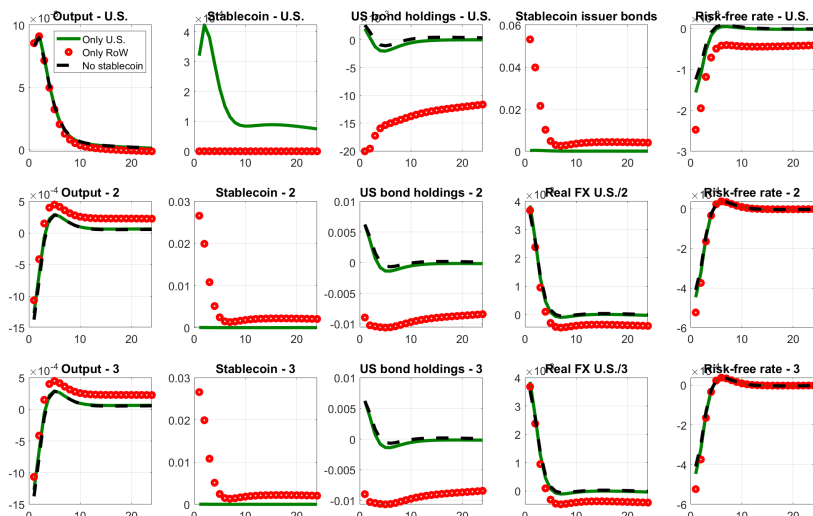


Figure B.14: Impulse responses to a U.S. TFP shock – U.S. vs foreign issuance.

Notes: the chart shows the impulse response to a 1 standard deviation U.S. TFP shock. The black line shows the responses of the model without stablecoins, the green line the responses of the model with stablecoins issued only in the U.S. while the red dots show the response of the model with stablecoins issued only in foreign countries. Both models are simulated for high steady-state holdings of stablecoins.

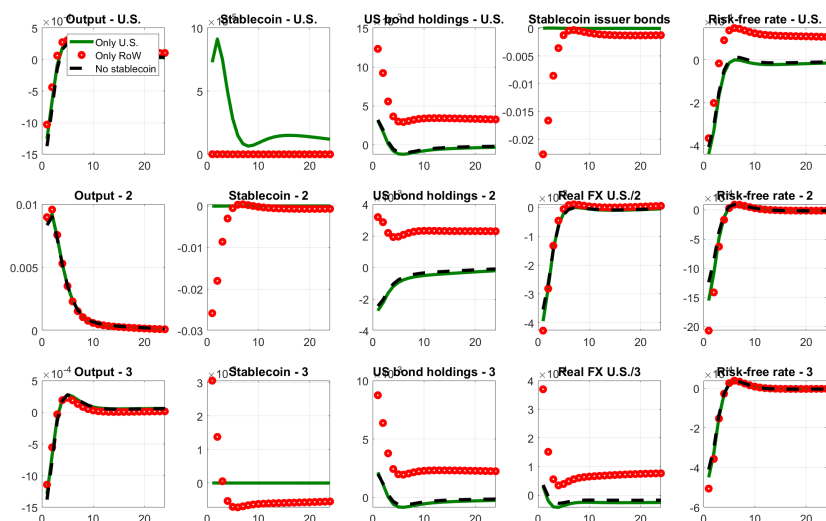


Figure B.15: Impulse responses to a TFP shock from country 2 – U.S. vs foreign issuance.

Notes: the chart shows the impulse response to a 1 standard deviation TFP shock from country 2. The black line shows the responses of the model without stablecoins, the green line the responses of the model with stablecoins issued only in the U.S. while the red dots show the response of the model with stablecoins issued only in foreign countries. Both models are simulated for high steady-state holdings of stablecoins.

B.4 Estimation of key parameters

We use the estimate of key parameters as in [Ferrari Minesso and Pagliari \(2023\)](#). The U.S. accounts for 42% of world GDP, the euro area (country 2) for 28 while rest of the world takes the remaining.

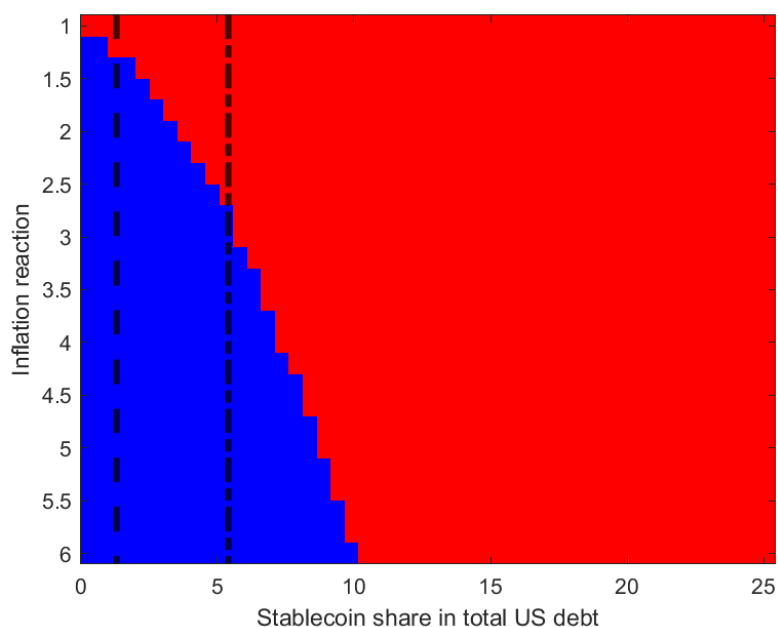


Figure B.16: Stability map.

Notes: the chart shows combinations of values for the US Taylor rule inflation reaction parameter (θ_{π}) and the stablecoin preference parameter (μ_{sc}). Blue areas indicate that the model is stable and red areas that it is unstable. Estimated parameters from [Ferrari Minesso and Pagliari \(2023\)](#) are used.

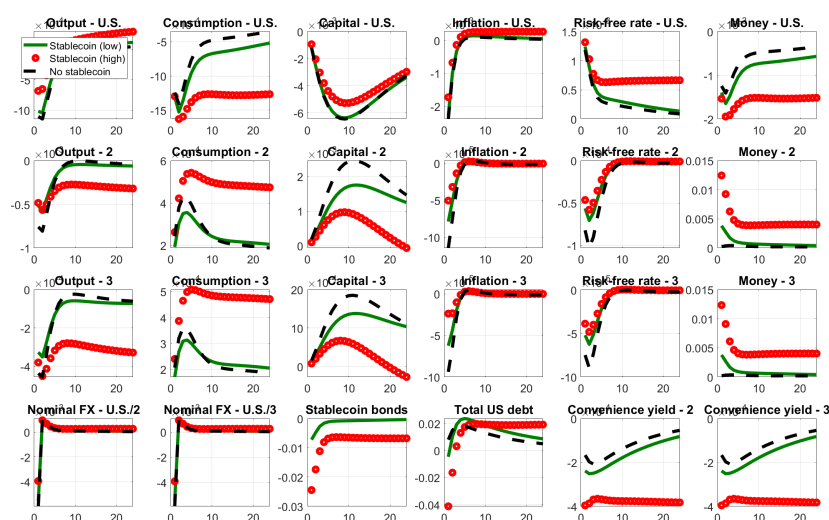


Figure B.17: Impulse responses to a U.S. monetary policy shock – estimated model.

Notes: the chart shows the impulse response to a 1 standard deviation U.S. monetary policy shock. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). We use parameter estimates and country weights as in [Ferrari Minesso and Pagliari \(2023\)](#).

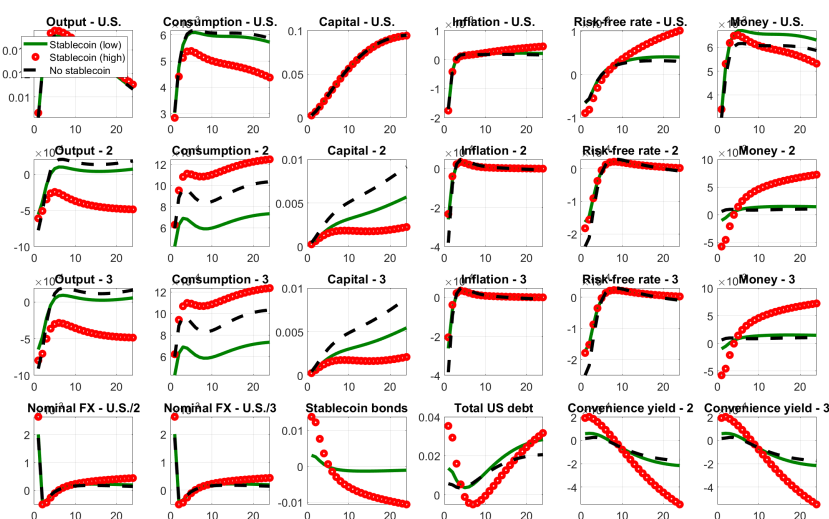


Figure B.18: Impulse responses to a U.S. TFP shock – estimated model.

Notes: the chart shows the impulse response to a 1 standard deviation U.S. monetary policy shock. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). We use parameter estimates and country weights as in [Ferrari Minesso and Pagliari \(2023\)](#).

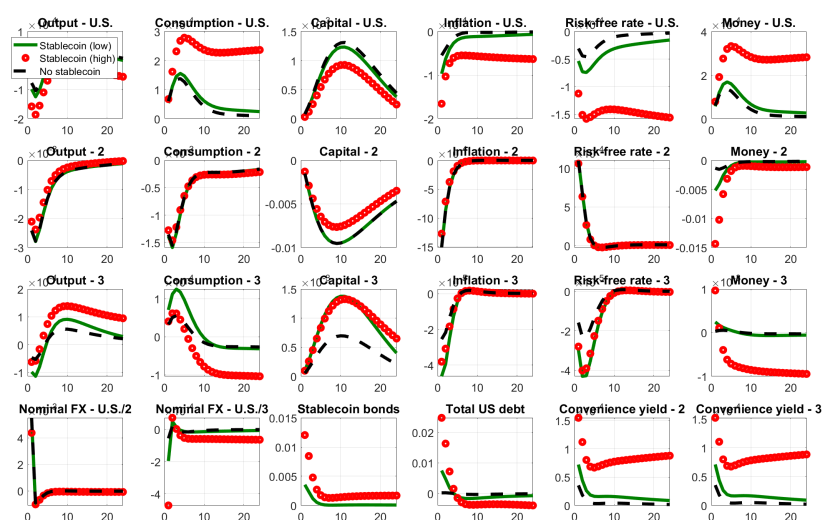


Figure B.19: Impulse responses to a monetary policy shock from country 2 – estimated model.

Notes: the chart shows the impulse response to a 1 standard deviation monetary policy shock from country 2. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). We use parameter estimates and country weights as in [Ferrari Minesso and Pagliari \(2023\)](#).

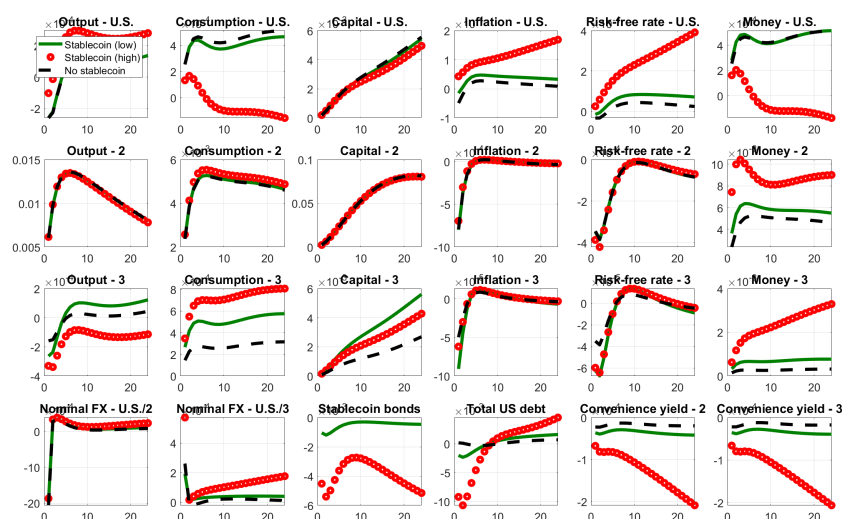


Figure B.20: Impulse responses to a TFP shock from country 2 – estimated model.

Notes: the chart shows the impulse response to a 1 standard deviation TFP shock from country 2. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). We use parameter estimates and country weights as in [Ferrari Minesso and Pagliari \(2023\)](#).

B.5 Redemption fee

We extend the model by including a very large redemption fee of 2%, which is 20 times larger than the one applied by Tether (while USD Circle does not have any). Expectedly, redemption fees reduce the quantitative impact of stablecoins on yields and exchange rate as households balance less out of stablecoins following each shock. The main mechanism however still holds.

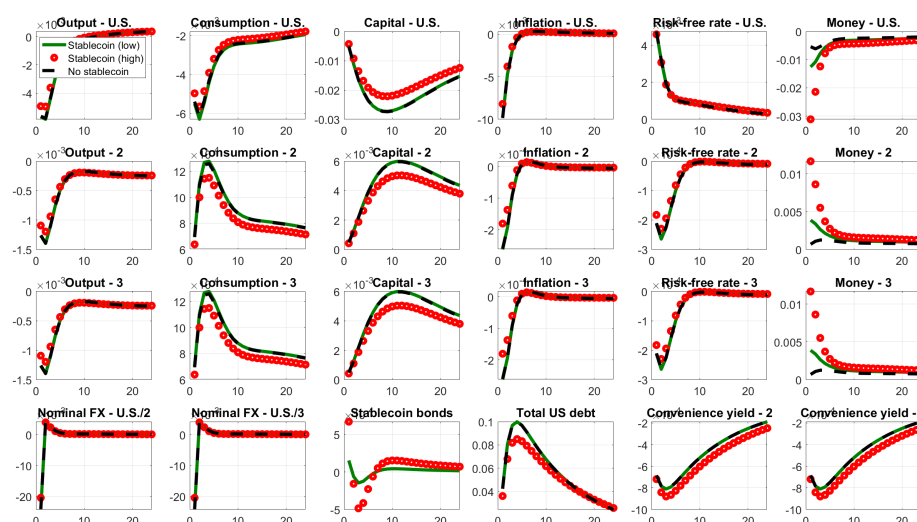


Figure B.21: Impulse responses to a U.S. monetary policy shock – model with redemption fee.

Notes: the chart shows the impulse response to a 1 standard deviation U.S. monetary policy shock. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). A 2% redemption fee on stablecoin holdings is imposed.

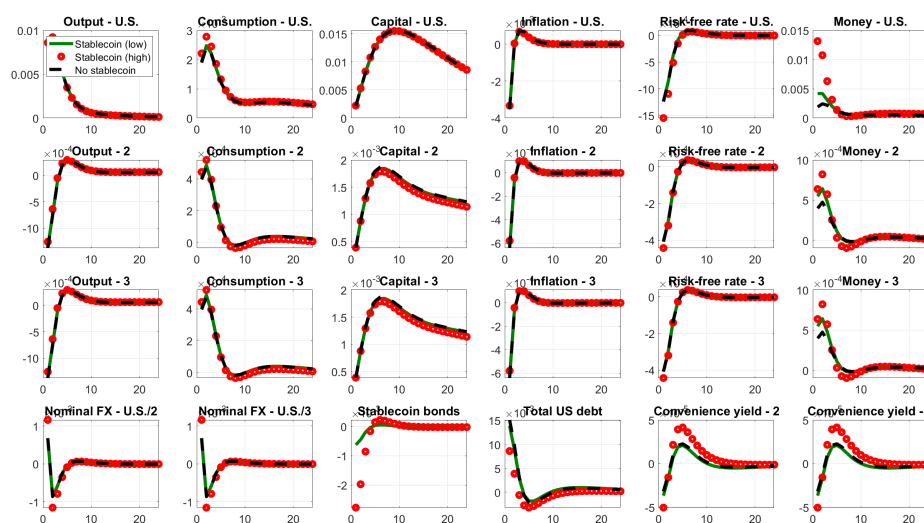


Figure B.22: Impulse responses to a U.S. TFP shock – model with redemption fee.

Notes: the chart shows the impulse response to a 1 standard deviation U.S. monetary policy shock. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). A 2% redemption fee on stablecoin holdings is imposed.

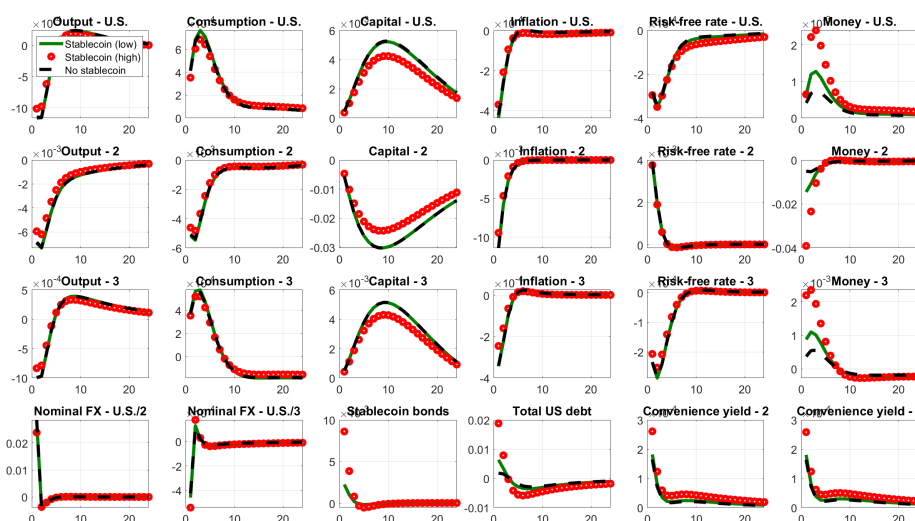


Figure B.23: Impulse responses to a monetary policy shock from country 2 – model with redemption fee.

Notes: the chart shows the impulse response to a 1 standard deviation monetary policy shock from country 2. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). A 2% redemption fee on stablecoin holdings is imposed.

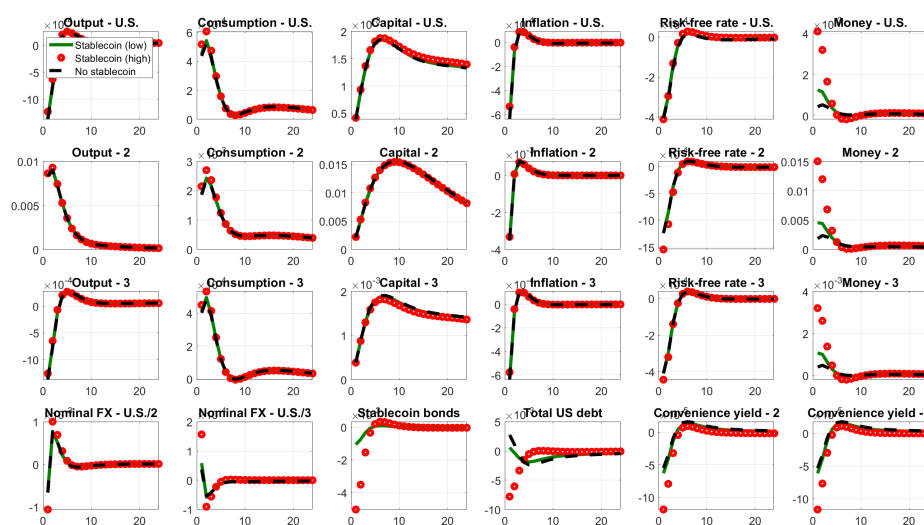


Figure B.24: Impulse responses to a TFP shock from country 2 – model with redemption fee.

Notes: the chart shows the impulse response to a 1 standard deviation TFP shock from country 2. The black line shows the responses of the model without stablecoins, the green line the responses of the model with low steady-state stablecoin market capitalization (about 500 USD bl, 1.3% of the U.S. bond market), the red dots the responses of the model with high steady-state stablecoin market capitalization (about 2 USD tr, 5.4% of the U.S. bond market). A 2% redemption fee on stablecoin holdings is imposed.

Acknowledgements

The authors are grateful to Carlo Altavilla, Michele Ca' Zorzi, Piero Cipollone, Luca Dedola, Sergio de Ferra, Johannes Gräß, Jean-François Jamet, Pierlauro Lopez, Arnaud Mehl, Fabrizio Perri, Ricardo Reis, Linda Rousoy, Julia Schmidt, Isabel Schnabel for early comments and to participants to the CAMEG 2025 Conference and seminars at the European Central Bank and Politecnico di Milano. The views expressed in this paper are those of the authors and do not necessarily represent the views of the European Central Bank or the Eurosystem.

Massimo Ferrari Minesso

European Central Bank, Frankfurt am Main, Germany; Complexity Lab in Economics. Catholic University of Milan, Milan, Italy; email: massimo.ferrari_minesso@ecb.europa.eu

Daniele Siena

Politecnico di Milano, Milan, Italy; email: daniele.siena@polimi.it

© European Central Bank, 2026

Postal address 60640 Frankfurt am Main, Germany

Telephone +49 69 1344 0

Website www.ecb.europa.eu

All rights reserved. Any reproduction, publication and reprint in the form of a different publication, whether printed or produced electronically, in whole or in part, is permitted only with the explicit written authorisation of the ECB or the authors.

This paper can be downloaded without charge from www.ecb.europa.eu, from the [Social Science Research Network electronic library](#) or from [RePEc: Research Papers in Economics](#). Information on all of the papers published in the ECB Working Paper Series can be found on the [ECB's website](#).

PDF

ISBN 978-92-899-7620-6

ISSN 1725-2806

doi:10.2866/1321559

QB-01-26-008-EN-N